

BODY BUILDERS GUIDE

GENERAL PRECAUTIONS

2015

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ISUZU MOTORS LIMITED

INTRODUCTION

The purpose of this **Body Builder's Guide** (hereinafter referred to as "guide") is to provide information and guidance to the body builders that take charge of the final stage of vehicle manufacture by mounting their vehicle bodies on the cab chassis (incomplete vehicles) provided by ISUZU MOTORS LIMITED. The body builders are encouraged to refer to the information in this guide in order to perform rear body mounting at suitable conditions with good use of their own experiences.

- This guide is a summary of a technical content for body mounting and remodeling according to mounting of ISUZU MOTORS incomplete vehicles.
- It is not that ISUZU MOTORS guarantee the quality of construction or the safety and running performance of the modified complete vehicle which remodeling done according to this guide.
- For body mounting and remodeling according to mounting, the body builders must design and manufacture so that there is no influence on safety, the running performance, and the function of genuine devices.
- The body builders are to comply with the national regulations with respect to their vehicle bodies, cab chassis modification incidental to rear body mounting, and completed vehicles after body mounting.
- The manufacturing responsibility of completed vehicles after body mounting is in the body builders. ISUZU MOTORS does not owe the obligation.
- ISUZU MOTORS is not responsible for any human injury or instrument damage due to using this guide.

Note: This guide is made commonly for various destination. There may be equipment or function described which is not developed depending on destination.

Follow the owner's manual for handling the vehicle or the equipments on vehicle.

The content of this guide might change without a previous notice. Contact ISUZU MOTORS LIMITED. and the ISUZU distributor or ISUZU dealership for exact details of latest information.

Chapter 1 GENERAL

1-1 PRECAUTIONS FOR INSTALLING BODY

1-2 PRECAUTIONS AT INSTALLING

1-3 PRECAUTIONS FOR BOLT TIGHTENING

1-1 PRECAUTIONS FOR INSTALLING BODY

When designing and examining body, consider regulation control in use country and agreement to standard, and safety and prevention of accidents of vehicle, and follow instructions below.

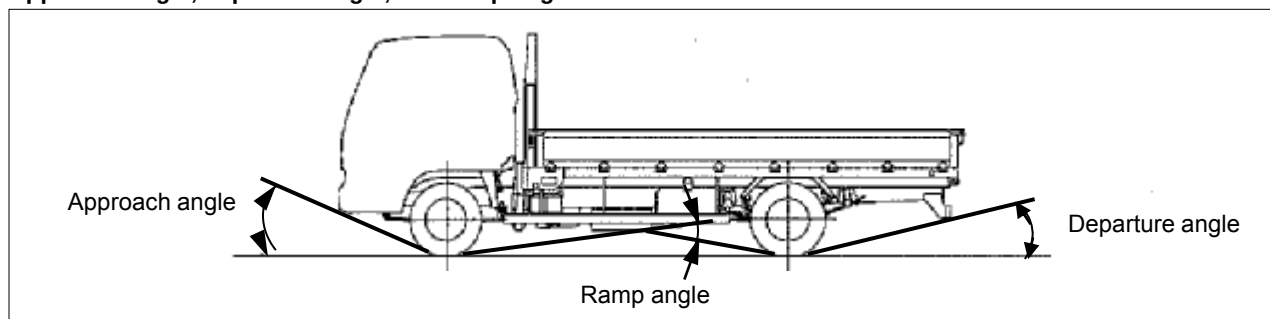
① Weight of body

- Design the rear body to evenly distribute the load of rear body on chassis frame.
 - Installation of special equipments on a chassis frame can change the said load distribution. Therefore, such special equipments must be installed, taking load distribution into account, otherwise stress concentration in particular part of the chassis frame may occur, possibly resulting in cracks in chassis frame members or other problems even if the total weight of these special equipments falls within allowable range. Confirm that there is no problem in safety in chassis frame member before the examination.
- If there is any equipment addition (ex. fuel tank addition) or weight trend to one side (ex. side door installation of the van body), adjust right and left weight difference on rear body.
 - Chassis frame ground height on "SPECIFIC INFORMATION" is described as standard equipment (full fuel) described in the shipment car type with there no right and left difference.

② Approach angle, departure angle, and ramp angle.

- When adding equipments on front overhang, rear overhang or lower part between axles, note that the approach angle, departure angle, and ramp angle do not go below the position of existing equipments.

Approach angle, departure angle, and ramp angle



③ Cab

- Give appropriate rust prevention treatment to the inside and surrounding the hole when drilling hole the inside or outside of the cab. Do similar processing when the surface of the cab is mistakenly damaged.
- When installing lever or switch to control rear body in the cab, maintain enough clearance from other levers and switches of existing and install it in the cab so as not to come in contact at operation.
- Place a plate or a label on the lever and the switch newly added to distinguish the purpose of use.
- When installing warning buzzer of the rear body device in the cab, install one with a buzzer sound different from the safe warning sound of other warning buzzers such as buzzers for the brake.

④ Consideration to chassis maintenance

- Rear bodies must be designed or mounted on chassis so as not to obstruct maintenance.
 - Checking, feeding and draining engine oil
 - Level checking, feeding and draining coolant
 - Checking, feeding and draining transmission oil
 - Checking, feeding and draining differential oil
 - Adjustment and detaching of fan belt
 - Checking and replenishment of battery fluid, detaching of battery or battery cover
 - Refueling to fuel tank, check and maintenance around fuel tank
 - Draining of air tank
 - Checking and replenishment of air dryer, relay valve, brake booster, and other valves related to brake system and accessories
 - Checking and air bleeding from brake system, clutch system, and servo unit
 - Operation of spare tire carrier including detaching spare tire
 - Detaching of DPD filter and piping
 - Checking of lamps of rear body and exchange of valves
 - Other check work

⑤ Other

- Make sure the rear body does not obstruct forward and the side views, nor the views such as rearview mirrors and under mirrors.
- Do not install components likely to drop out by the vibration of the vehicle in the vicinity of the driver's seat, since it might obstruct the driving operation.
- Do not install rear body (especially, components impact or the outside force may be applied such as side guard) on to chassis accessories such as fuel tank support and air tank support, etc.
- No projection or sharp edges on iron plates of rear body which may give scratch.

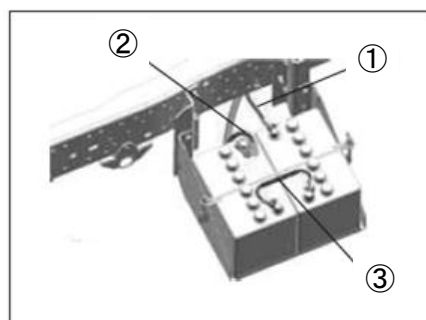
1-2 PRECAUTIONS AT INSTALLING

Note the following instructions when installing, and work safely. Please refer to the owner's manual of a vehicle concerned for the handling of the vehicle and devices.

① Preparation

- Stop the engine, apply the parking brake, and set sprag on the tire, to prevent the vehicle from moving.
- When installing, make sure no inclination of rear body at horizontal place.
- Observe the instruction of the caution plate stuck on the cab chassis strictly.
- When you disconnect the battery cables, to prevent short circuit, do according to the following procedure. Connect battery cables following the procedure for cable disconnection in reverse order.

- Turn off all the switches.
- Disconnect the negative cable ①.
- Disconnect the positive cable ②.
- Disconnect the battery cable ③.
- When you connect the battery cable, to prevent battery terminals from corroding, clean them and lightly apply grease on them.

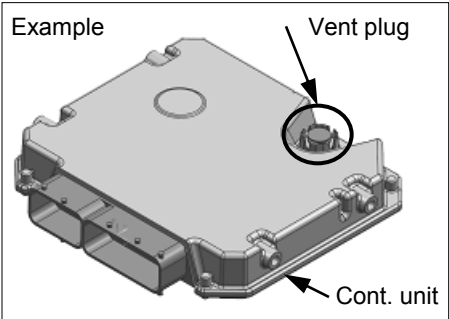


- After turning off the starter switch, wait about 1 minute and disconnect negative cable.
- After disconnecting negative terminal, wait more than 15 seconds before starting to work.
- When all the work is done, restore (connect) the battery negative terminal. The terminal should be tightened with specific torque.
- Partially protect the chassis if necessary.

② At work

- Do not get on valves and equipments related to the brake, the brake pipes, the fuel tanks, the fuel pipes, the air tanks, the brake boosters, the batteries, wire harnesses, the transmission control cable and the relay boxes, etc. to lead to damage and the breakdown of the equipment.
- Use the grip and the step, etc. to prevent from slipping and falling when getting on and off of the cab.
- When tilting the cab, give a necessary safety precaution like certain fixation of the cab stopper stay etc.
- In the case of the tilt cab car, when heavy load is installed on the cab roof or inside the cab, descending speed quicken. Be careful not to get caught in between.

- When installing additional components, be careful not to damage the harness due to jamming or smashing. If the harness is damaged, have it repaired.
- Do not leave tools or waste cloth, etc. behind in the engine room.
- Do not touch the vent plug of electronic unit (control unit).
- The vent plug might come off from the unit when external force is applied.
- If the vent plug came off, replace it with new one.
- If the vent plug is reused or left off, foreign substance (water, waste, etc.) may enter inside the electronic unit.



③ Starting up engine

- Confirm whether the gear shift lever is in neutral or parking (when there is parking range).
- Start up an engine while pressing the clutch pedal.
- Turn off PTO if there is.
- Do not put the combustible in the direction of the exhaust and the vicinity of the hot section.
- If starting engine inside, install exhaust device to release the exhaust gas outdoor.
- Do not work in the direction of the exhaust or place where the exhaust gas collect easily.
- When the engine is on, be careful with the rotation part and the high temperature section, and not to touch, get caught in, or get burnt.

Example of rotation part and high temperature section

Rotation part	High temperature section
Cooling fan and belt Pulley PTO output shaft and flange Alternator and belt Cooler compressor, and belt etc.	Exhaust manifold Turbo-charger Intake manifold Intercooler Oil pan Radiator and radiator cap Water hose Exhaust pipe and exhaust silencer/DPD Brake drum and retarder immediately after running etc.

④ Moving the vehicle

- Do not put anything obstacle of the driving operation such as tools and parts.
- Idle rotate the engine, and move the vehicle after confirming all the warning lamps have gone off.
- Follow the instruction of owner's manual when warning lamp does not go off.

⑤ Storing the vehicle

- When storing the vehicle for long term, do not leave the key in starter switch. Small amount of electric current is flowing to each system while the key is in, and keeping this for long term will cause flat battery.

1-3 PRECAUTIONS FOR BOLT TIGHTENING

① Fixing with bolt

- Do not fix together the accessories and equipments for rear body with same bolt.
- Use the bolts of the same strength class when using the bolt with same size and different length.
- Follow the tightening torque on next page when re-installing each part on the chassis.
- Follow the work book when re-use prohibition and tightening torque are specified.

Bolt tightening torque

unit: N·m {kgf·m}

Bolt head shape Strength class Bolt head Nominal of thread (Diameter × Pitch)	Hexagon head bolts			Hexagon bolt with flange		
	8.8		9.8	8.8		9.8
	Thermal refining	None-thermal refining		Thermal refining	None-thermal refining	
						
						
						
M6 × 1	5.6 to 11.2 {0.6 to 1.1}		—	6.6 to 12.2 {0.7 to 1.2}		—
M8 × 1.25	13.4 to 25.7 {1.4 to 2.6}		16.7 to 30.4 {1.7 to 3.1}	15.3 to 28.4 {1.6 to 2.9}		18.1 to 33.6 {1.8 to 3.4}
M10 × 1.25	31.3 to 52.5 {3.2 to 5.4}		37.3 to 62.8 {3.8 to 6.4}	35.4 to 58.9 {3.6 to 6.0}		42.3 to 70.5 {4.3 to 7.2}
*M10 × 1.5	31.3 to 51.4 {3.2 to 5.2}		36.3 to 59.8 {3.7 to 6.1}	34.5 to 57.5 {3.5 to 5.9}		40.1 to 66.9 {4.1 to 6.8}
M12 × 1.25	69.3 to 104.0 {7.1 to 10.6}		75.5 to 113.8 {7.7 to 11.6}	77.7 to 116.5 {7.9 to 11.9}		85.0 to 127.5 {8.7 to 13.0}
*M12 × 1.75	64.8 to 96.1 {6.6 to 9.8}		71.6 to 106.9 {7.3 to 10.9}	71.4 to 107.2 {7.3 to 10.9}		79.5 to 119.2 {8.1 to 12.2}
M14 × 1.5	106.2 to 158.8 {10.8 to 16.2}		113.8 to 170.6 {11.6 to 17.4}	114.9 to 172.3 {11.7 to 17.6}		123.4 to 185.1 {12.6 to 18.9}
*M14 × 2	100.6 to 149.8 {10.3 to 15.3}		106.9 to 160.0 {10.9 to 16.3}	108.2 to 162.2 {11.0 to 16.5}		115.5 to 173.3 {11.8 to 17.7}
M16 × 1.5	154.3 to 232.5 {15.7 to 23.7}		160.0 to 240.3 {16.3 to 24.5}	171.1 to 256.7 {17.4 to 26.2}		176.9 to 265.3 {18.0 to 27.1}
*M16 × 2	147.6 to 221.4 {15.1 to 22.6}		153.0 to 229.5 {15.6 to 23.4}	162.5 to 243.8 {16.6 to 24.9}		168.5 to 252.7 {17.2 to 25.8}
M18 × 1.5	222.5 to 334.3 {22.7 to 34.1}		229.5 to 345.2 {23.4 to 35.2}	—		—
*M18 × 2.5	223.6 to 335.4 {22.8 to 34.2}		230.5 to 346.2 {23.5 to 35.3}	—		—
M20 × 1.5	307.4 to 461.7 {31.3 to 47.1}		293.2 to 440.3 {29.9 to 44.9}	—		—
*M20 × 2.5	284.0 to 472.1 {29.0 to 48.1}		293.2 to 440.3 {29.9 to 44.9}	—		—
M22 × 1.5	413.6 to 620.5 {42.2 to 63.3}		424.6 to 636.5 {43.3 to 64.9}	—		—
*M22 × 2.5	385.7 to 578.0 {39.3 to 58.9}		394.2 to 592.3 {40.2 to 60.4}	—		—
M24 × 2	490.8 to 810.5 {50.0 to 82.6}		554.1 to 830.6 {56.5 to 84.7}	—		—
*M24 × 3	462.8 to 693.1 {47.2 to 70.7}		520.7 to 781.6 {53.1 to 79.7}	—		—

* Used for internal thread of soft material such as casting.

Chapter 2 MODIFICATION OF CHASSIS

- 2-1 DRILLING OR WELDING A CHASSIS FRAME**
- 2-2 REINFORCING A CHASSIS FRAME SIDE MEMBER**
- 2-3 MODIFYING REAR OVERHANG OF A CHASSIS FRAME**
- 2-4 BOLT AND NUT FOR CROSS MEMBER**
- 2-5 PRECAUTIONS FOR WELDING**
- 2-6 PRECAUTIONS FOR HYDRAULIC OR AIR PIPING**
- 2-7 NYLON AIR TUBES AND PUSH-IN CONNECTOR**
- 2-8 NYLON TUBE ROUTING AND FIXING**
- 2-9 AIR EXTRACTION**
- 2-10 RELOCATION OR ADDITION OF A FUEL TANK**
- 2-11 PRECAUTIONS FOR TOWING HOOK**
- 2-12 PRECAUTIONS FOR MODIFICATION OF THE ELECTRICAL SYSTEM**
- 2-13 PRECAUTIONS FOR PAINTING**
- 2-14 EMBLEMS**
- 2-15 DECALS**

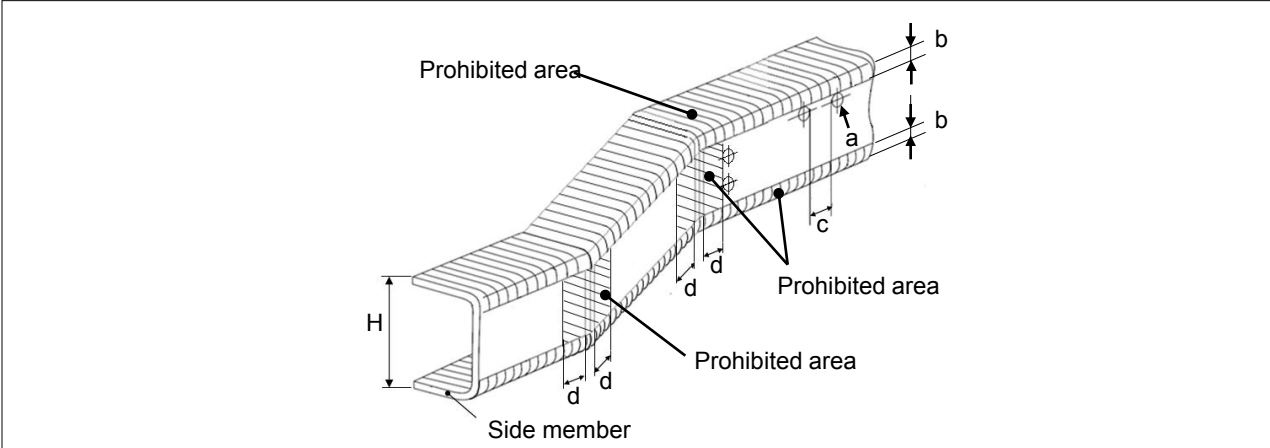
2-1 DRILLING OR WELDING A CHASSIS FRAME

Drilling holes and welding fluctuate stresses in various parts of a chassis frame and negate the frame’s load condition, possibly resulting in deformation or cracks. To install any special equipment, follow the instructions below.

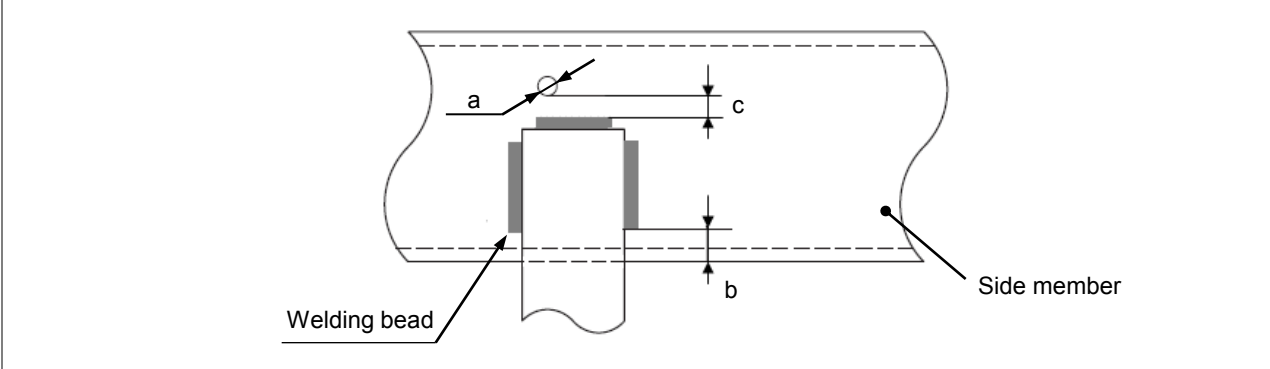
① Side member

- Do not drill a hole or make a notch in or weld the shaded areas in the below figure.
- Use drill to drill holes. Gas cut prohibited.

Working prohibited areas of side member (1)



Working prohibited areas of side member (2)



- Maintain the following measurements at working.

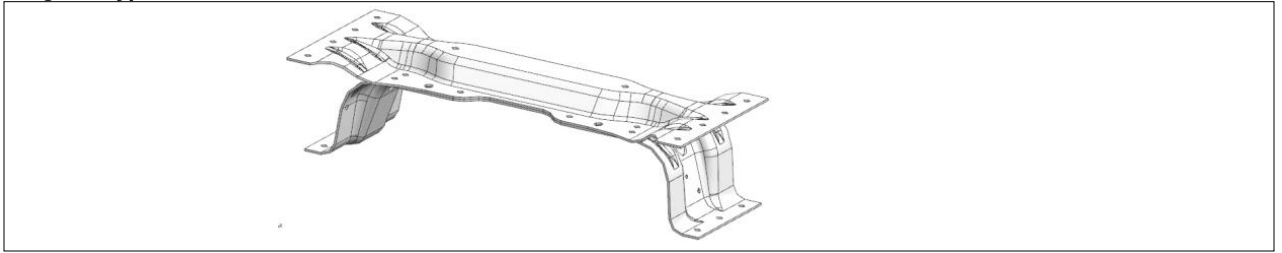
	"a" in the above figure (Allowable maximum hole diameter)	"b" in the above figure (Working-prohibited area)		"c" in the above figure (Minimum required distance)	"d" in the above figure (Minimum required distance)
Heavy duty vehicle	17mm	H/5 (Welding)	H/7 (Hole)	40mm	30mm
Medium duty vehicle	15mm				
Light duty vehicle	11mm			25mm	

Note: Measure “d” from end of bending radius.

② Cross member

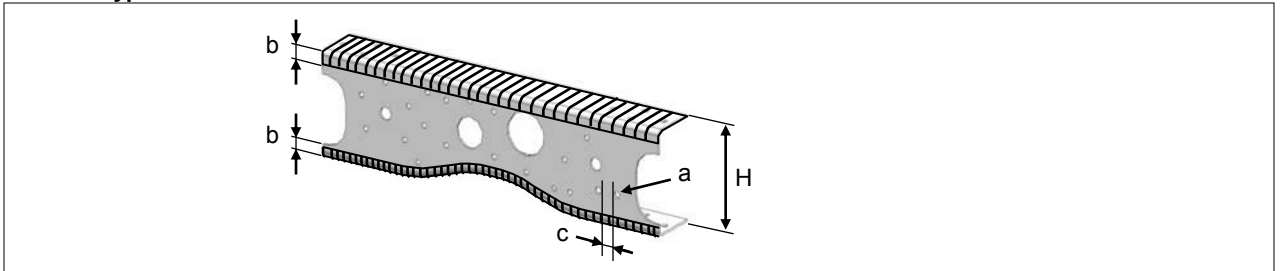
- For alligator type, hole drilling, notch making and welding are prohibited.

Alligator type cross member



- For channel type, allowable maximum hole diameter “a” is 9mm, and this hole should be used only for piping or harness routing. “b” and “c” shown in figure above depend on the side member.

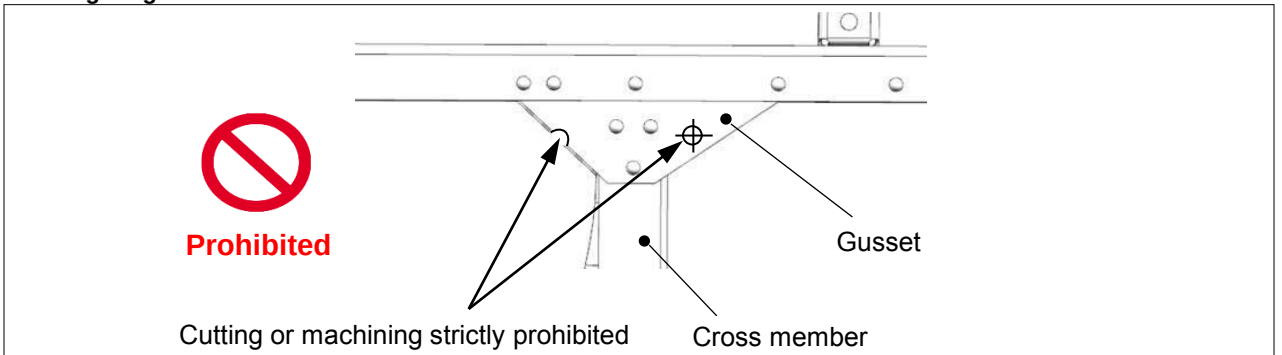
Channel type cross member



③ Gusset

- Hole drilling and notch making are prohibited.

Working on gusset



④ Other cautions at work

- When making a hole, use drilling machine.
- Length of a weld should be 30 to 50mm, and welds should be spaced a minimum of 40mm.
- Be sure that weld beads do not protrude over holes. Welding should be done away from the edge of holes by the distance “c” in the table in previous page.
- When making a notch by using gas, finish the cut surface by grinding.
- Cold rivet only.
- Apply rust proofing after drilling hole, making notch or welding.

2-2 REINFORCING A CHASSIS FRAME SIDE MEMBER

When reinforcing a chassis frame side member, follow the instructions below to prevent an acute change in the section modulus and stress concentration due to welding, since it might cause deform or crack.

① Reinforcement material

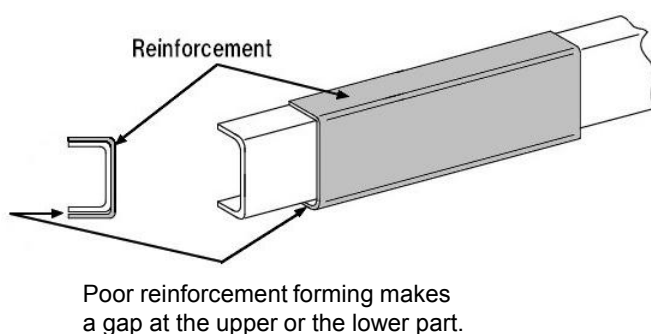
- The reinforcement material should be equivalent to that of side members. Refer to "SPECIFIC INFORMATION" of the target vehicle model.
- Thickness and shape of the reinforcement material should be selected according to rear body mounting and vehicle usage condition.
- Use L-shaped reinforcement material.
- Channel-shaped reinforcement is not recommended.
 - It is difficult to match the dimension of a channel-shaped reinforcement with a chassis frame side member, and difference in shape between a reinforcement and a side member results in slacking and reinforcing effect.

Reinforcement shape

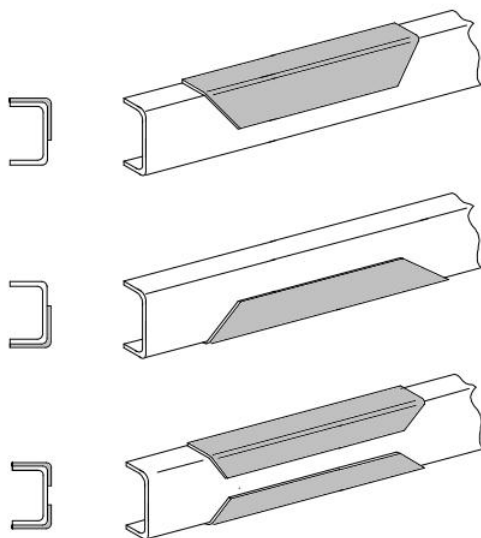
Poor example



Prohibited



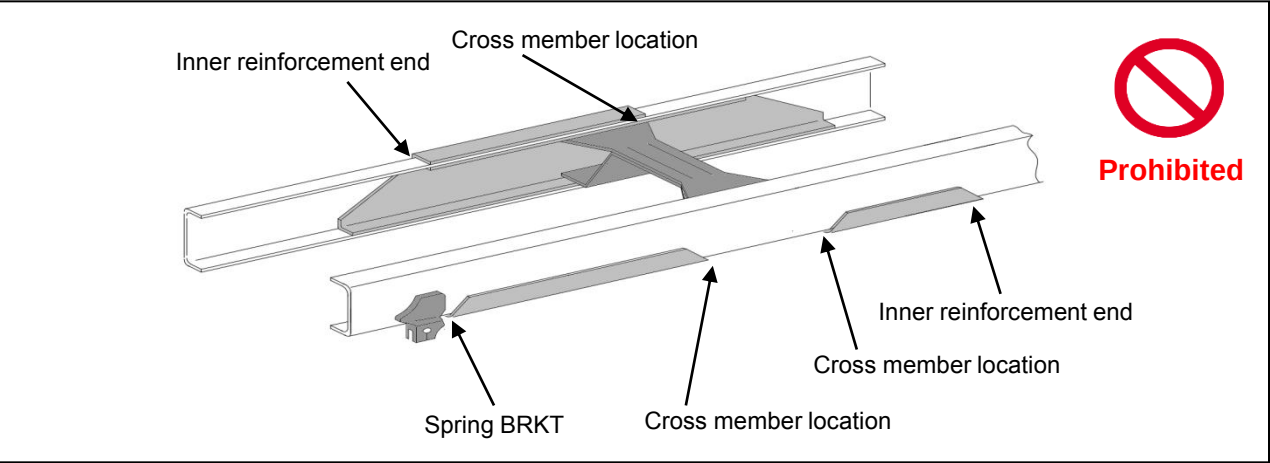
Good example



② Installation position of reinforcement material

- Ends of outer reinforcement and inner reinforcement should not overlap.
- An end of outer reinforcement and cross member should not overlap.
- An end of outer reinforcement and spring bracket should not overlap.

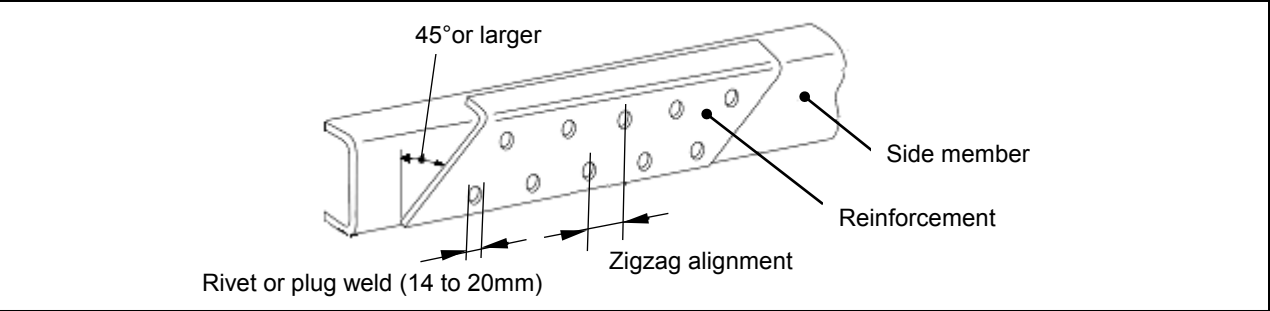
Prohibited areas for installing reinforcement



③ Installing reinforcement material

- Install a reinforcement by riveting or plug welding. Place them in zigzag alignment.
- When performing plug welding, be sure that electrical components such as electric harnesses on the inner side of a chassis frame side member are a minimum of 50mm apart from welding site.
- When inserting a rivet in a hole from which another rivet has previously been removed, the rivet should be 1 or 2mm larger in diameter than the removed one. Cold rivet only.

Example of installing reinforcement



④ Size of rivet

Rivet size detail

Vehicle type	Rivet Diameter (d)	Diameter (D)	Unit ; mm
			Height (H)
Light duty vehicle	10	16	7
Medium duty vehicle	11	18	7.7
Heavy duty vehicle	13	21	11

The diagram shows a cross-section of a rivet. Labels indicate dimensions: 'D' for the diameter of the rivet head, 'H' for the height of the rivet head, and 'd' for the diameter of the hole in the material.

2-3 MODIFYING REAR OVERHANG OF A CHASSIS FRAME

If a rear body constrains a sub frame to necessarily protrude outward from the rear end of a chassis frame by 300mm or more, lengthen rear overhang of the chassis frame according to need. If it is absolutely necessary to cut a side member of a chassis frame for modification, make sure that the cut line does not split existing holes.

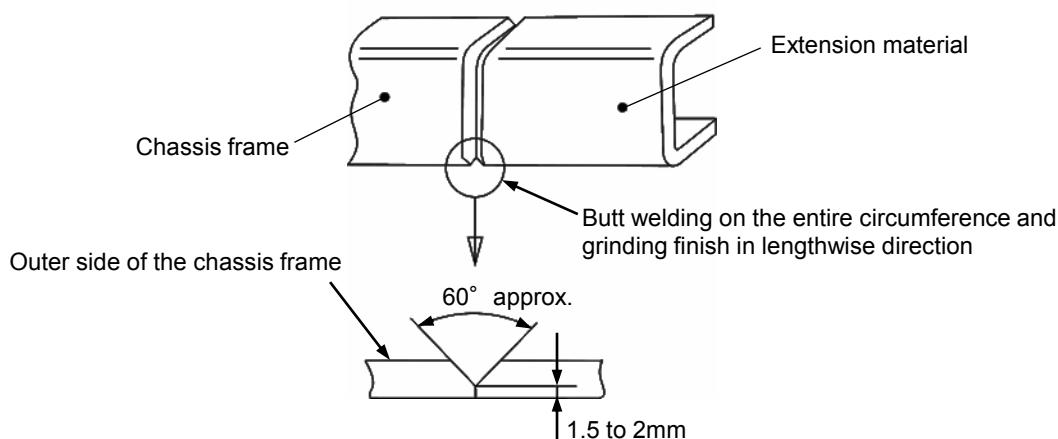
① Extension material

- The extension material should be equivalent to that of side members. Refer to " SPECIFIC INFORMATION " of the object vehicle model.
- Thickness and bending radius of the extension material should be the same as that of side members. Refer to " SPECIFIC INFORMATION " of the object vehicle model.

② Installing extension material

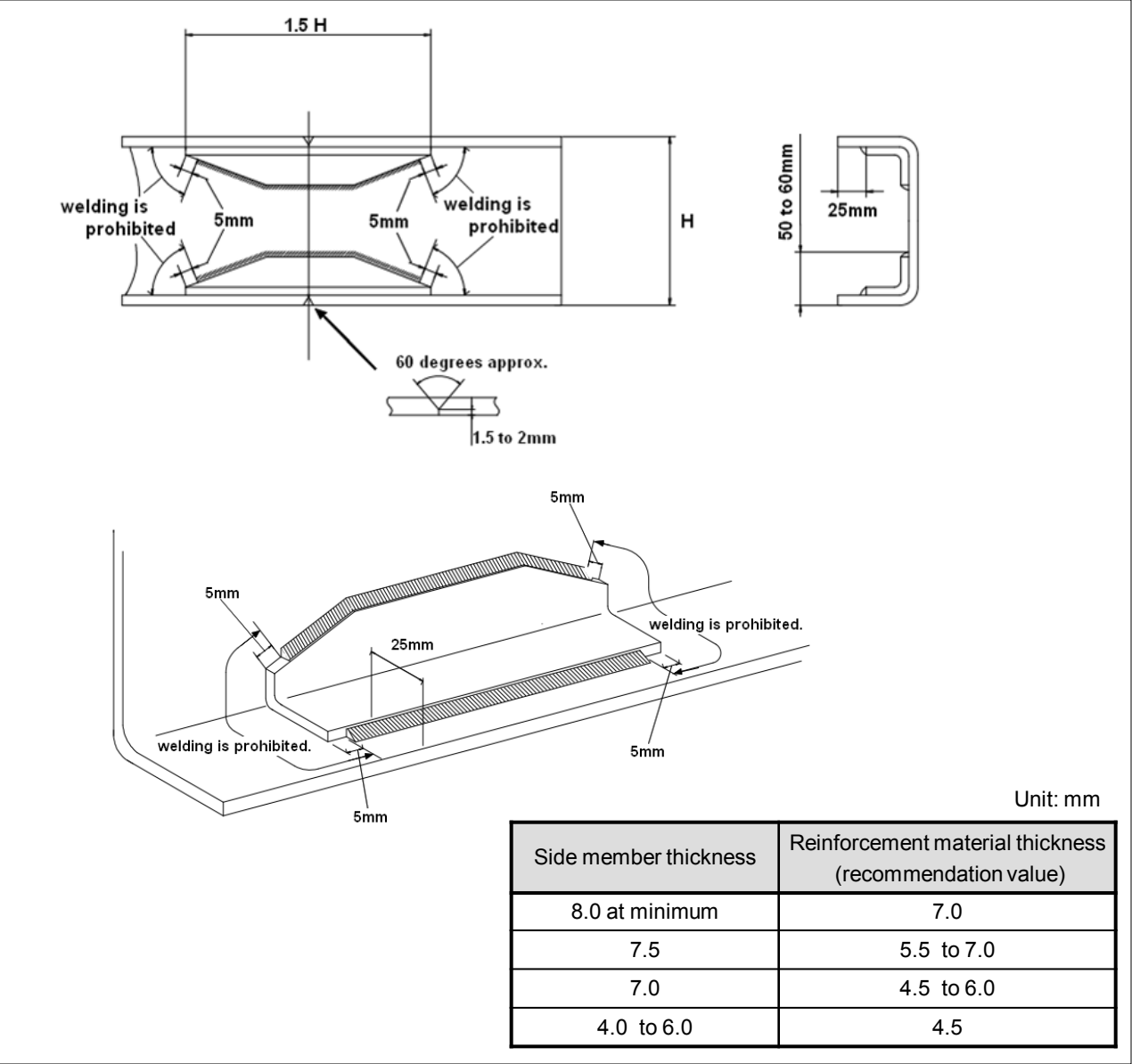
- In case that extension material is 300mm or shorter:
 - Join an extension material and a chassis frame by continuous butt welding on the entire circumference. After welding, grind finish weld surface.

Installing reinforcement (1)



- In case that extension material is longer than 300mm:
 - Join an extension material and a chassis frame by butt welding, and then fit a reinforcement on the inner side of them.

Installing reinforcement (2)



2-4 BOLT AND NUT FOR CROSS MEMBER

When the bolt and nut fixing cross member to side member is removed, re-fix with tightening torque in following chart.

Nominal of thread (Diameter × Pitch)	Identification (Bolt head shape and strength class)	Tightening torque N·m {kgf·m}	Target location
M10 × 1.25	Hexagon bolt with flange 8 or 8.8	44 to 59 {4.5 to 6.0}	Bolt and nut fixing cross member , including those fixing fuel tanks, battery, etc. together
M10 × 1.25	Hexagon head bolts 8 or 8.8	39 to 53 {4.0 to 5.4}	
M12 × 1.25	Hexagon head bolts 8 or 8.8	85 to 104 {8.7 to 10.6}	
M12 × 1.25	Hexagon head bolts 9 or 9.8	93 to 114 {9.5 to 11.6}	
M14 × 1.5	Hexagon head bolts 9 or 9.8	139 to 171 {14.2 to 17.4}	

2-5 PRECAUTIONS FOR WELDING

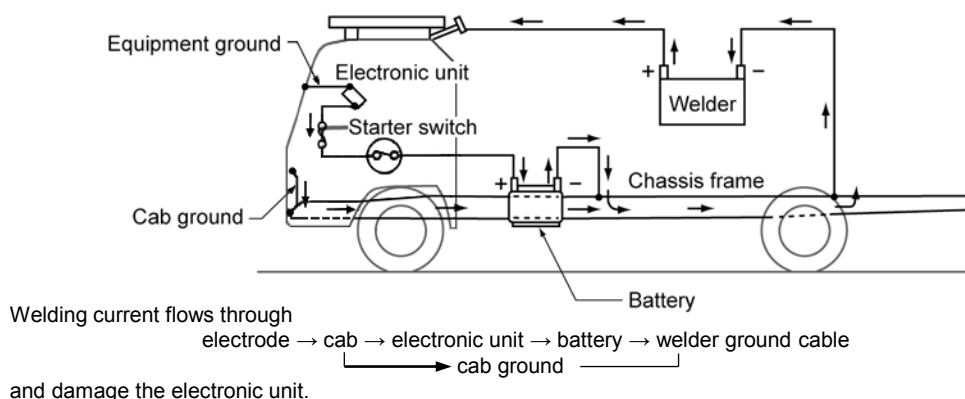
Be cautious of following when electric welding. Welding current may damage electronic devices installed in a vehicle.

① Before welding

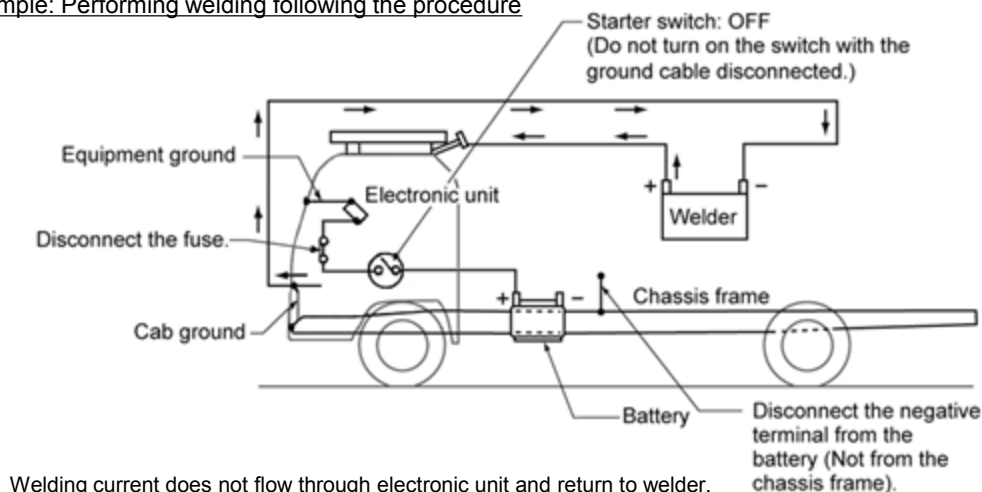
- Turn off a starter key switch.
- Disconnect a negative battery terminal.
- Disconnect fuse for air bag. (Confirm the fuse label for location of fuse.)
- Disconnect all wire harnesses from electronic unit (control unit).
- Cab welding
 - Attach the welder's ground cable to a plated bolt or metallic part of the cab in the vicinity of welding site. Before securing the ground cable on such a metallic part, peel off the paint on the area to reveal its base.
- Chassis welding
 - Attach the welder's ground cable to a plated bolt or a chassis frame member in the vicinity of welding site. Before securing the cable on the chassis frame member, peel off the paint on the area to reveal its base.

Welding current flow direction (Reference)

Poor example: Performing welding without following the procedure



Good example: Performing welding following the procedure



- Do not attach the ground cable to a chassis spring, otherwise the spring may break.
- Remove the paint coating and dry the area where welding is to be performed.
- To prevent the following components from being damaged by welding spatters or heat, cover them up or temporarily relocate them to somewhere reasonably distant. In particular, if welding any area within 200mm from a fuel tank, remove the fuel tank beforehand.
 - Plastic parts
 - Air intake duct
 - Intercooler
 - Tire
 - Rubber parts
 - Radiator hose
 - Pipes
 - Electric harnesses
 - Radiator
 - Chassis springs
- To minimize impact of heat on peripheral area and to assure welding quality, select a suitable welding torch and perform welding at suitable welding conditions.
- No weld defect such as undercuts, overlaps, and pin holes.

② After welding

- Welded area must not be rapidly cooled.
- Apply rust-proof paint with the same color, if the paint is removed from chassis frame or cab.
- After welding, put the disconnected electronic units such as control unit connectors and fuses back in place exactly as before. After that, connect negative battery terminal securely.
- Check operation of equipments. Contact ISUZU distributor or ISUZU dealership for checking method.

③ Welding electrode

- Use welding electrode depending on parts in following chart.

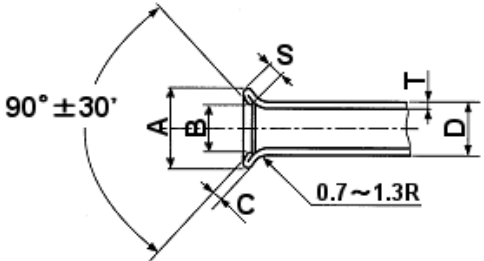
Welding part	Details	Recommended welding electrode
High tensile steel	HT540 (ISUZU standard) Tensile strength: 540MPa {55kgf/mm ² }	JIS Z3211 E4916 / AWS A5.1 E7016 or JIS Z3211 E4316 / AWS A5.1 E7016 equivalent
Steel for general frame	JSH440 (ISUZU standard) Tensile strength: 440MPa {45kgf/mm ² }	

2-6 PRECAUTIONS FOR HYDRAULIC OR AIR PIPING

Since the air pressure and the hydraulic piping of the brake system and the steering system are important parts, modification is prohibited. If there is necessity to add pipes for body building, follow the instructions below for safety assurance.

- ① Working efficiency
- Checking and detaching of equipment (hydro master, air master, brake oil tank, etc.) installed at remodeling area should not be obstructed.
 - Do not interfere with movable body of special purpose vehicle.
- ② Pipe forming
- Use one full pipe to form, do not connect pipes.
 - When damaged, do not repair, use new pipe.
 - Do not bend the same portion of a pipe more than once. Do not perform heat flare forming and heat bending.
 - Before and after forming a pipe, for safety assurance, check the surface of the pipe for damage, deformation, or dent, and make sure of no foreign substance inside..
 - Clean inside the pipe after forming.

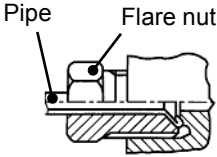
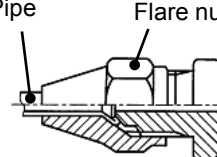
Flare forming and dimensions Unit: mm

Type	Double flare						
Shape	 Dimension 'S' is the width of flat seat (inside flare face) coming into contact with mating seat						
Nominal Diameter	D	A	B	T	C		S (at minimum)
4.76	4.76	6.6 to 7.1	3.0 to 3.7	0.7	1.4	±0.2	1.0
6.35	6.35	8.6 to 9.1	4.5 to 5.2	0.7	1.4		1.0
8	8	10.5 to11.0	6.2 to 6.9	0.7	1.4		1.6
10	10	13.0 to13.5	8.2 to 8.9	0.7	1.4		1.6
12	12	15.0 to15.7	9.8 to 10.5	0.9	1.8		1.6
15	15	18.1 to18.8	12.7 to 13.4	1.0	2.0		1.6

Remarks:

- The decentering with the center of the seat side and the tube is assumed to be 0.15mm or less.
- For the double walled steel pipe, peeling in the inside part of the flare must be 1/3 or less of the dimension “S”.

Tightening torque

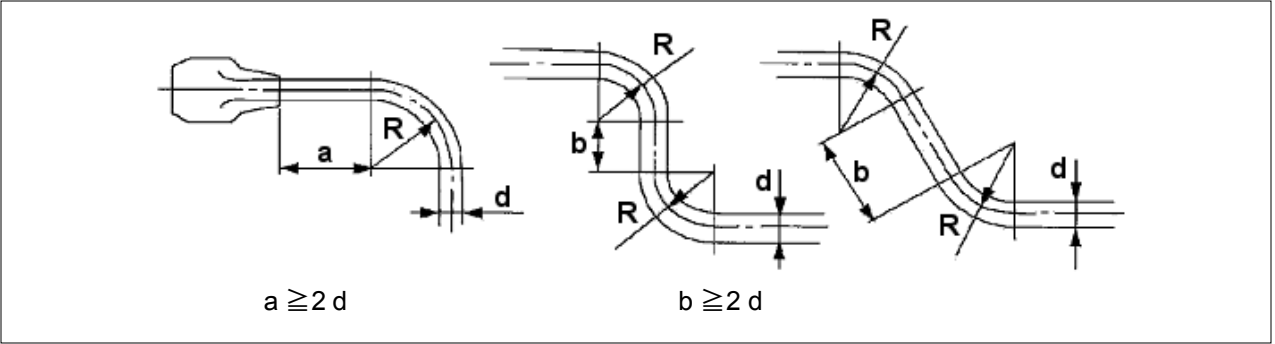
Joint type	Nominal Diameter of applied pipe	Flare nut Width across flats (mm)	Screw size	Tightening torque value N·m{kgf·m}	Remarks
A-type flare nut 	4.76	10	M10 x 1.0	Iron:13 to 18 {1.3 to 1.8} Non-iron:9 to 15 {0.9 to 1.5}	Tightening torque values selected depending on mating materials. Non-iron: Aluminum and brass
	6.35	12	M12 x 1.0	25 to 35 {2.6 to 3.6}	
	8	14	M14 x 1.5	29 to 39 {3.0 to 4.0}	
	10	17	M16 x 1.5	45 to 55 {4.6 to 5.6}	
C-type flare nut 	4.76	14	M10 x 1.0	13 to 18 {1.3 to 1.8}	
	6.35	17	M12 x 1.5	25 to 35 {2.6 to 3.6}	Mating materials: iron
	8	17	M14 x 1.5	25 to 39 {2.6 to 4.0}	Mating materials: iron
	10	19	M16 x 1.5	45 to 80 {4.6 to 8.2}	Mating materials: iron
	12	24	M20 x 1.5	65 to 115 {6.6 to 11.7}	Mating materials: iron
	15	30	M22 x 1.5	80 to 130 {8.2 to 13.3}	Mating materials: iron

• Minimum bending radius should be as shown below.

Unit: mm

Pipe Nominal Size	4.76	6.35	8	10	12	15
Bending Radius	15	20	25	30	35	40

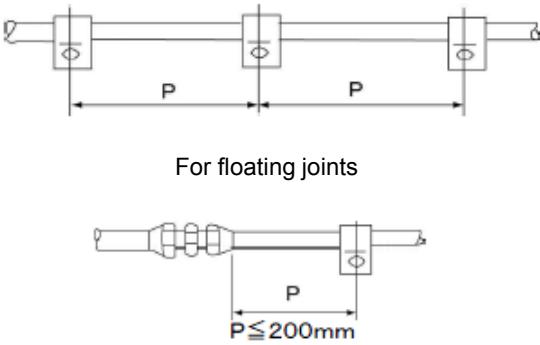
Straight portion of the pipe



③ Routing steel pipes

- Pipes should be considered not to contact with other components, heat influence, and vibration.
- Protect pipes from sand, water, and other foreign substances fly up from road.
- Clip fix pipes at straight piping line. Refer to following

Clip distance



For floating joints

$P \leq 200\text{mm}$

Pipe diameter	P
4.76 / 6.35	400 at max
8 / 10	500 at max
12 / 15	600 at max

Unit: mm

- For piping between air compressor and air drier, do not fix by shape that moisture collects because moisture in the tube might freeze.

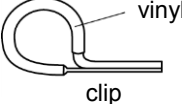
Poor example of fixing pipe

Pipe bended at an inappropriate angle



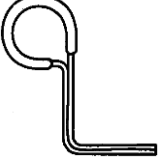
Moisture in the compressed air accumulates here.

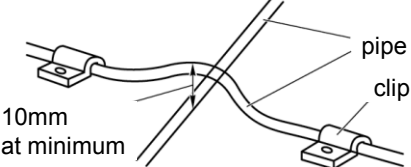
- Apply rust-proofing on clips. Apply vinyl or rubber coating to the part of the clips where pipes may contact. (Usage of genuine ISUZU clips is recommended.)
- When the steel pipes intersect, secure 10mm or more at crossing point, and clip fix at straight part and point off the bending radius.



vinyl

clip





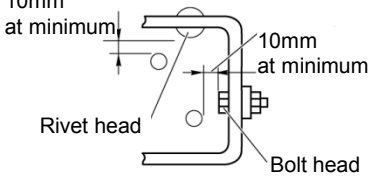
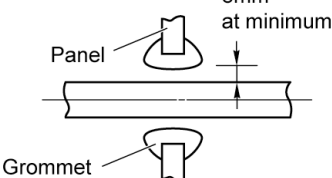
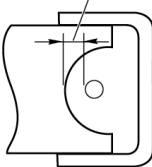
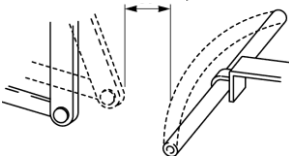
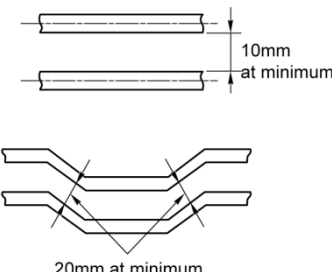
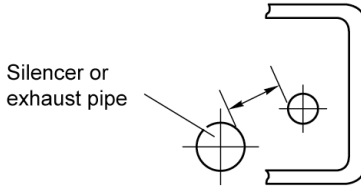
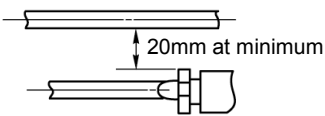
pipe

clip

10mm at minimum

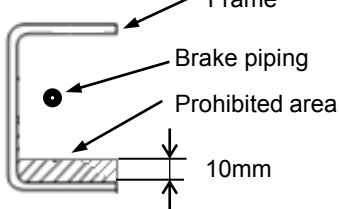
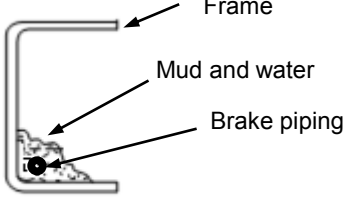
- Secure the distance between steel pipes or a pipe and other components according to the following Fig.

Distance between steel pipe and other components

Bolt head or rivet head on chassis frame member 10mm at minimum Rivet head Bolt head 10mm at minimum 	Hole in the panel 8mm at minimum Panel Grommet 
Edge 10mm at minimum 	Moving components such lever, link and arm: 30mm (moving range +30mm) 30mm at minimum 
Pipes arranged in parallel 10mm at minimum 20mm at minimum 	Silencer and exhaust pipe ① In case that the pipe and the component intersect with each other -- 80mm at minimum Silencer or exhaust pipe 
Flare nut 20mm at minimum 	② In case that the pipe and the component are placed in parallel -- 200mm at minimum (180mm if a heat insulation panel is provided.) There may be individual values provided. Refer to " II SPECIFIC INFORMATION" of target vehicle type.

- Do not route pipes over, under or outside the frame. Pipes may be damaged due to , stepping on, flying gravels, etc.
- Do not route pipes in lower part inside the frame. Pipes may be damaged due to piled up or freezing mud or water.

Piping location

Good example Frame Brake piping Prohibited area 10mm 	Poor example Frame Mud and water Brake piping 
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2-7 NYLON AIR TUBES AND PUSH-IN CONNECTOR

Since the air pressure piping of the brake system is important parts, remodeling is prohibited. If there is necessity to add piping for body building, note the instructions below.

① Nylon air tube

- Nylon tube is weak to heat (93°C). Remove before welding.
 - Nylon tube is weak to acid. Do not use waste cloth or globes with battery fluid.
 - Keep distance from high-pressure cleaner nozzle (MIN 100mm).
 - When removing it, make sure not to bend the tube. When damaged, use new tube.
 - Do not use any nylon tubes other than the genuine ISUZU nylon tubes.
- Followings are specifications of nylon tubes for reference.

Standard	Manufacturer	Type
DIN Standard	Nitta Moore or Reflex & Allen	Push-in type

Outer Diameter (mm)	Inner Diameter (mm)	Minimum bending Radius (mm)	Purpose example
6	4	50	Air suspension seat
8	6	80	Accessory line
12	9	100	Brake line

② Nylon tube connector

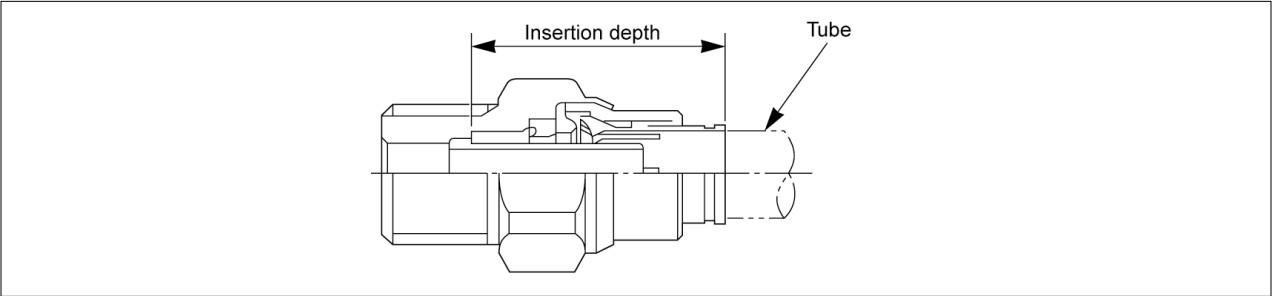
- To disconnect nylon tube, push the release collar and pull out the nylon tube.
- If the nylon tube is frequently connected and disconnected, purchase genuine ISUZU products from ISUZU distributor or ISUZU dealership.

Part name	Part number
Nylon tube remover	185239-0080

- When using steel connector, use connectors with following treatment to prevent pipes from corrosion.
 - Surface treatment: Zinc chromate plating
 - Plating thickness: 4μm (MIN)
 - Rust proofing test with salt water spray: no rust for 240 hours

- Check the inside of the connector. If there is any extraneous substance, blow air into it to remove it.
- If there is friction marks from connector on nylon tube, cut off that portion. Confirm there is no scratch on tube after cutting. If there is, cut off that part.
- Check the insertion length before inserting the nylon tube. Paint the position and insert to the mark. Insertion depth differs by connector makers. Firmly insert the nylon tube to the end.

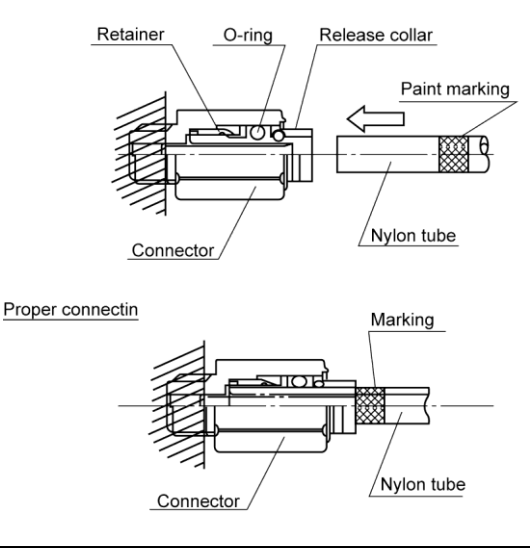
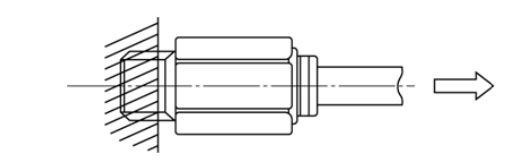
Connecting nylon tubes



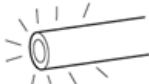
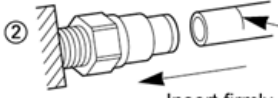
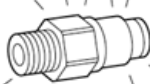
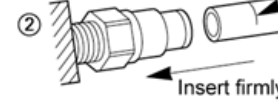
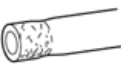
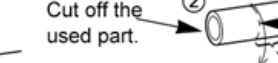
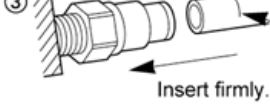
Disconnecting nylon tube

	• Before removing a tube, check a joint for adhesion of dirt, and remove it if there is any. Note: If the joint is so dirty that a release collar is stuck, replace both the tube and the connector with new ones. • Push the release collar by using a dedicated tool (nylon tube remover) to pull out the nylon tube <u>by hand</u>. • When using a tool other than the above dedicated tool, be sure to use a tool that does not scratch the nylon tube (e.g. Monkey wrench). Dedicated tool (nylon tube remover) part number: 185239-0080
Remove dirt from the inserted part	• Check the inserted part of the removed nylon tube for adhesion of dirt. If there is any, remove it. Thoroughly blow air into the connector to remove extraneous substance. • Cover the removed nylon tube and connector with vinyl bags (lint-free bag) to protect them from scratches and dirt.

Assembling new nylon tube

	• Confirm that the part of a nylon tube, which is to be inserted into a connector is free of scratches, warping and dirt. • Lightly apply lubricant (e.g. Engine oil) on the outer surface of the end of the nylon tube. • Mark up an insertion depth on the tube with paint or something, and then insert the tube into the connector up to the marking. Note: Some resistance may be encountered when the tube passes the O-ring and the retainer inside the connector. Be sure to insert the tube up to the marking.
	• After completing the tube insertion, hold the connector and at the same time pull the tube to confirm secure connection. • Confirm that there is no air leakage.

Connecting nylon tube and connector

	Combination of tube and connector	How to connect
1	New tube  Reused connector 	①  Check inside for cleanness. Remove dirt by blowing air onto it. ②  Mark up an insertion depth. Insert firmly.
2	Reused tube  New connector 	①  Cut off the used part. Check for scratches. Cut off. ②  Mark up an insertion depth. Insert firmly.
3	Reused tube  Reused connector 	①  Check inside for cleanness. Remove dirt by blowing air onto it. ②  Cut off the used part. Check for scratches. Cut off. ③  Mark up an insertion depth. Insert firmly.

③ Band clips

- Use 66-nylon band clip.
- Use clips only to hold nylon tube, not to restrict the direction of nylon tube.
- Fix the clip directly on to chassis frame. Brackets are allowed.
- Do not use galvanized zinc brackets. Otherwise, chemical change will damage nylon clips.

④ Protection corrugated tube

- When nylon tube contacts chassis frame or other components, cover the nylon tube with corrugated tube.
- Wrap tape on both end of corrugated tube to prevent from moving.

2-8 NYLON TUBE ROUTING AND FIXING

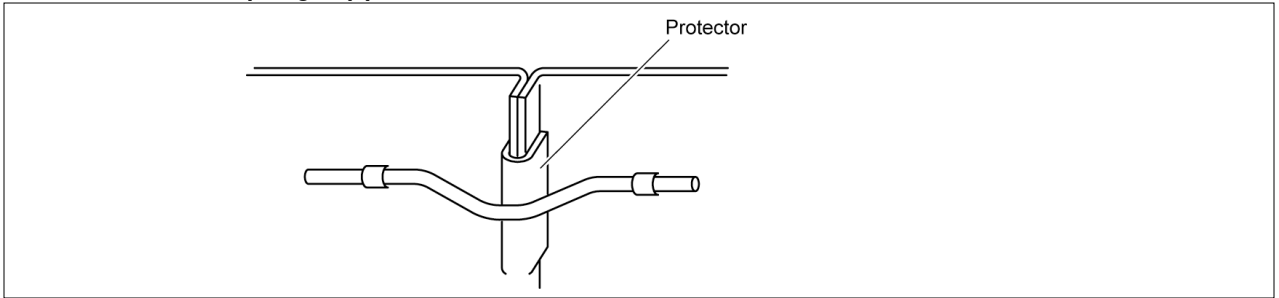
① Precautions

- To prevent wearing and damaging, nylon tube should not interfere with other tube or component.
- At routing, prevent the nylon tube from twisting and note to suppress the vibration of the nylon tube to the minimum.
- Ambient temperature of nylon tube routing area should be from -40°C to +93°C.
- When nylon tube is needed to be routed in areas where mechanical damage, heat damage, chemical damage may be received, provide a protector in this area to protect the tube.
- When routing around high temperature are such as engine, place heat insulator and confirm that the ambient temperature is in between tolerance temperature. Areas enclosed for such a noise insulation requires extra attention because the temperature of these areas tend to be high.
- Route the nylon tube where exhaust emission would not hit even when the gas leaked from exhaust pipe by corrosion or damage.
- Maintain sufficient clearance from vibrating components such as transmission.
- Do not bundle nylon tube with starter cable.
- When nylon tube is clipped together with harness, consider the clip distance so tube and harness would not be rubbed by vibration.
- Clip distance to be 600mm or less.
- Do not cross nylon tube and wire harness as possible.
- When routing nylon tube around battery, be sure that battery fluid (or leaked battery fluid) does not adhere to them while checking the vehicle is running.
- Multiple tubes bundled together can control the vibration effectively than single tube.
- When single tube branches out from a bundle, fix the bundle at branch point to prevent from rubbing.
- When nylon tubes intersect, clip fix each tube individually to avoid interference.

② Measure around sharp edges

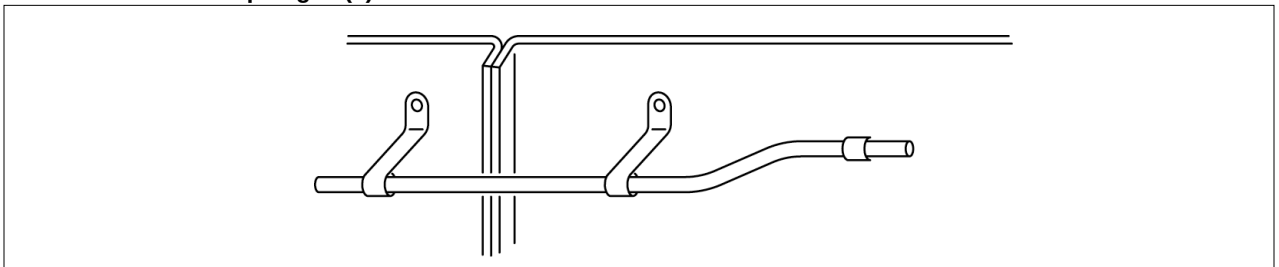
Do measure appropriately by referring to the following example.

Measure around sharp edges (1)



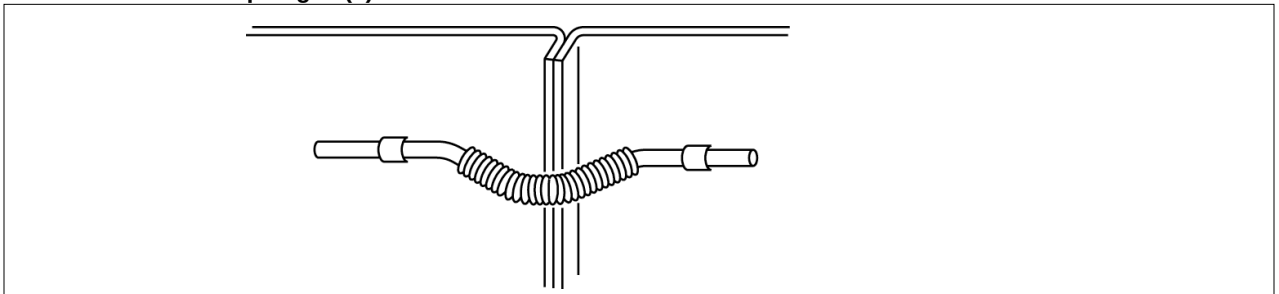
- Cover the sharp edge with protection.

Measure around sharp edges (2)



- Use F-clip to avoid from contact.

Measure around sharp edges (3)

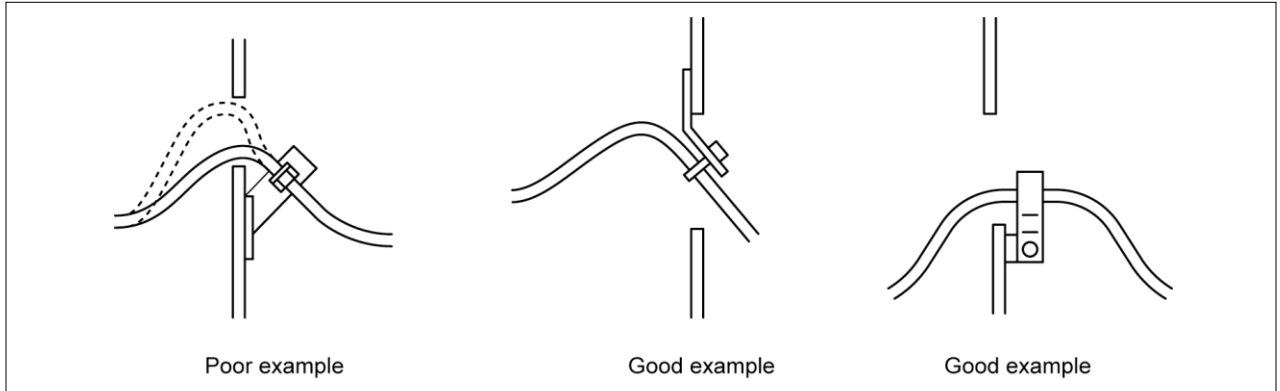


- Cover the tube with corrugated tube and clip fix the tube to avoid corrugate tube to be scratched.

③ Measure through holes

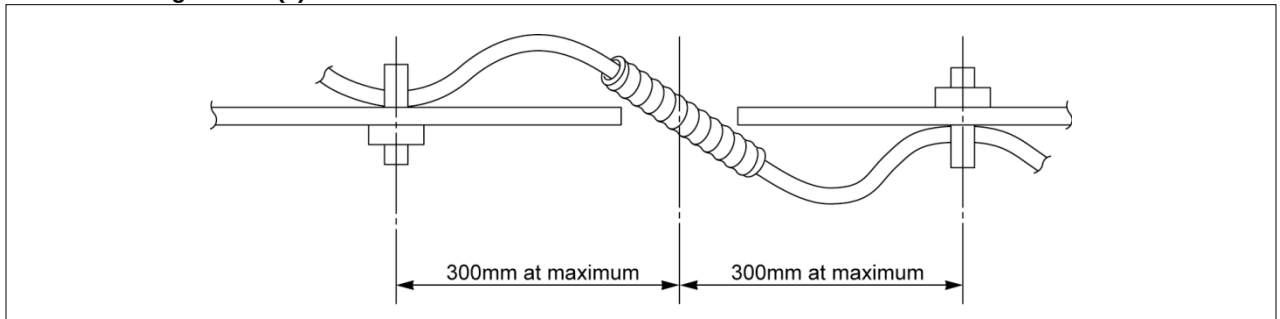
- Prevent the tube from interfering the edge of the hole when you pass the tube through the hole of the cross member.

Measure through holes (1)



- When passing the tube through a hole of chassis frame member, protect the tube in the corrugated tube, and fix the nylon tube so that the tube does not contact with the edge of the hole.

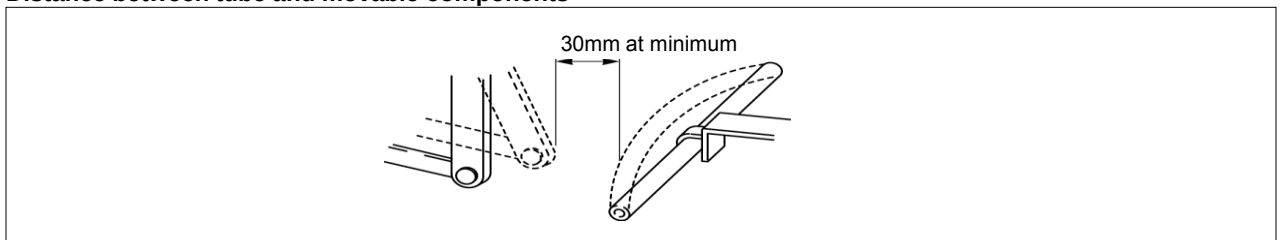
Measure through holes (2)



④ Distance between tube and movable components

- The tube near the movable component such as the lever, the drag links, and arms must be fixed with distance of 30mm or more from them.

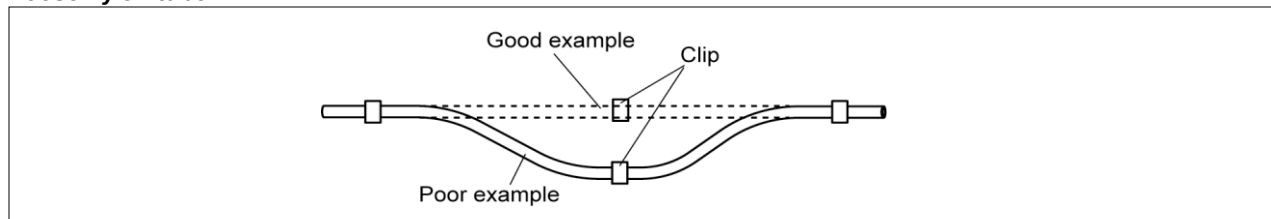
Distance between tube and movable components



⑤ Freeze proofing

- Do not fix by shape that moisture collects because moisture in the tube might freeze.

Loose nylon tube

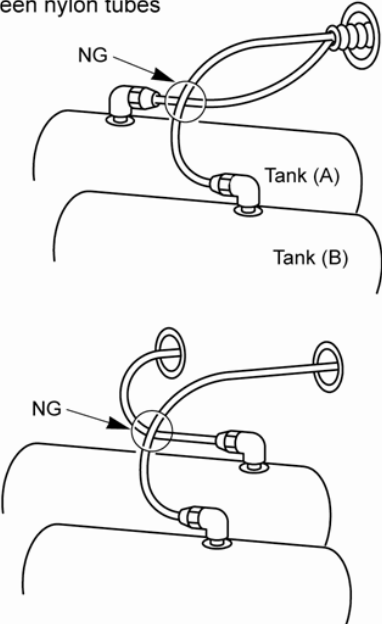
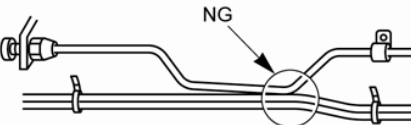
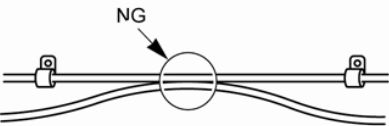
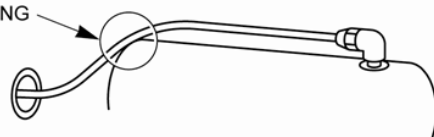


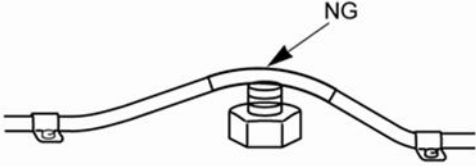
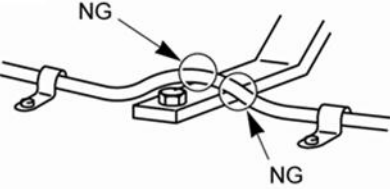
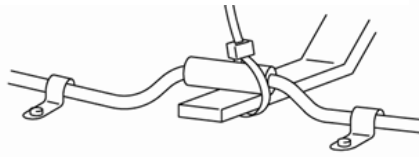
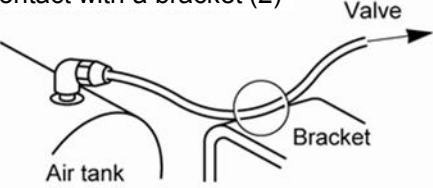
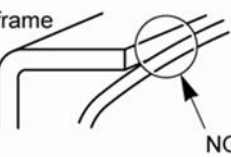
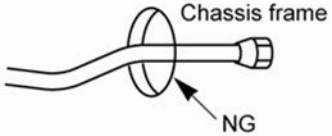
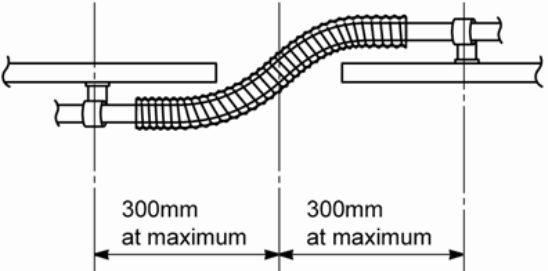
⑥ Installation length

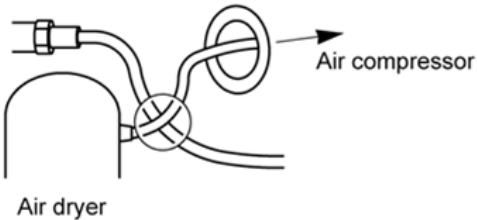
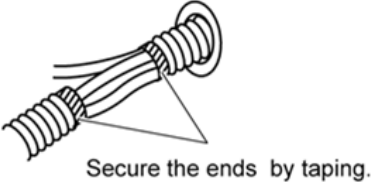
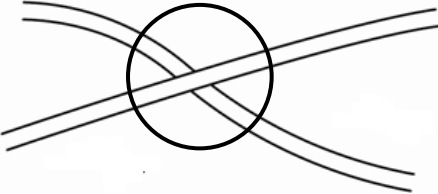
- The tube expands and contracts according to the temperature change. Install it with enough length.

⑦ Example of defective installation

- Poor example is shown below. Do necessary preventive measures referring to the example of measures.

Poor example	Preventive measures
Contact between nylon tubes 	Wrap the nylon tubes with the rubber sleeve, and clamp a with band clip.
Contact with a steel pipe 	Wrap the steel pipe with a rubber sleeve, and clamp with band clip.
Contact with a rubber hose 	Wrap the nylon tube with a rubber sleeve and clamp it with a band clip, provide a minimum of 30mm clearance between the nylon tube and the rubber hose.
Contact with a cable 	Provide a minimum of 10mm clearance between the nylon tube and the cable, or clamp the nylon tube with band clips.
Contact with the air tank 	Provide a minimum of 10mm clearance between the nylon tube and the air tank, or wrap the nylon tube with a rubber sleeve and clamp with a band clip.

Poor example	Preventive measures
Contact with a bolt or a nut 	Provide a minimum of 10mm clearance between the nylon tube and the bolt or the nut, or wrap the nylon tube with a corrugated tube.
Contact with a bracket (1) 	Wrap the pipe with rubber sleeve and fix with band clip to a bracket. 
Contact with a bracket (2) 	Provide a minimum of 10mm clearance between the nylon tube and the bracket or connector, or wrap the nylon tube with a corrugated tube.
Contact with an edge of a cross member or a chassis frame member 	Provide a minimum of 10mm clearance between the nylon tube and the bracket, or wrap the nylon tube with a corrugated tube.
Through hole 	Wrap the nylon tube with a corrugated tube, and secure it at the points within 300mm from the center of the through hole. 

Poor example	Preventive measures
Contact with a hot object such as air charge pipe, power steering oil pipe, and battery cable. 	Provide a minimum of 20mm clearance and fix with clip. Otherwise, place pipes in another route.
Contact with an edge of a corrugated tube 	To prevent the nylon tube from being damaged, split the corrugated tube and tape its ends. 
Others Contact between vibrating nylon tubes 	Wrap the nylon tubes with rubber sleeves and clamp them with band clips.

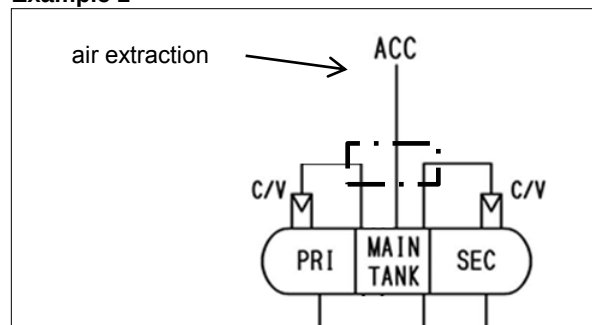
2-9 AIR EXTRACTION

For air to accessories, branch from accessory (ACC) line. Branching from other lines (especially from brake line) is prohibited, since it may affect braking performance. Accessory line differs in vehicle specifications. Refer to the followings.

① Without multi protection valve (MPV)

- Extract air from accessory line including main tank.

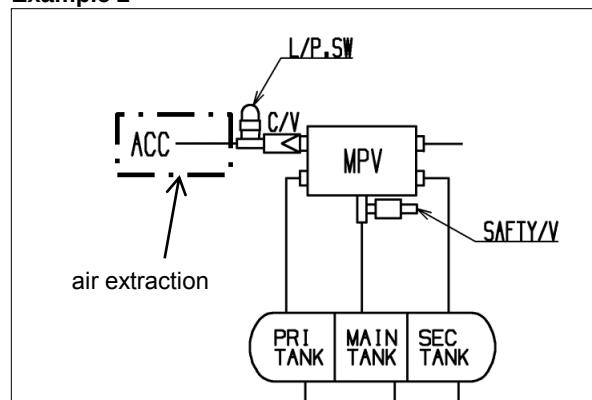
Example 1



② With multi protection valve without accessory tank

- Extract air from accessory line downstream of low pressure switch.

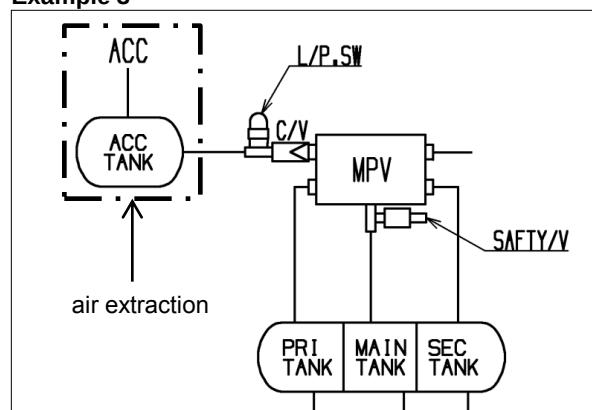
Example 2



③ With multi protection valve with accessory tank

- Extract air from accessory line downstream of accessory tank.

Example 3



For the vehicles with air extraction details, refer to section in "SPECIFIC INFORMATION".

2-10 RELOCATION OR ADDITION OF A FUEL TANK

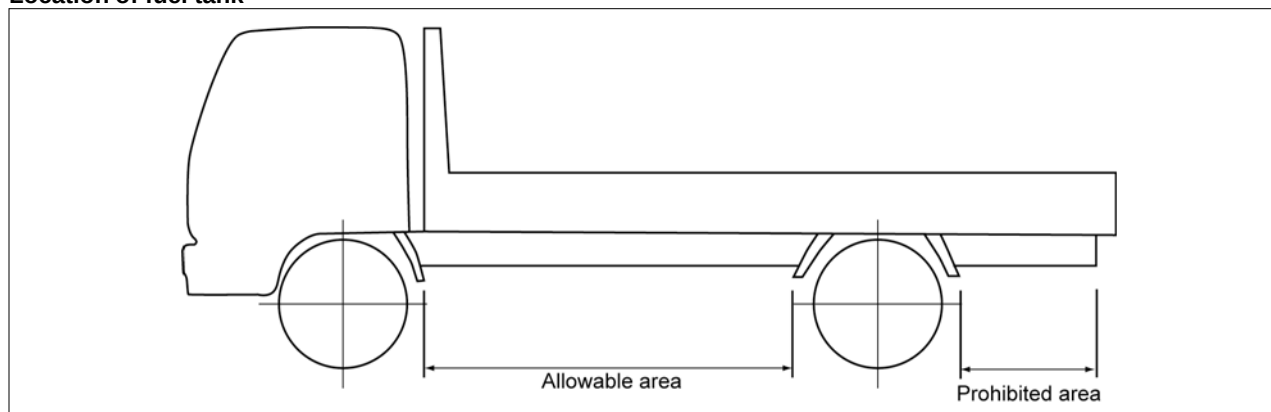
When relocating or adding a fuel tank is necessary, follow the local regulations.

When installing an additional fuel tank, be sure that the total amount of fuel does not exceed the regulation.

① Location of a fuel tank

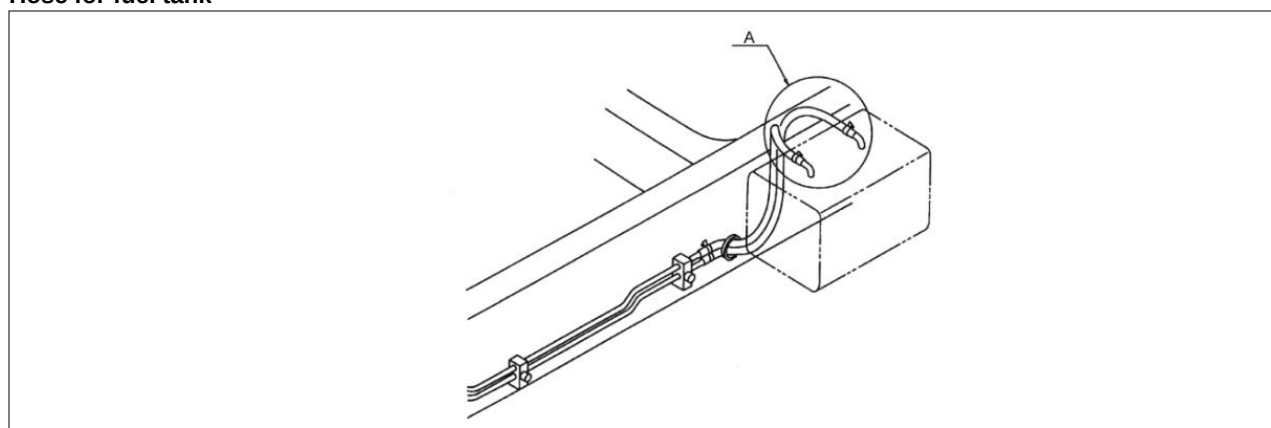
- Place a fuel tank somewhere between wheelbase. Do not place it in rear overhang area.

Location of fuel tank



- When lowering the position of the fuel tank, carefully determine the ground height of a fuel tank not to contact the ground with particular attention to the conditions of use.
- Be sure that the breather opening of the relocated fuel tank will not be subjected to water. There are two types of the breather: One of them is fitted on a fuel tank body, which can be opened with a hose, and the other type is at a filler cap.
- Maintain sufficient clearances between the hoses for a fuel tank ("A" in the figure below) and the object to be mounted on a chassis frame such as sub frame, U-bolt and double ended bracket. If contact is anticipated, wrap the hoses with corrugated tubes or protectors.

Hose for fuel tank



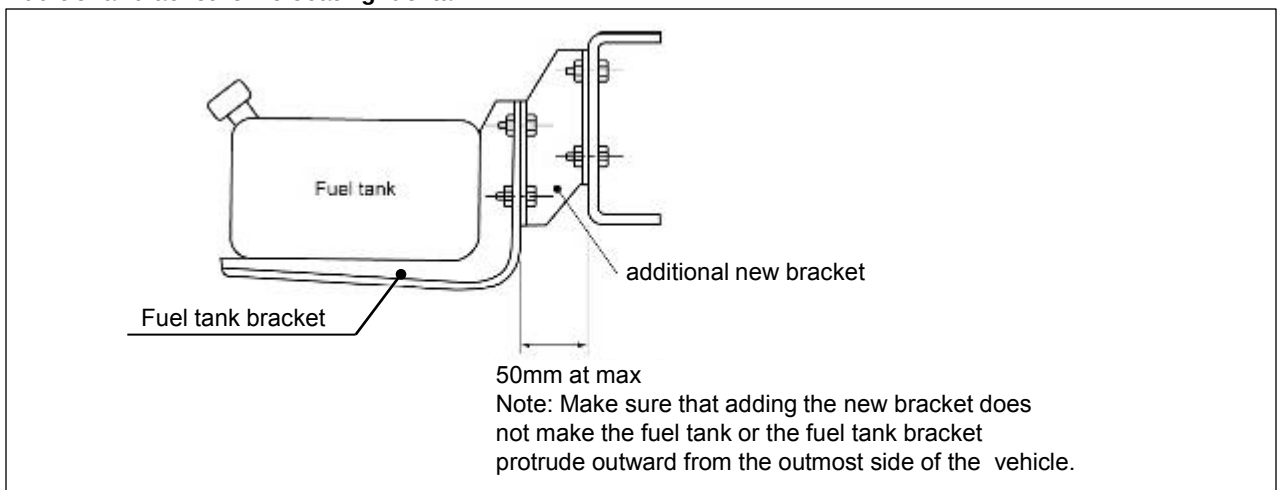
- Be sure that the relocated tank is sufficiently apart from any movable or projecting object.
- The fill opening of a fuel tank should be easily accessible so that its cap can be removed or attached and fuel can be fed without problems.
- Maintain a reasonable clearance between an exhaust system component (front pipe, silencer and tail pipe) and a fuel tank or a fuel tube.

② Installation of fuel tank

Individual method such as the reinforcement of the intraframe might be provided by the vehicle type. Confirm "SPECIFIC INFORMATION" of the targeted vehicle type.

- When relocating a fuel tank, make sure whether there are free holes in a chassis frame side member adjacent to the area to which the fuel tank is to be relocated, so that these holes can be used for installing a bracket. Using these holes is recommended rather than making new holes.
- If more holes are required, see the section " I -2-1 DRILLING OR WELDING A CHASSIS FRAME" for details.
- When moving the fuel tank vertically, make bracket referring below drawing.
 - Do not install the bracket together with other equipment by sharing same hole.
 - Do not make hole pitch smaller than the original bracket hole pitch vertically or horizontally.

Additional bracket for relocating fuel tank



- Be sure to air bleed after detaching tank or hose.
- If nylon lock nut is used for band fixing fuel tank, do not reuse it, change to new one. Refer to " SPECIFIC INFORMATION" for cautions at installing fuel tank.
- When installing an additional fuel tank, follow the instructions below to prevent vehicle fire.
 - Use a genuine ISUZU fuel tank.
 - Fuel must be supplied from the main tank.
 - In cold regions, to prevent a fuel hose from sagging due to snow, secure the fuel hose on any retaining component such as stay attached to a tank hanger bracket.
 - Never branch from engine supply line. It may affect fuel supply to engine.

③ Fuel hoses and pipes

Follow the below cautions when relocating or adding fuel tank.

- When changing a fuel hose, use a fuel hose of the same material and size as the genuine fuel hose or steel pipe.

Feed side fuel hose

Unit: mm

Vehicle type	Nominal size	Inner diameter	Outer diameter
Light duty vehicle	09	9.3±0.4	17.0±0.7
Medium duty vehicle	10	9.5±0.4	14.5±0.7
Heavy duty vehicle	12	11.3±0.5	19.0±0.8

Return side fuel hose

Unit: mm

Vehicle type	Nominal size	Inner diameter	Outer diameter
Light duty vehicle	05	5.8±0.4	12.2±0.7
Light duty vehicle (Q-Series)	06	5.8±0.4	13.5±0.7
Medium duty vehicle	08	7.5±0.3	13.5±0.7
	09	9.3±0.4	17.0±0.7
Heavy duty vehicle	10	9.5±0.4	14.5±0.7

- Use genuine ISUZU rubber hose. Use of non-genuine hose may cause problems such as fire.
- Use steel pipes in engine room.
- Follow the below chart for bending radius.

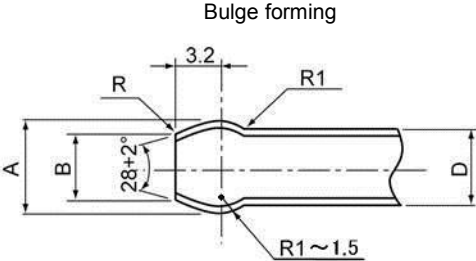
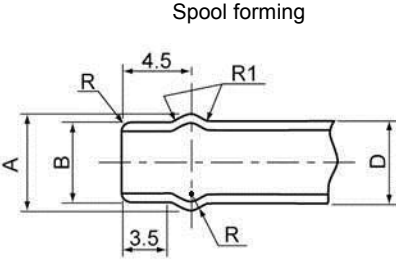
Unit: mm

Hose diameter	12.2	13.5	14.5	17.0	19.0
Min bending Radius	70	75	80	95	105

Note: Form pipes referring to above chart, without wrinkles at inner bent or blockage.

- Use fuel pipes size shown below.

Detail of end portion

Bulge forming		Spool forming		Unit: mm
				
Hose Nominal Size	D	A	B	
05/06	6.35	7.4±0.3	6.1±0.3	
08	8	9.3±0.3	7.9±0.3	
10	10	11.5±0.3	10.0±0.3	
12	12	13.5±0.3	12.0±0.3	
Hose Nominal Size	D	A	B	
05/06	6.35	7.4±0.3	6.35	
08	8	9.3±0.3	8	
10	10	11.5±0.3	10	
12	12	13.5±0.3	12	

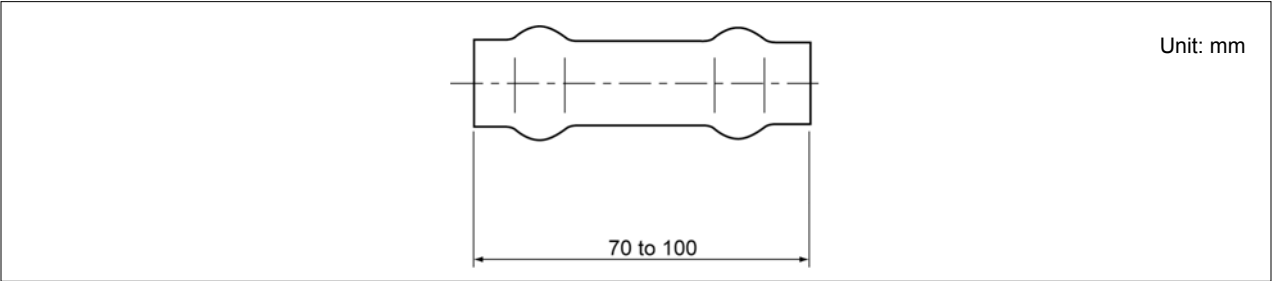
Inner surface	Copper plating (Thickness: 3μ at minimum)
Outer surface	Galvanization (Thickness: 13μ at minimum) and Organic coating (*)

Note: * Corrosion resistance: 2,500hrs. at minimum before reddish brown corrosion appears after salt spray test.

④ Precautions for connecting pipes

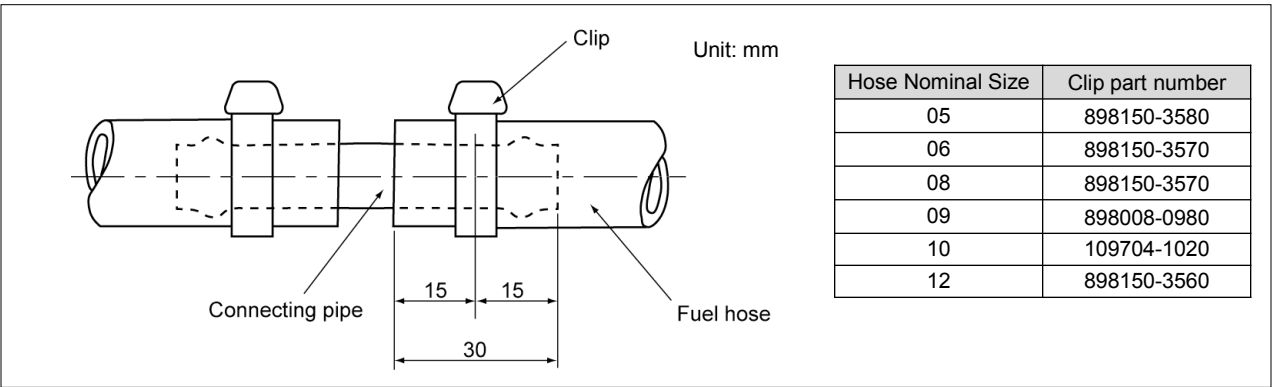
- Do not connect two or more hoses to lengthen a single hose.
- Fabricate a connecting pipe from a steel pipe. Perform bulge forming or spool forming on both ends of this pipe.

Length of connecting pipe



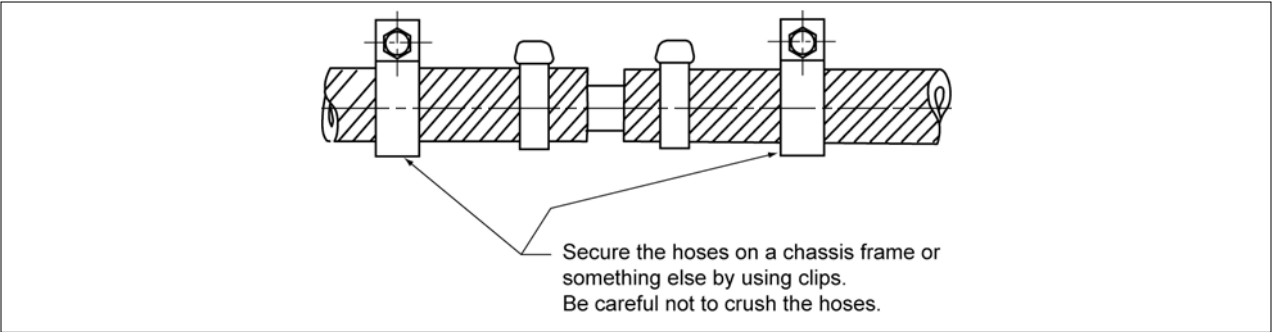
- Insert this connecting pipe into hoses by more than 30mm, and secure with clips.

Connecting hose and pipe



- If bulge forming and spool forming are impossible, the insertion depth should be 35mm or more.
- Secure the hoses on a chassis frame or something else by using clips.

Hose fixing



- Hose joints must not be located above an exhaust system.
- After completing the hose connection, confirm that there is no fuel leakage.
- Do not join a fuel hose to the factory-installed one to take out fuel. Fuel must be taken out directly from the fuel tank.

2-11 PRECAUTIONS FOR TOWING HOOK

Follow the instructions below when relocating towing hook installed at delivery.

① Location

- For straight frame: locate towing hook leveled with frame.
- For kick-up frame: locate towing hook incline 15° downward from frame to avoid effect to rear body or poor operativity from the hook located higher.

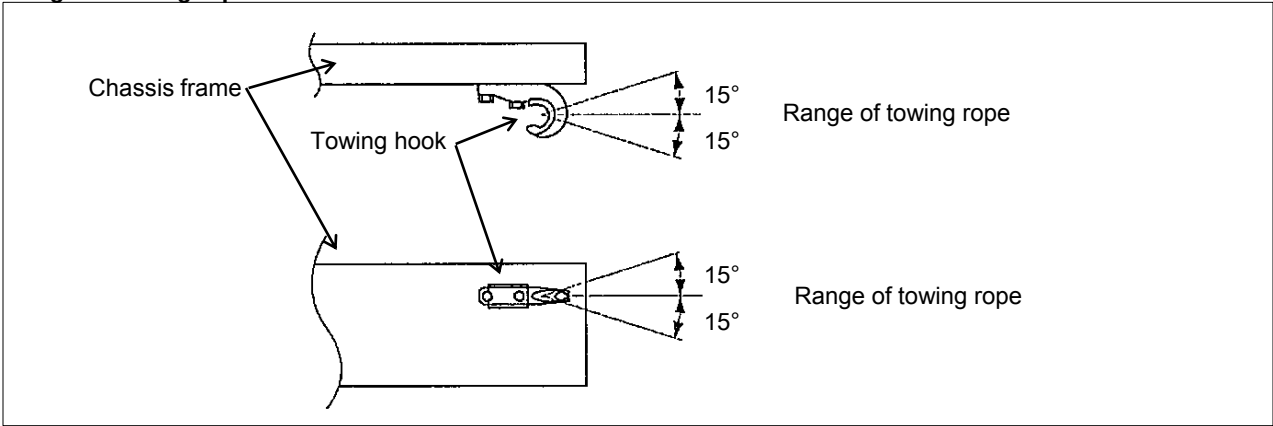
② Installation

- For drilling holes on frame, see the section " I -2-1 DRILLING OR WELDING A CHASSIS FRAME" for details.
- Hole diameter and tightening torque is as follows.

Vehicle type	Location	Hole diameter mm	Torque N·m {kgf·m}	Remarks
Light duty vehicle	Lower	11	51±7 {5.2±0.7}	—
	Side	15	157±32 {16.0±3.3}	—
Medium duty vehicle	Side	15	133±26 {13.6±2.7}	FR, FS series
		17	193±39 {19.7±4.0}	FT, FV series
Heavy duty vehicle	Side	17	193±39 {19.7±4.0}	—

The rear body posterior of towing hook must not be in range of towing rope.

Range of towing rope



2-12 PRECAUTIONS FOR MODIFICATION OF THE ELECTRICAL SYSTEM

① General Precautions

- Careless modification may result in fire or equipment breakdown. Follow the below cautions.
- Modification to the factory-installed electrical components such as wires, fuses and relays are prohibited.
- Do not tear protective corrugated tubes and pull out inside wires to connect additional equipments because this may damage the remaining harnesses.
- Be careful not to damage the wire harness and other components. When they are damaged, have, them repaired.
- Never modify the harness around electric unit (control unit).
- Restore the terminals, connectors, clips, and harnesses removed at work where they were originally. If the terminal polarity or connector partner is confused, it may lead to defect or damage the component.
- If the original clip cannot be used due to additional components, add alternative clip to fix harness.
- When a clip for fixing harness is broken, replace it with new clip same as the broken one. Metallic clip has coating (rubber or vinyl chloride) at harness converging part. Clip with tear on coating, hole, crack is considered damaged, and replace it with new clip.
- Exposing connectors, relay box, and electric equipments to such water may cause water intrusion into electric components or internal condensation, resulting in problems such as operational malfunction, erosion, and in the worst case, a vehicle fire.
- When mounting rear body on chassis frame, be careful not to have harness jammed.
- When there is melting fuse, investigate the cause, repair it, and change it with new one of same capacity. Never increase the fuse capacity, it will burnout components and wires, and lead to fire.

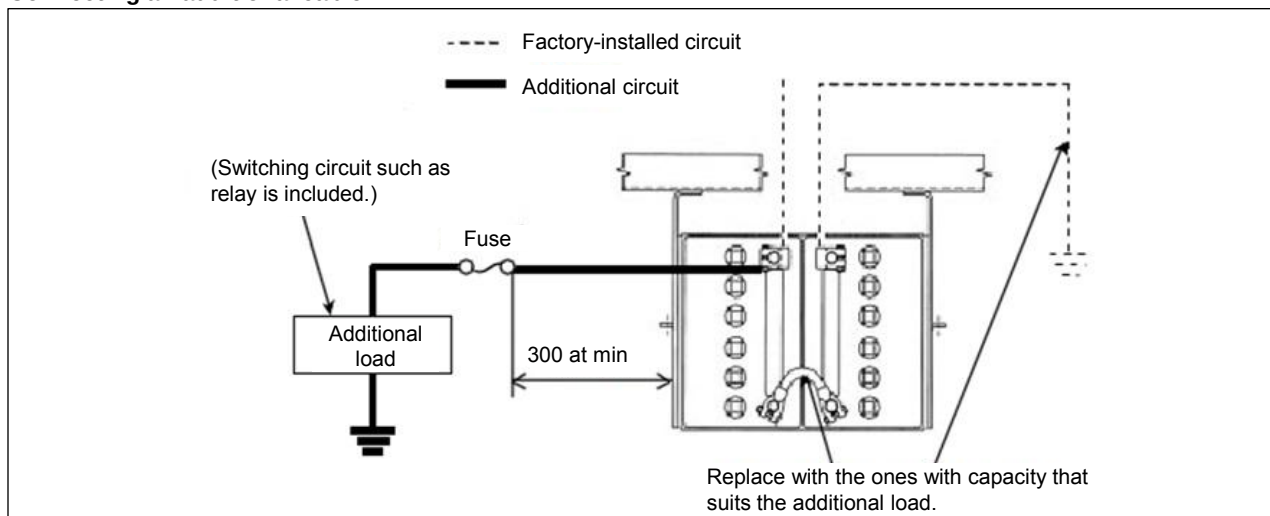
② Additional power supply (standby power supply) and ground circuit

Sizes and shapes of terminals and electric wires used for the increase must use the one that corresponded to the increased load. Select the battery and ACG that corresponds to the electric load when installing body with high electric load (fire engine and refrigerator car, etc.). Refer to "SPECIFIC INFORMATION".

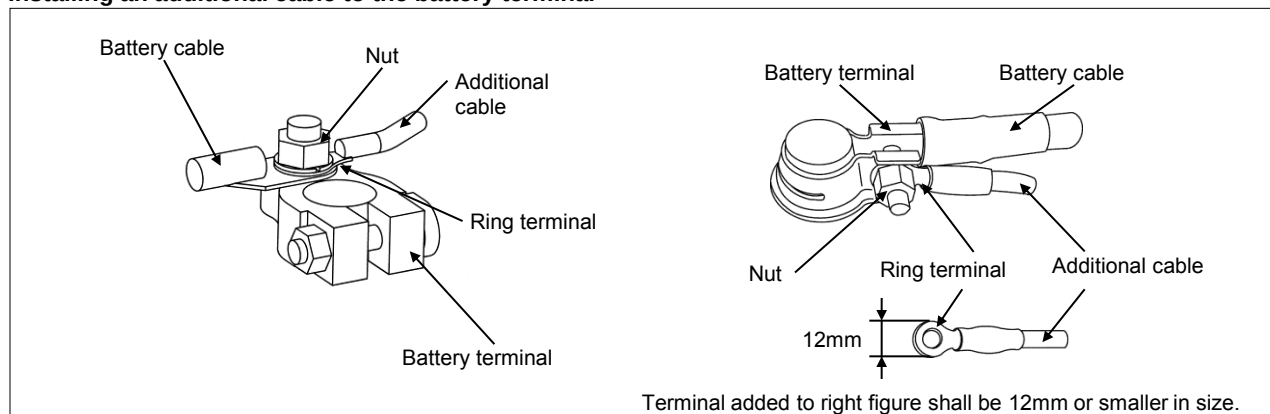
- How to extract power from the factory-installed wires (when the electrical load of rear body does not exceed the allowable capacity of standby power supply)
 - Do not extract power other than from the extraction of the standby power supply prepared in the vehicle.
 - Maintain the power below the ratings capacity of the standby power supply.
 - Refer to "SPECIFIC INFORMATION" of the targeted vehicle type for the position and the rating capacity of the standby power supply.
 - Do not connect ground circuit of additional power supply with the existing harness. Connect it with either the ground for standby power supply, negative terminal of the battery, or the frame.

- How to extract power from the positive battery terminal (when electrical load of rear body exceed the allowable capacity of standby power supply)
 - Provide a fuse in each power extraction circuit.
 - Fuses must be a minimum of 300mm apart from the battery (volatile gas may come out of the battery) and protected from water.
 - Do not put a fuse directly on the battery body or on the reverse side of the battery cover, otherwise the battery will be damaged or explode.
 - Use the ring terminal to extract from battery, fix it together with battery terminal to prevent loosening.
 - Only one terminal may be added to prevent loosening.
 - Do not connect ground circuit of additional power supply with the existing harness. Connect it with negative terminal of the battery.
 - When using a cable of 5mm² or larger, solder the pressured part of the wire terminal, and then wrap the area with a heat shrink tubing.
 - Extracted power from battery is always energized, does not depend on the state of the key switch on the underbody. This can cause the battery go flat. Take the necessary measures such as appending to the caution label or the owner's manual.

Connecting an additional cable



Installing an additional cable to the battery terminal



③ Ground point

Careless modification may result in fire or equipment breakdown.

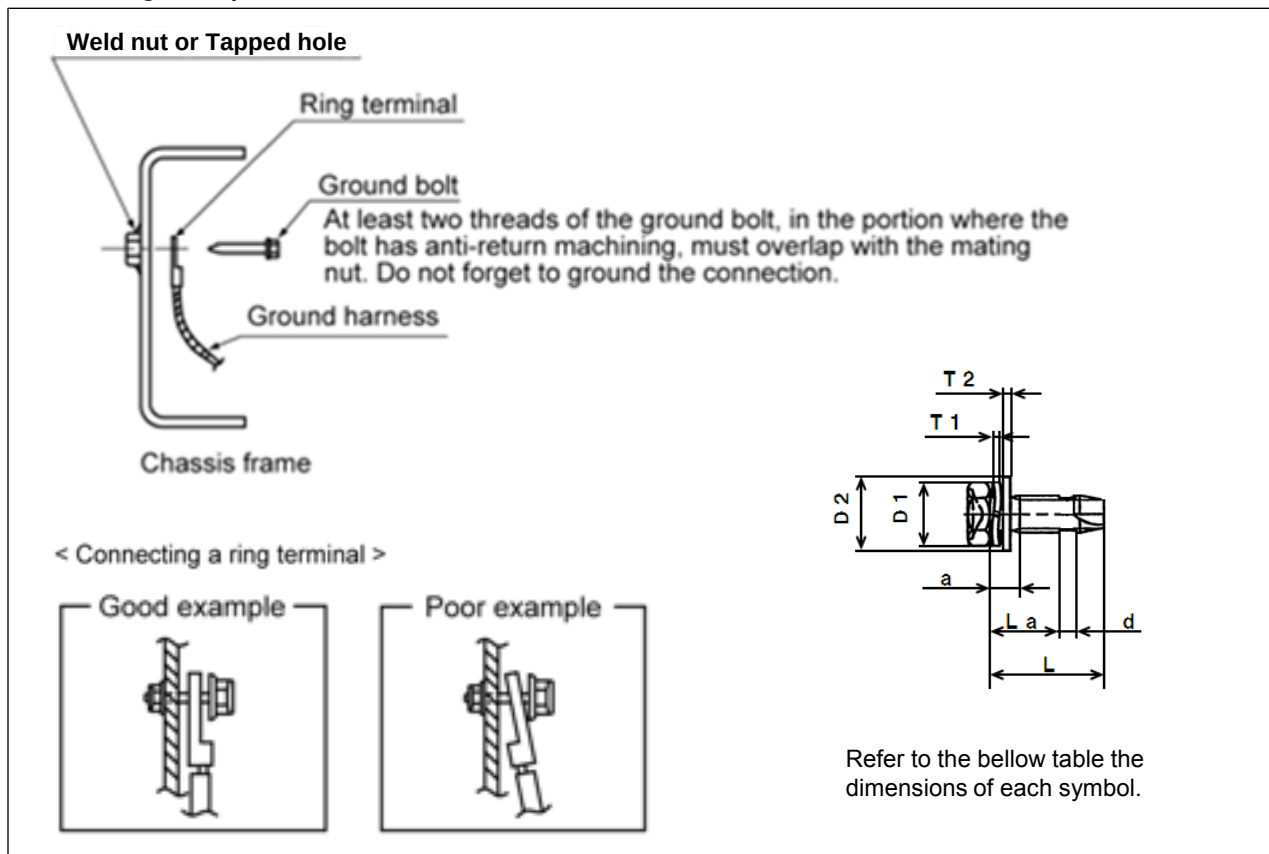
Relocating the ground points of the factory-installed ground circuit and lengthening or shortening these wires or cables is prohibited, because doing any of these may result in breakdown of electrical equipments or electronics devices due to surge, etc.

Follow the bellow instructions when adding ground circuit for bodies or additional equipments.

• How to connect the ground circuit to a chassis frame

- The ground point of an additional equipment should be connected to the chassis frame member to which the negative battery cable is connected.
- Do not secure the ground point of additional equipment together with a factory installed ground circuit because the fixing bolt will loosen and contact failure will occur.
- Use ring terminal and secure it without looseness. Jamming foreign substance at connecting will be a cause of looseness.
- Bolt for fixing ground circuit on frame and cab panel is exclusive use. When the bolt is damaged at work, replace it with specified exclusive ground bolt.
- Weld a mating nut on the chassis frame or directly tap the chassis frame. To drill holes for directly tapping chassis frame, follow "I -2-1 DRILLING OR WELDING A CHASSIS FRAME".

Addition of ground point



Ground bolt part list

Part No.	Nominal Diameter	Pitch	a (Max) mm	Length L mm	Spring washer T1×D1	Plane washer T2×D2	Lock length d mm	Lock position La mm	Tightening torque N·m {kgf·m}
898164-8360	M6	1	2.6	14	None	1.6×12	4	5	7.8±2.0 {0.8±0.2}
897086-8220	M6	1	3.9	14	1.2×11.3	1.6×12	3.5	6	7.8±2.0 {0.8±0.2}
897086-8230	M6	1	5.4	16	1.2×11.3	1.6×12	4	7.5	7.8±2.0 {0.8±0.2}
897086-8240	M6	1	5.4	20	1.2×11.3	1.6×12	4	10	7.8±2.0 {0.8±0.2}
897087-2010	M6	1	5.4	25	1.2×11.3	1.6×12	4	16.5	7.8±2.0 {0.8±0.2}
897087-6170	M6	1	5.4	40	1.2×11.3	1.6×12	4	31	7.8±2.0 {0.8±0.2}
897119-4790	M6	1	3.9	14	1.2×11.3	1.6×12	3.5	6	7.8±2.0 {0.8±0.2}
897206-4070	M6	1	3.9	14	1.2×11.3	1.6×12	3.5	6	7.8±2.0 {0.8±0.2}
898164-8370	M8	1.25	3	14	None	1.6×16	4	5	18.6±4.9 {1.9±0.5}
898141-2550	M8	1.25	3	18.5	None	1.6×16	6.5	6	18.6±4.9 {1.9±0.5}
897086-8250	M8	1.25	6.5	20	1.6×14.3	1.6×16	4	10	18.6±4.9 {1.9±0.5}
898141-2561	M8	1.25	3	24	None	1.6×16	10	7	18.6±4.9 {1.9±0.5}
897087-0550	M8	1.25	4.7	16	1.6×14.3	1.6×16	4	6	18.6±4.9 {1.9±0.5}
897087-2030	M8	1.25	6.5	25	1.6×14.3	1.6×16	4	15	18.6±4.9 {1.9±0.5}
897087-2040	M8	1.25	6.5	30	1.6×14.3	1.6×16	4	20	18.6±4.9 {1.9±0.5}
897264-7660	M8	1.25	2	23	None	None (flange diameter 16mm)	5	4	18.6±4.9 {1.9±0.5}
898164-8380	M10	1.25	4.5	20	None	2×20	6.5	6	40.2±5.9 {4.1±0.6}
897087-2050	M10	1.25	5.9	25	1.8×17.4	2×20	5	10	40.2±5.9 {4.1±0.6}
897087-2060	M10	1.25	7.2	30	1.8×17.4	2×20	5	17.5	40.2±5.9 {4.1±0.6}

④ Wire size and fuse capacity

- Improper use of wire harnesses may result in fire or equipment breakdown.
- If additional wire harnesses are required, select appropriate wires according to the load current to be added, the circuit length, and the harness route.
- Use electric wires for automobiles (JIS-certified wires such as AV, AVS, AVX, AEX, or equivalent).
- When using an AEX wire, parts used with it such as protection tube, corrugated tube, vinyl tape, etc. should also be heat-proofed.
- When using a wire whose size is 5mm² or larger, solder the pressured part inside the wire terminal to prevent it from heating.

Permissible temperature and current of electric wire

Wire type	Allowable temperature	Allowable current (ampere)							Remarks
		0.5 mm ²	0.85 mm ²	1.25 mm ²	2 mm ²	3 mm ²	5 mm ²	8 mm ²	
AV	80°	9A	11A	14A	20A	27A	36A	47A	Standard
AVS	80°	8A	11A	14A	19A	26A	35A	No spec.	Standard
AVX	90°	8A	10A	13A	17A	24A	33A	43A	for high temp.
AEX	110°	7A	9A	12A	17A	23A	32A	42A	for high temp.

- Fuse capacity, wire size and circuit length
 - Fuse capacity on the power source side determines usable wire size and circuit length.
 - Select appropriate wire size in consideration of the fuse capacity on the power source side and the wire route. If a wire is connected to an equipment that produces relative movement, the size of the wire should be 0.85mm² or larger so as to prevent wire breaking.

Permissible current of fuse, wire size and circuit length

Fuse			Wire size and circuit length						
Type	Rating capacity	Allowable continuous current	0.5 mm ²	0.85 mm ²	1.25 mm ²	2 mm ²	3 mm ²	5 mm ²	8 mm ²
Blade fuse	5A	3.5A	Max.25m	Max.40m	–	–	–	–	–
	7.5A	5.2A	Max.15m	Max.25m	Max.40m	–	–	–	–
	10A	7.0A	Max.10m	Max.20m	Max.30m	–	–	–	–
	15A	10.5A	Max.5m	Max.10m	Max.15m	Max.30m	Max.45m	–	–
	20A	14.0A	×	Max.5m	Max.10m	Max.20m	Max.30m	–	–
	25A	17.5A	×	×	Max.10m	Max.15m	Max.25m	Max.40m	–
SBF (Slow-blow fuse)	30A	15.0A	×	×	×	Max.10m	Max.15m	Max.30m	Max.45m
	40A	20.0A	×	×	×	×	Max.10m	Max.20m	Max.30m
	50A	25.0A	×	×	×	×	×	Max.15m	Max.25m
	60A	30.0A	×	×	×	×	×	Max.15m	Max.20m
	80A	40.0A	×	×	×	×	×	×	Max.15m
	100A	50.0A	×	×	×	×	×	×	Max.10m

Legend: ×: Cannot be used –: 50m at max

⑤ Lengthening or shortening an electric wire or a cable

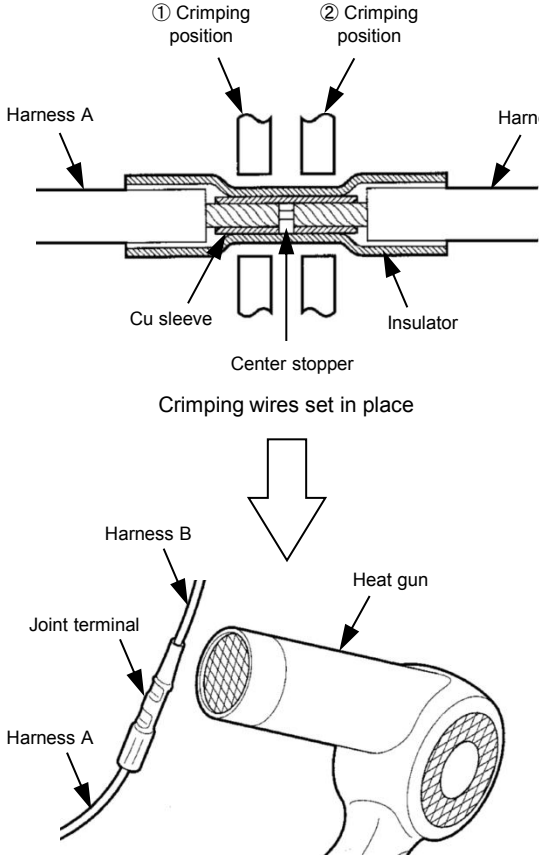
- Splicing of battery cable is prohibited. When high current is energized, it may generate heat at splicing point and cause fire. (Replace it with new cable.)
- When lengthening a wire, splice a new wire with the same section area and color as those of the wire to be lengthened.
- When splicing a wire (adding a wire to an end of another wire), use specified joint terminals, soldering or crimping terminal for secure connection.
- Do not splice the wires by only twisting them together.
- For the area where wires may be subject to vibration such as above engine or transmission, splicing using specified joint terminals or with crimping terminals is recommended rather than soldering because solder may seep into the inside of the wires, making them harden and lose flexibility, resulting in separation.
- When splicing a wire with soldering or crimping terminals, remove burrs and wrap with vinyl tape to insulate the area. If the insulation is insufficient, it might short or cause vehicle fire.
- When splicing wires with soldering or crimping terminal outside a cab or on the chassis frame where water proofing is necessary, wrap butyl rubber at splicing point for waterproofing. If the water proofing is insufficient, the conductor (copper wire) will corrode and lead to disconnection or conduction defective.

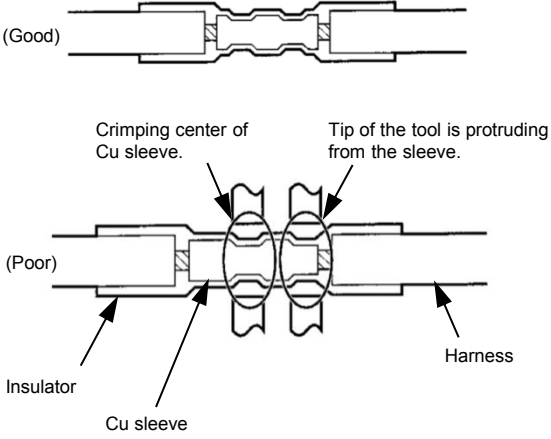
Parts used for wire splicing (joint terminal)

Joint terminal	Color	Wire size mm ²	Wire strip length mm
897169-4730	Yellow	0.3 to 0.5	6.0 to 7.0
897169-4740	Red	0.75 to 1.25	7.0 to 8.0
897169-4750	Blue	2.0	7.0 to 8.0

Note: Use specialized tool (587410-6470) for crimping tool.
For wires with size larger than 2mm², follow the instruction on next page.

Splicing of wire harness (joint terminal)





Heat the tube with a heat gun or the like to shrink it.
Keep heating until translucent plastic ooze out whole circumference from both ends of tube.

Roughly heating temperature ; 120°C

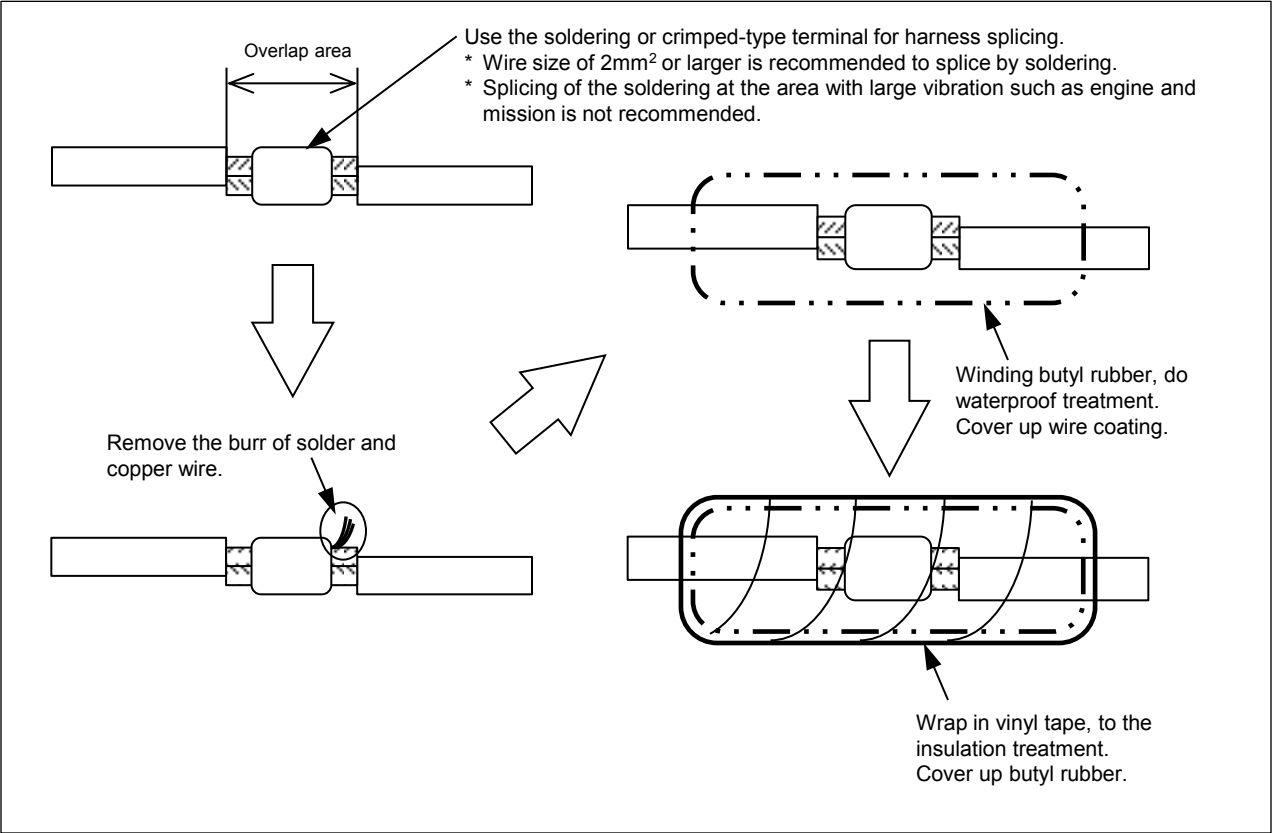
* At the crimping work, the tube(insulator) of the joint terminal may be damaged and leave a hole. However, when it is possible to cover the entire hole with the heated translucent plastic, rework is not needed. Because it is considered to satisfy the waterproof performance.

Parts used for wire splicing (butyl rubber)

Purpose	Parts name	Parts number	Remarks
Splice	Crimped-type terminal	(Commercially)	Work with tools and working methods that are recommended in the used crimping terminal.
	Solder	(Commercially)	—
Water proof	Butyl rubber	182698-2630	Not required in waterproof unnecessary area(such as area with out water splash).
Insulator	Vinyl tape	(Commercially)	—

Note: For the additional wire, consider the overlap length of splicing area to the wire cut off. If there is recommended length for the used crimped-type terminal, prepare additional wire with the length added.

Splicing of wire harness (butyl rubber)

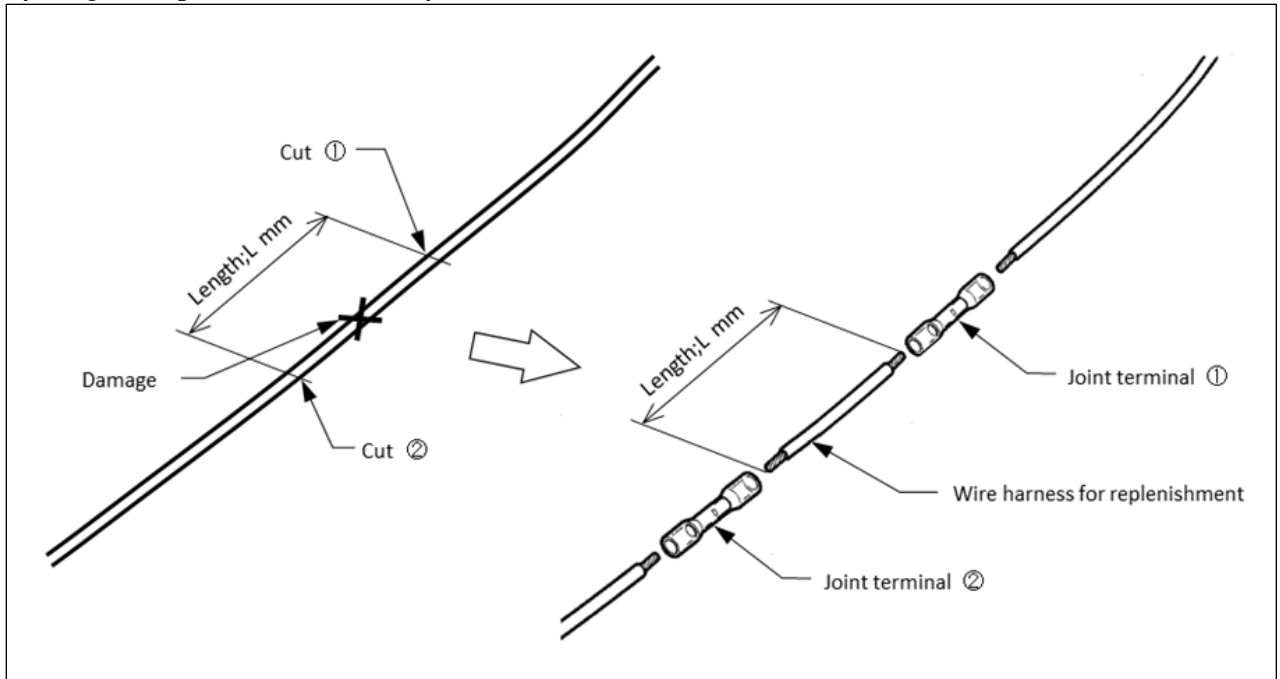


- When splicing wires, do not splice at vibrating position. It may be disconnected
- When shortening a wire or a harness, do not cut it but tie its redundancy with a vinyl tape, and then bind it to the factory-installed harnesses or something else.

⑥ Repairment of damaged wires

- For repairing damaged wires, cutoff the damaged portion, add wire, and perform splicing.
- Below figure shows joint terminal method. Follow the same procedure for butyl rubber method.
- When using specific joint terminal for splicing, the length of wire cutoff and added must be the same.
- When splicing with soldering or using crimping terminal, length of the additional wire must be the total of length of wire cutoff and overlap of joint portion (2 places) added. If there is recommended length for used crimping terminal, add the overlap length (2 places) to it.

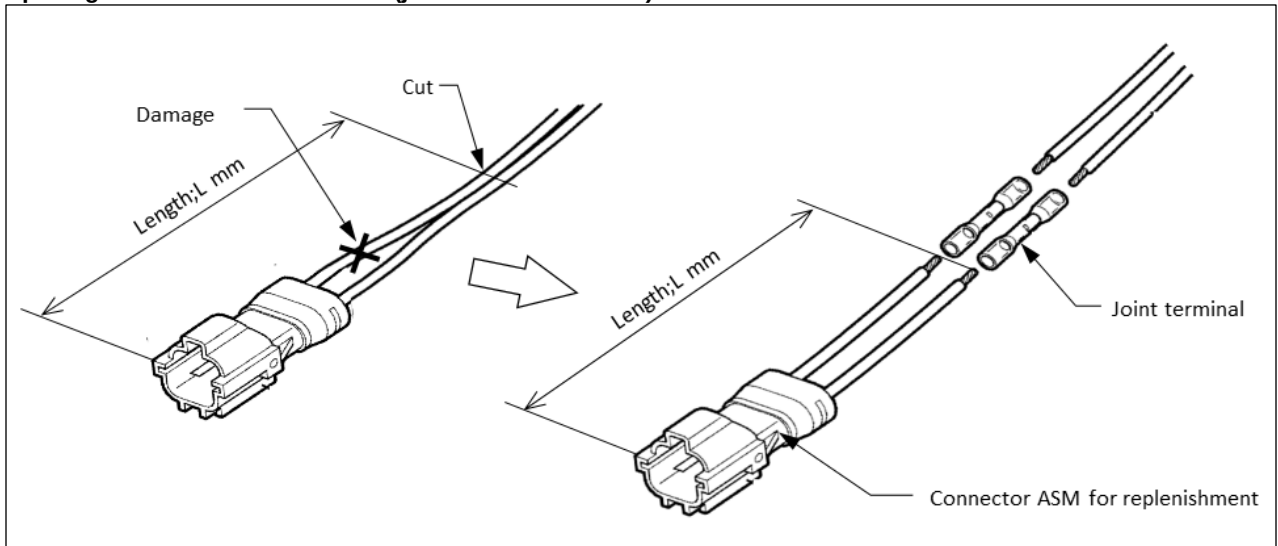
Splicing wires (joint terminal method)



- When rerouting the damaged circuit due to working space etc., follow the above directions except for the additional wire length.
- For the rerouted wires, perform damage prevention treatment.

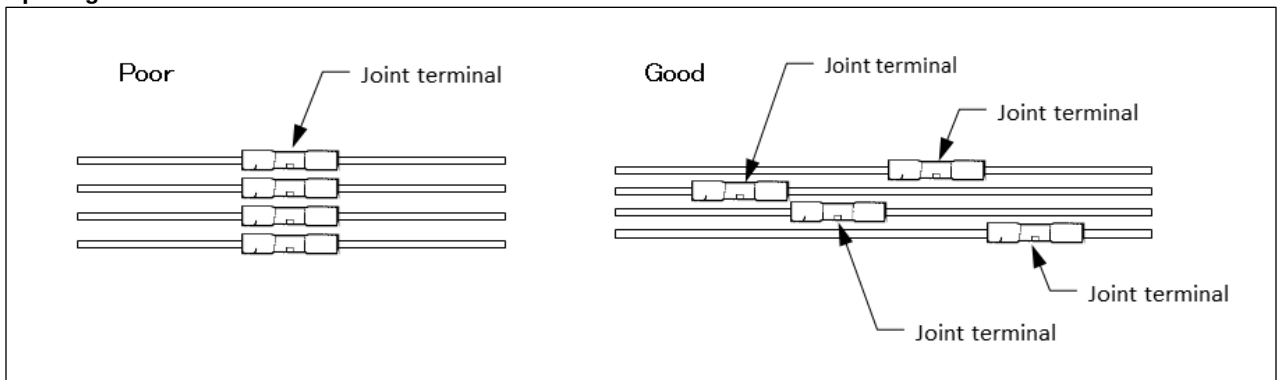
- When the connector or the wire close to connector is damaged, prepare connector ASM for replenishment and perform splicing.
- Below figure shows joint terminal method. Follow the same procedure for butyl rubber method.
- When using specific joint terminal for splicing, the length of wire cutoff and connector ASM for replenishment must be the same.
- When splicing with soldering or using crimping terminal, length of the additional wire must be the total of length of connector ASM and overlap of joint portion added. If there is recommended length for used crimping terminal, add the overlap length to it.

Splicing wires close to connector (joint terminal method)



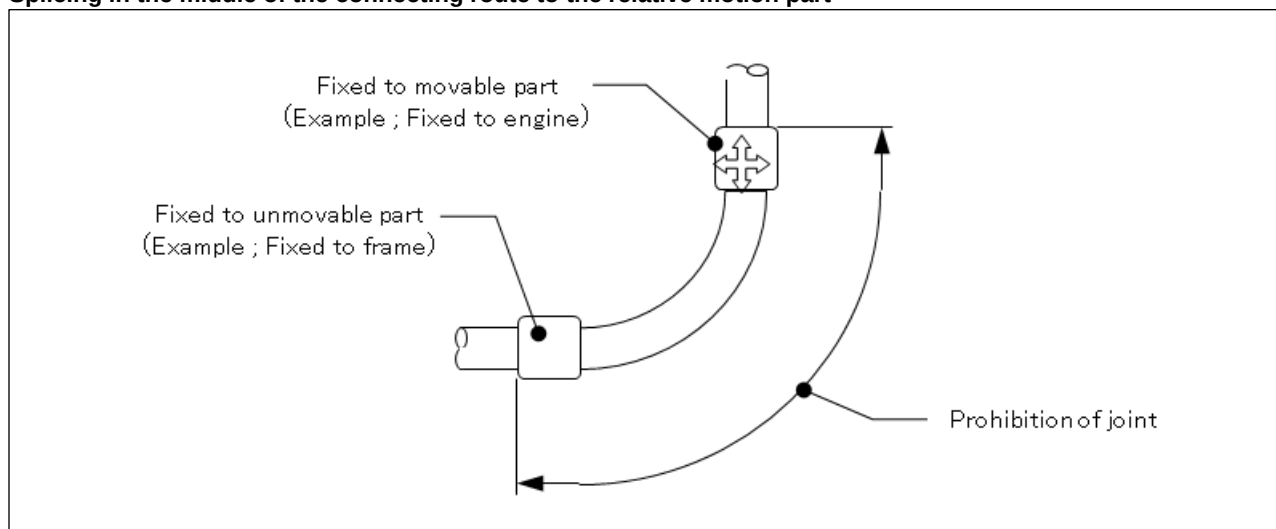
- Additional wires and connector ASM for replenishment must be the same size (diameter of copper wire). Considering re-repairing, use of same color wire is recommended.
- When splicing 4 or more wires, adjust splice position by shifting back and forth to avoid overlapping to keep the harness diameter enlargement in minimum. Do the above same with 3 or less, if they do not fit in the protective material with splicing at the same position.

Splicing 4 or more wires in the same area



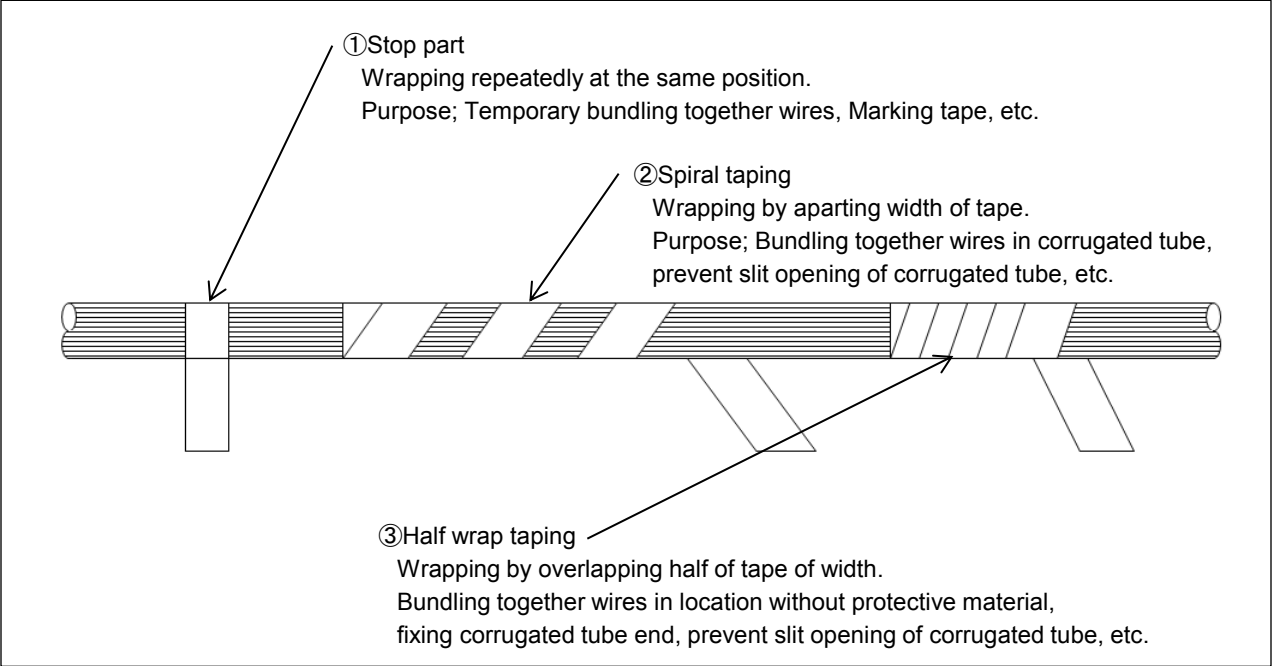
- Do not splice in the following area; bent location of harness, area of large vibration, and area of high temperature.
 - Since the flexibility is reduced at splicing portion as compared with other portions, it is possible that harness route may not return to the original.
 - Possibility of disconnection occurrence will be higher, because flutter increases due to larger weight.
 - Decide the position of the splice carefully
- In case of performing the splicing in the middle of the connecting route to the relative motion part (cab to frame, engine/mission to frame), there is a risk of breakage caused by bending input. Perform the splicing out of the connection area by replacing the lengthened wire all spans the area.

Splicing in the middle of the connecting route to the relative motion part

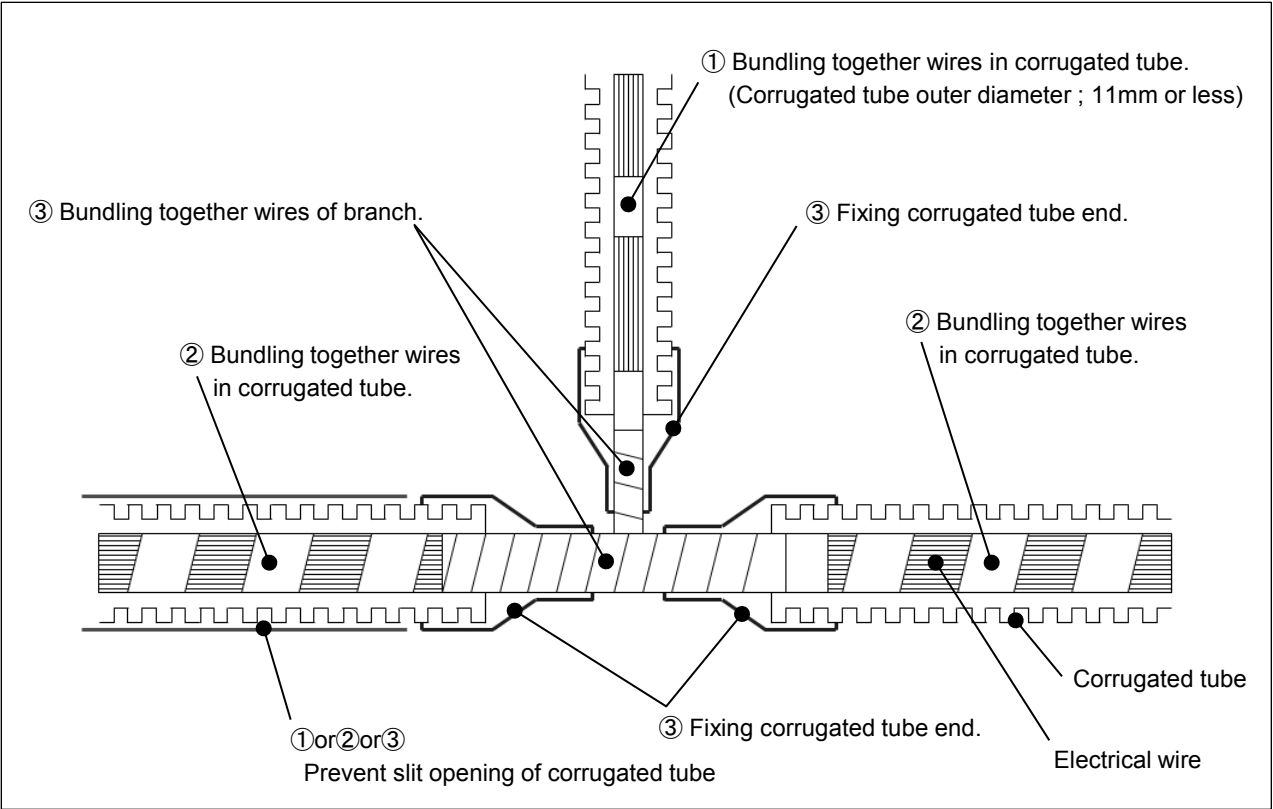


⑦ Wrapping tapes

Wrapping tapes



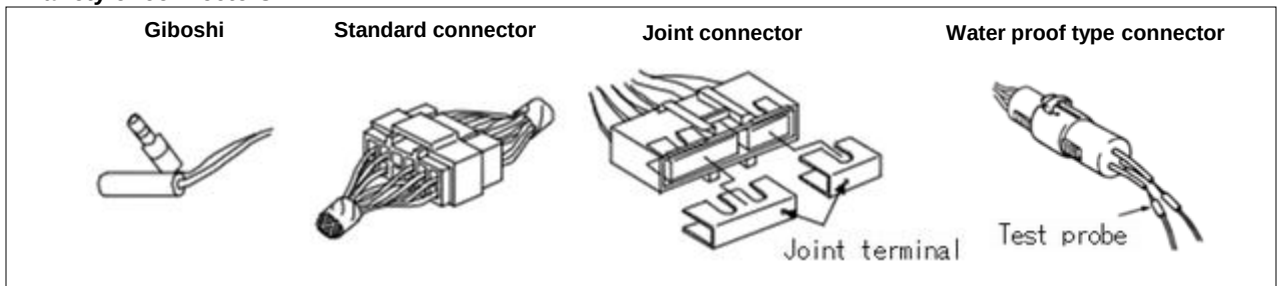
Wrapping tapes at branch



⑧ Connectors

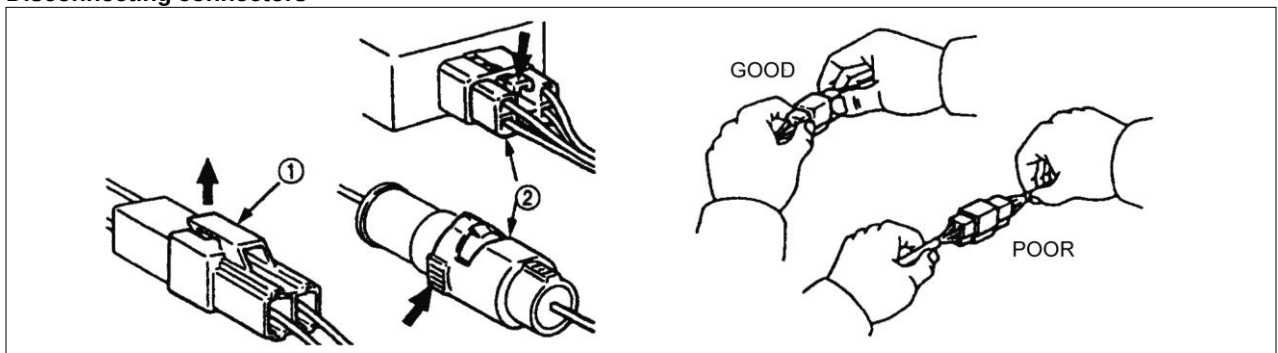
- Use water-proof connectors for the harnesses placed outside a cab.
- Do not unnecessarily disconnect these waterproof connectors, otherwise continuity failure will occur.
- When disconnecting waterproof connectors, to prevent foreign substances such as water and metal chips from getting into the joint, or to protect the sealed part from being damaged, cover the disconnected connectors with vinyl bags or something. If water or any foreign substance get into the joint, dry up the joint or remove such foreign substance before connecting the connectors.

A variety of connectors

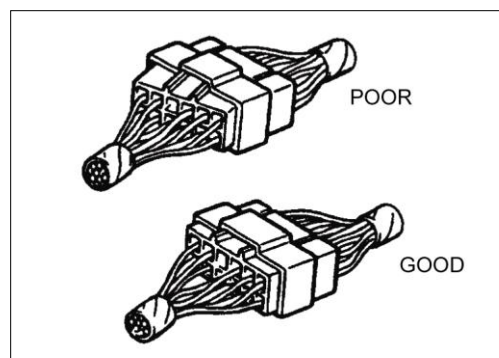


- As for the area where connectors may be subjected to vibration, use connectors with a detach-safe lock.
- When using a "GIBOSHI" type plug, be sure that its socket connector (female) with insulation cover is connected to the wire from the power source, to prevent it from shorting when the plug comes off and contacts with vehicle body.
- Disconnecting connectors
 - Certain types of connectors have a tang lock that prevents these connectors from coming off when vehicle is running.
 - Some tang locks can be released by pulling (①) while the others by pushing in (②). Check which type of tang lock the connectors have.
 - To prevent harnesses from coming off from the connector or breaking, when disconnecting connectors, do not pull the harnesses but securely hold both connectors (male and female), release the tang lock, and then separate the connectors with care.

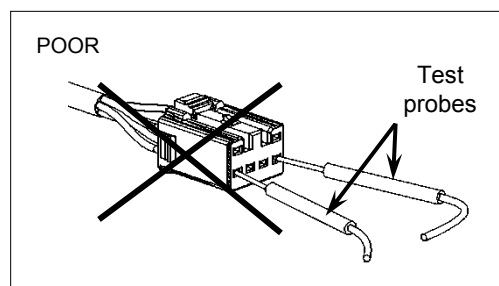
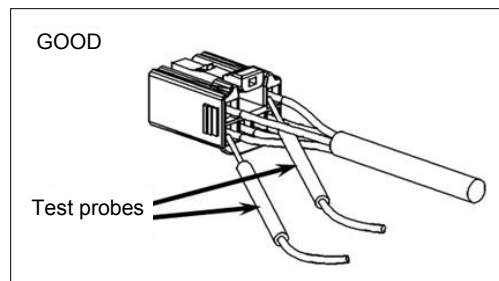
Disconnecting connectors



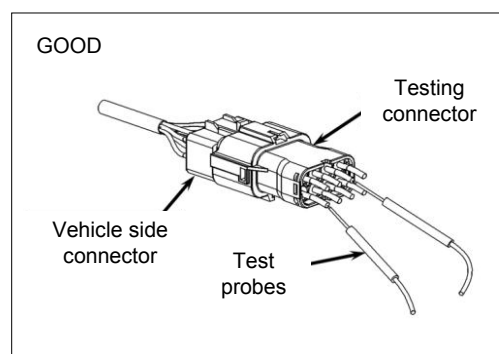
- Connecting connectors
 - Securely hold both connectors (male and female) and check whether the connector pins and the cavities match. Also, check whether the connectors are aligned.
 - Securely hold the connectors and carefully engage them by pushing until the connectors click.



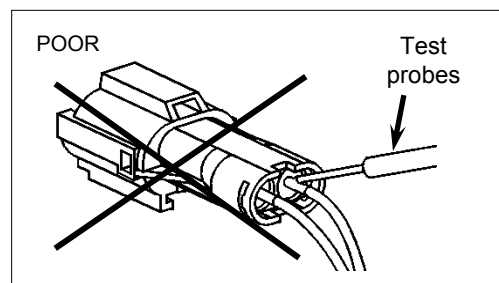
- Checking connectors
 - When using circuit tester to test conduction of connector, insert test probes of circuit tester from harness side of the connector.
 - Do not insert test probes of the tester in female terminal of connector. It may deform the terminal and lead to poor contacting.



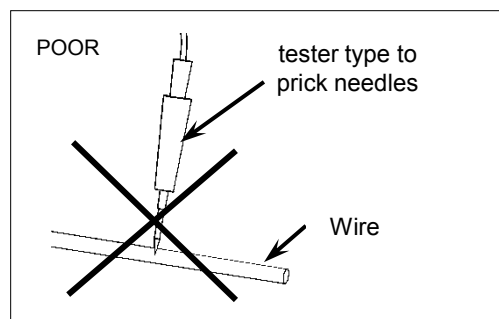
- For the water proof connector, the test probes cannot be inserted from the harness side structurally. Use testing connector prepared beforehand to check. Connect test connector to the vehicle side connector and connect test probes to the harness of the testing connector.



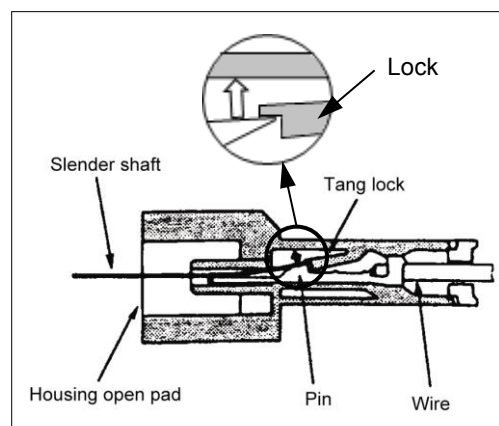
- Do not insert test probes to protection rubber of water proof connector. It may decrease the water proof performance and became cause of corrosion.



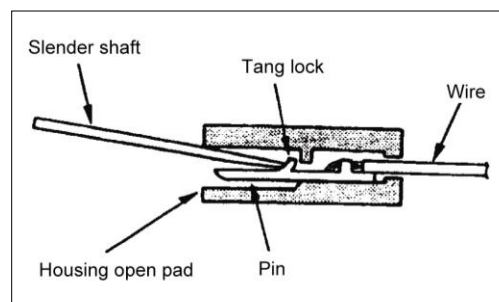
- Do not use tester type to prick needles to cover of harness. It may become cause of corrosion of wires and short.
- Confirm there is no damage to lead line or connector from test probes after checking.



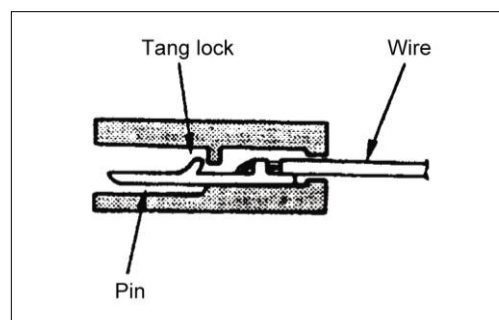
- Removing a connector pin (Built-in tang lock)
 - Insert a slender shaft into the opening of the connector housing.
 - Push up the tang lock in the direction of an arrow in the figure. Pull out the wire and the connector pin from connector.



- Pin-type tang lock
 - Insert a slender shaft into the opening of the connector housing.
 - Push the tang lock toward the wire so as to make it flat. Pull out the wire and the connector pin from the connector.



- Inserting a connector pin
 - Check if the tang lock is upright.
 - Insert a connector pin into the connector housing for wires. Push the pin until the tang lock is securely engaged.
 - Gently pull the wire to check if the pin is securely in place.



- Using a "GIBOSHI" type plug
 - When using a "GIBOSHI" type plug, be sure that its socket connector (female) with insulation cover is connected to the wire from the power source and its plug connector (male) to the wire from the electric load. This prevents disconnected "GIBOSHI" type plug from shorting when it comes off and contacts with vehicle body.

⑨ Harness routing

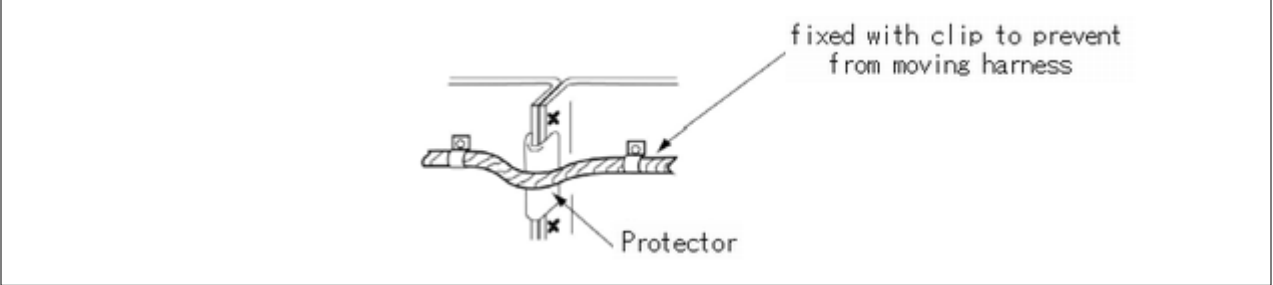
- When laying additional wires, be sure that they will not restrict inspection or maintenance of the vehicle components.
- To increase reliability of wires, protect all additional wires and harnesses with corrugated tubes or polyvinyl chloride tubes.
- When handling the wires, do not pull them hard.
- Additional harnesses should be laid along and secured on rear body members or on chassis frame. Do not pass them in the air.
- Securely fix the harnesses in place so that they will not contact with moving parts or sharp objects.
- Fix the harnesses so there is no jamming at rear body mounting.

Distance between harness and object

Object	Required distance between harness and object	
Moving part	most approached	10mm at min
Sharp object	minimum distance	10mm at min

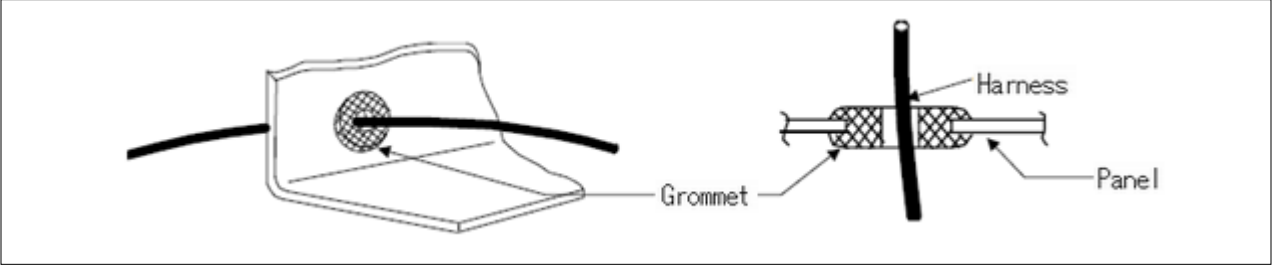
- If there is possibility that a harness contacts with a sharp edge of a metal part, protect the harness with corrugated tubes and clip fix to avoid contact the sharp edge, or cover the sharp edge with a protector, otherwise vibration will damage the harness coating.

Harness routing (1)



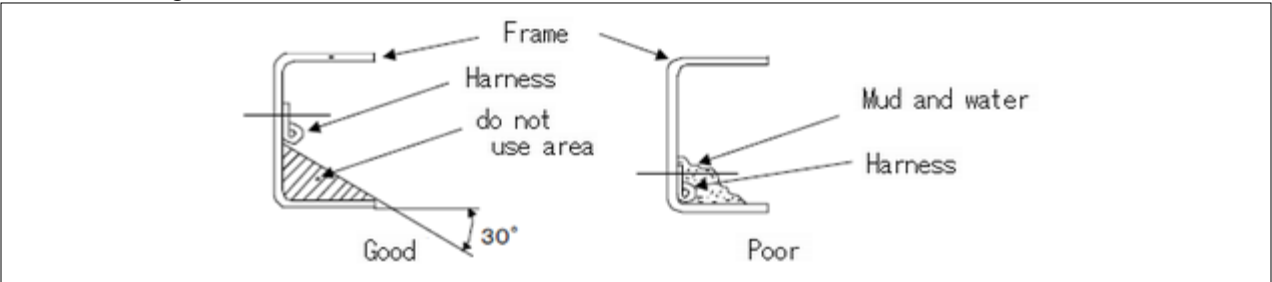
- To prevent harness coating from being damaged, attach grommets to the through holes in steel plates.

Harness routing (2)



- Do not put any harness on top, underside or outer side of chassis frame members because someone may step on it or flying gravels may damage it.
- Do not put any harness in the lower part of the inner side of chassis frame members because harness may be damaged or freeze due to mud, snow, water, etc. that may be built up in this area.

Harness routing of chassis frame members



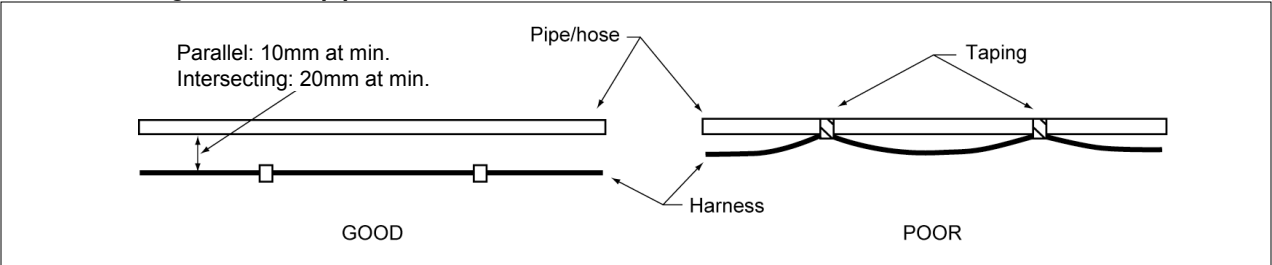
- If there are factory-installed harnesses nearby, secure additional harness on them with tapes or nylon clips. Do not tape wrap an additional harness directly on a metal pipe or a rubber hose including the brake pipe and the fuel pipe.
- Provide sufficient clearance between such pipe, hose and a harness.

Distance between harness and pipe/hose

Positional relation	Clearance between harness and pipe/hose
Parallel	10mm at min
Intersecting	20mm at min

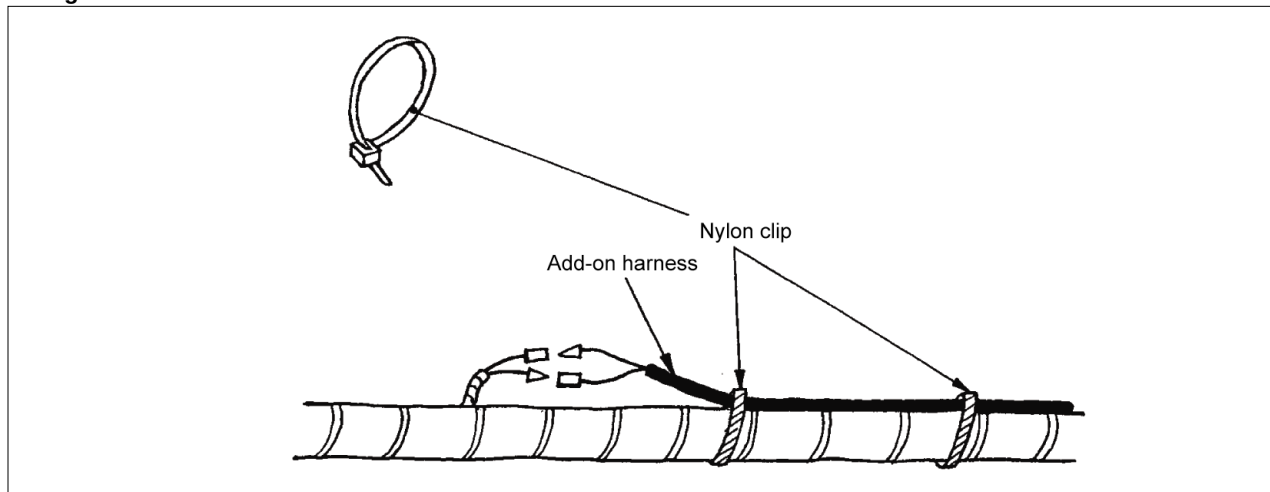
- When the Corrugated tube and the air tube vibrate while interfering, the air tube will damage. Arrange the tube maintaining enough space.

Harness routing of near the pipe/hose



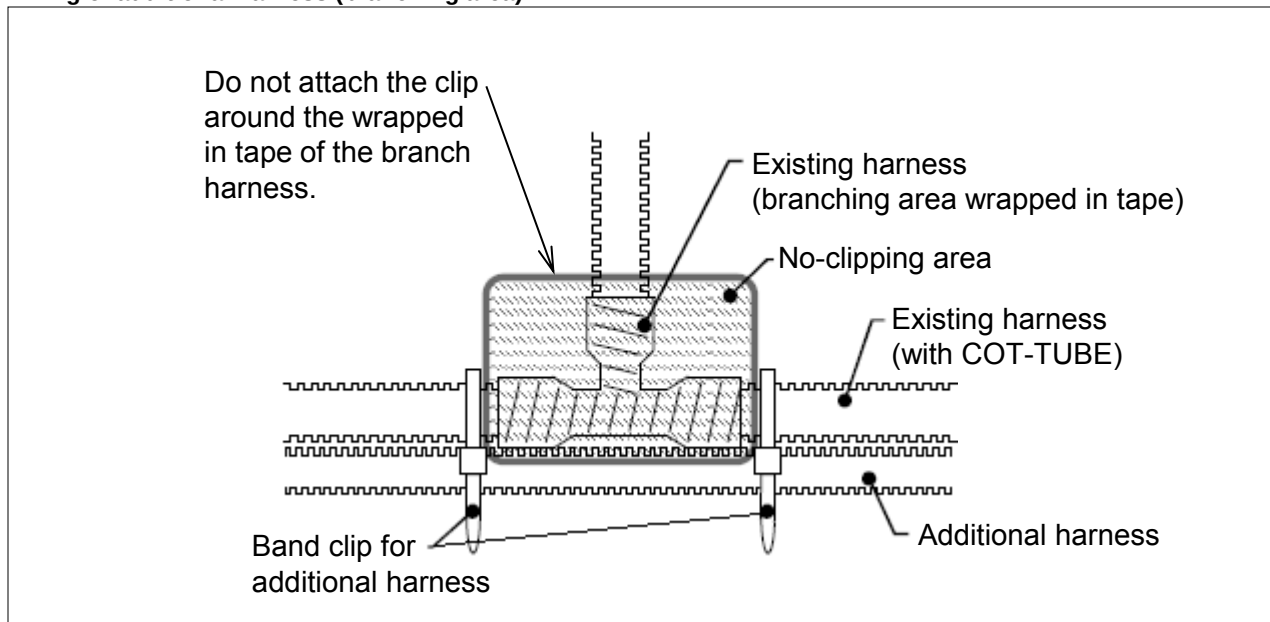
- Note the following when fixing together to factory-installed harness with band clips.
 - Put the additional harness in the vinyl chloride tube or the corrugate tube, use nylon clip to fix along the wire harness for the vehicle.

Fixing of additional harness



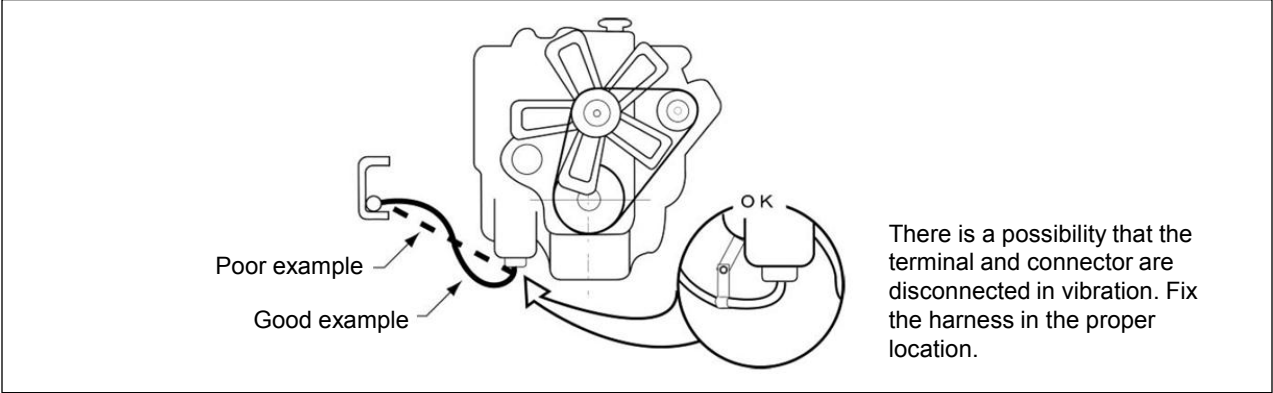
- If an additional harness is to be laid along the factory-installed harnesses, it should be firmly secured so as to avoid flip-flop movement induced by vibration of engine and vehicle, protecting factory-installed harnesses from being damaged. Fix clips at a the portion of the factory-installed harness which is protected by the materials such as COT-TUBE, polyvinyl chloride tube, hard tube, and protector, should be chosen to do so.

Fixing of additional harness (branching area)



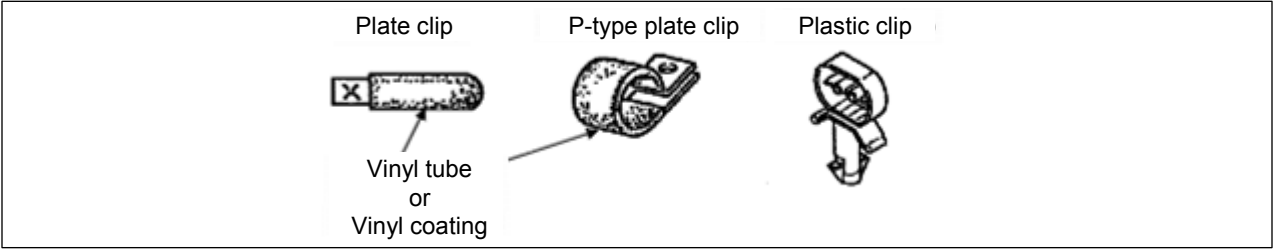
- Moving harness
 - When connecting an additional harness to any equipment that produces relative vibration against chassis frame, such as engine and transmission, reasonably slacken the harness and lay it along the factory-installed harnesses so as to absorb the relative vibration. Be sure that the additional harness does not contact with adjacent parts or equipments.
 - As for the factory-installed harnesses connected to the equipments that produce relative vibration against chassis frame, such as engine and transmission, do not change the way how they are secured, the clipping points, and the sagging volume.

Harness root of moving



- Heat protection
 - Harnesses must sufficiently be distant from heat source (e.g. Exhaust pipe). Be sure that the temperature of the harnesses does not exceed their allowable temperature limit (80° C in principle). There may be for an individual value provided according to the auto vehicle exhaust emission device kind . Refer to "SPECIFIC INFORMATION".
- Clip
 - Use plastic clips, or vinyl- or rubber-coated clips whose base material is rust-proofed.

A variety of clips



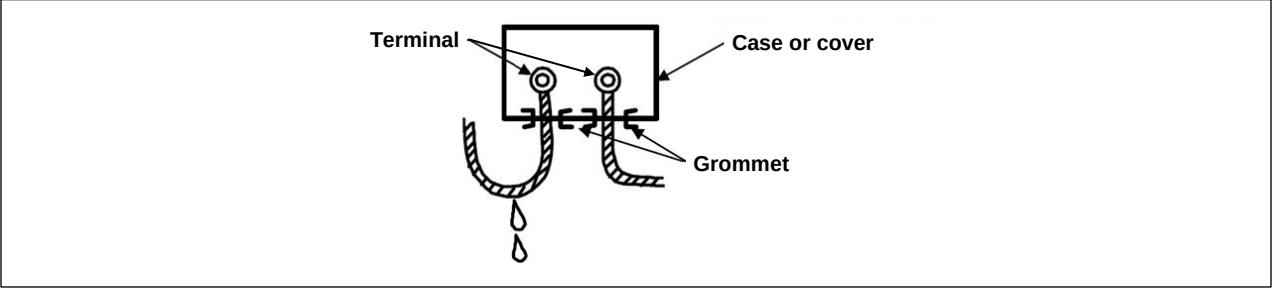
- Clip distance are as follows:

Distance of harness clip		Unit: mm
Harness diameter	Clip distance	
5 at max	300 at max	
5 to 10	400 at max	
10 at min	500 at max	

- If the wire vibrate due to engine or driving vibration, shorten the clipping intervals.

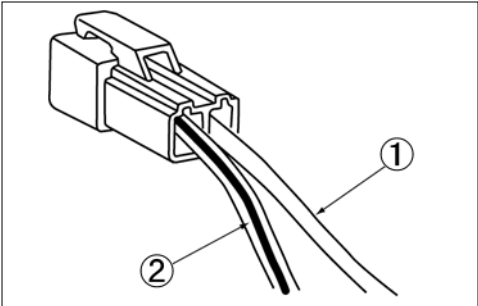
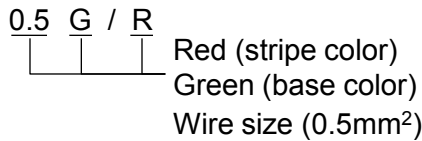
- Adhesive clips should be used only as supplemental parts. (Do not use at places with relative displacement or vibration)
- Especially, in such a place like the engine room where ambient temperature is high, use rubber- or vinyl-coated plate clips or band clips. Do not use non-heat-proof vinyl tape or adhesive clips to bind additional harnesses to the factory-installed harnesses because they will come unstuck due to heat aging.
- Seal with grommet so the water dripping along the wire does not enter the case, and route the harness so the terminals become higher than the harness entrance. Consider that water from tire splash or at cleaning does not enter.

Water proof of case



⑩ Wire color and size

Every wire is covered with insulation material with specific color code. The main wires used for systems are in solid color ① while the sub wires are with stripe ②.

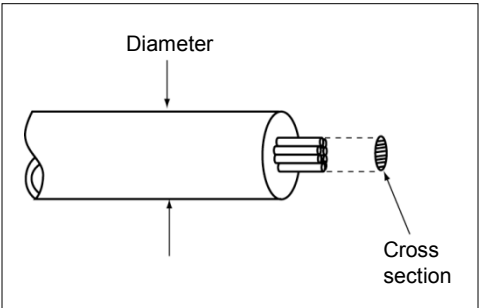


Color code of wire harness

Color code	Color	Color code	Color	Color code	Color
B	Black	L	Blue	P	Pink
W	White	O	Orange	LB	Light Blue
R	Red	BR	Brown	V	Violet
G	Green	LG	Light Green		
Y	Yellow	GR	Grey		

- Use the JIS-certified electric wires for automobiles, or equivalent
 - For common area: AV (Low-voltage wire for automobile) or AVS (Thin-walled low-voltage wire for automobile)
 - For high temperature area: AVX (Cross-linked vinyl insulated heat resistant low-voltage wire for automobile) or AEX (Cross-linked polyethylene insulated heat resistant low-voltage wire for automobile)

- The size of each wire to be used in a circuit is determined by the current amount (ampere) that will flow into the wire, the circuit length, the allowable voltage drop amount, etc.
- The following table shows sizes and allowable current of AVS, AV wires (Thin-walled low-voltage wire for automobile).



Wire size

Wire size (*)	Cross sectional area (mm ²)	External diameter (mm)	Allowable current (A)
0.3	0.372	1.8	9
0.5	0.563	2.0	12
0.85	0.885	2.2	16
1.25	1.287	2.5	21
2.0	2.091	2.9	28
3.0	3.296	3.6	37.5
5.0	5.227	4.4	53
8.0	7.952	5.5	67
15.0	13.36	7.0	75
20.0	20.61	8.2	97

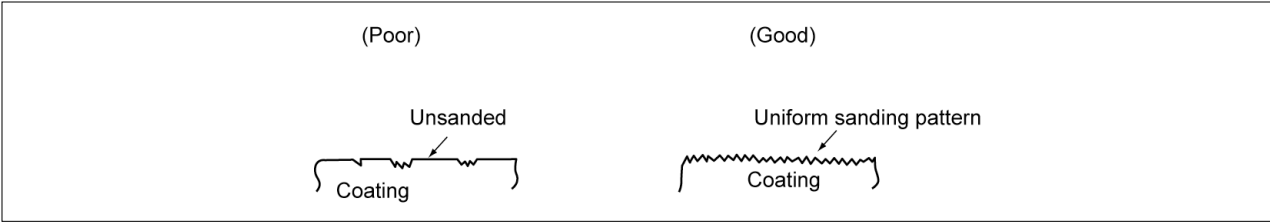
Note: * Rough sectional area

2-13 PRECAUTIONS FOR PAINTING

There are cab and the chassis component which marketability is dropped due to the heat from painting and drying, and likely to become the cause of a serious accident.

- ① Cab painting
 - Precautions before and after painting
 - When cleaning a cab, do not use thinner and gasoline but IPA (isopropyl alcohol).
 - To enhance coating adhesion, thoroughly sand undercoat surface. Be sure that the sanding pattern is uniform, and the sealant at the joints of steel plates must not be removed.

Good and poor sanded surface conditions



- ② Paint
 - Urethane paint is recommended for its high luster and distinctness-of-image gloss, and it allows low-temperature baking.
 - Contact ISUZU distributor or ISUZU dealership for cab color.

③ Components that should be removed before painting

- Remove the following components before painting to secure painting quality. Be sure to re-attach them after painting.
- For the components installed with the bolt with the adhesive such as assist grip, do not reuse the bolts because it causes slacking.
 - Assist grip on front panel
 - Mirrors (except for Stay)
 - Air intake duct
 - Fender step covers
 - Fender rubber
 - Radiator grill
 - Side panel covers
 - Wiper
 - Nozzle
 - Turn signal lamp (side)
- When re-attaching components after completion of painting, follow the instructions below.
 - Re-attach the components after the paint completely hardens.
 - For the scratches that can be made during unfastening and fastening screws, bolts, or nuts, touch them up with a brush soaked in clear lacquer, etc.
 - When re-attaching removed mirror, be sure to tighten the bolt with appropriate torque.

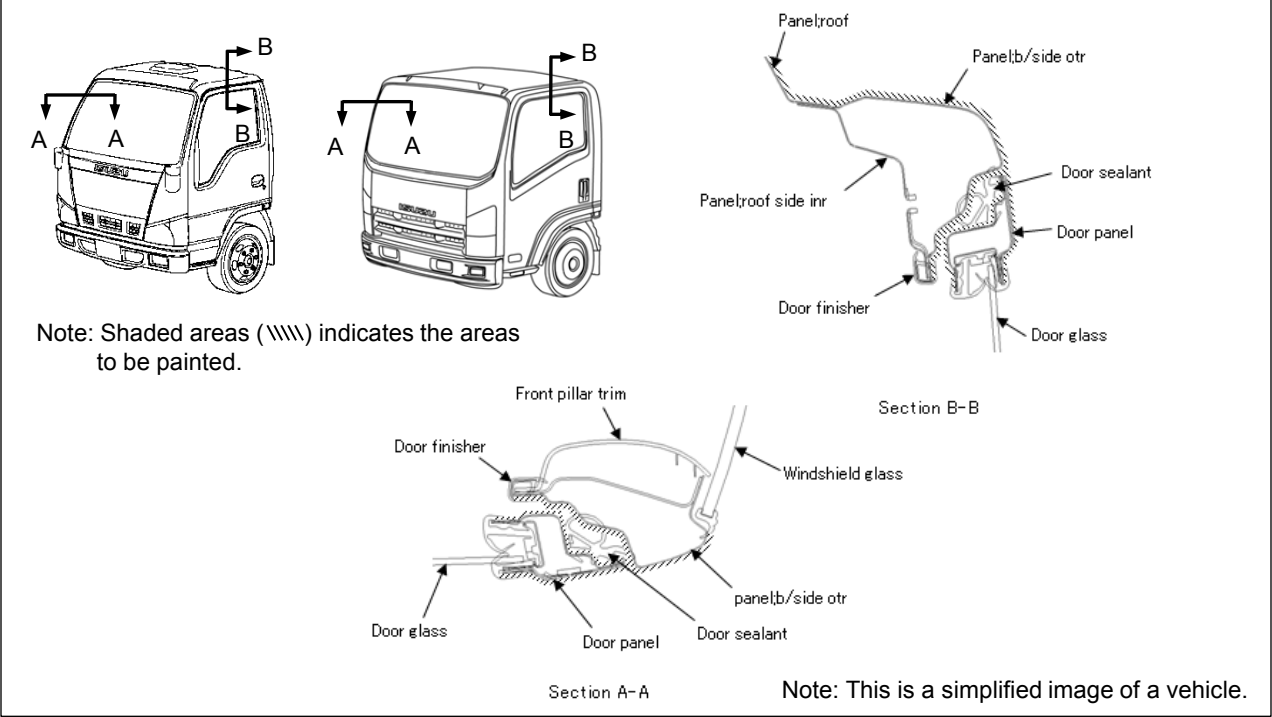
④ Rust prevention with Ziebart's

- The United States' Ziebart's rust-proofing agent is globally popular in protection from road snow melting agent, salty wind, etc. Apply this Ziebart's rust-proofing agent on cab under floor area, inner sides of fenders, outer side of a front panel, lower parts of doors, and joints of panels. Follow the instructions below.
 - Before applying the agent, cover the drain holes in the doors and in other areas with tape in order to prevent them from being plugged up with the agent. Also cover the air bleeding holes in air valves, etc. with tape to protect them from blockage.
 - After completion of the agent application, remove the tapes and check the drain holes for blockage. Since a cab is made from thin steel plates, if the drain holes are plugged up by something such as rust-proofing agent, water or salty water may consequently rust the panels, in the worst case making holes in them or weakening the strength of the members that are assuring the strength of the cab.

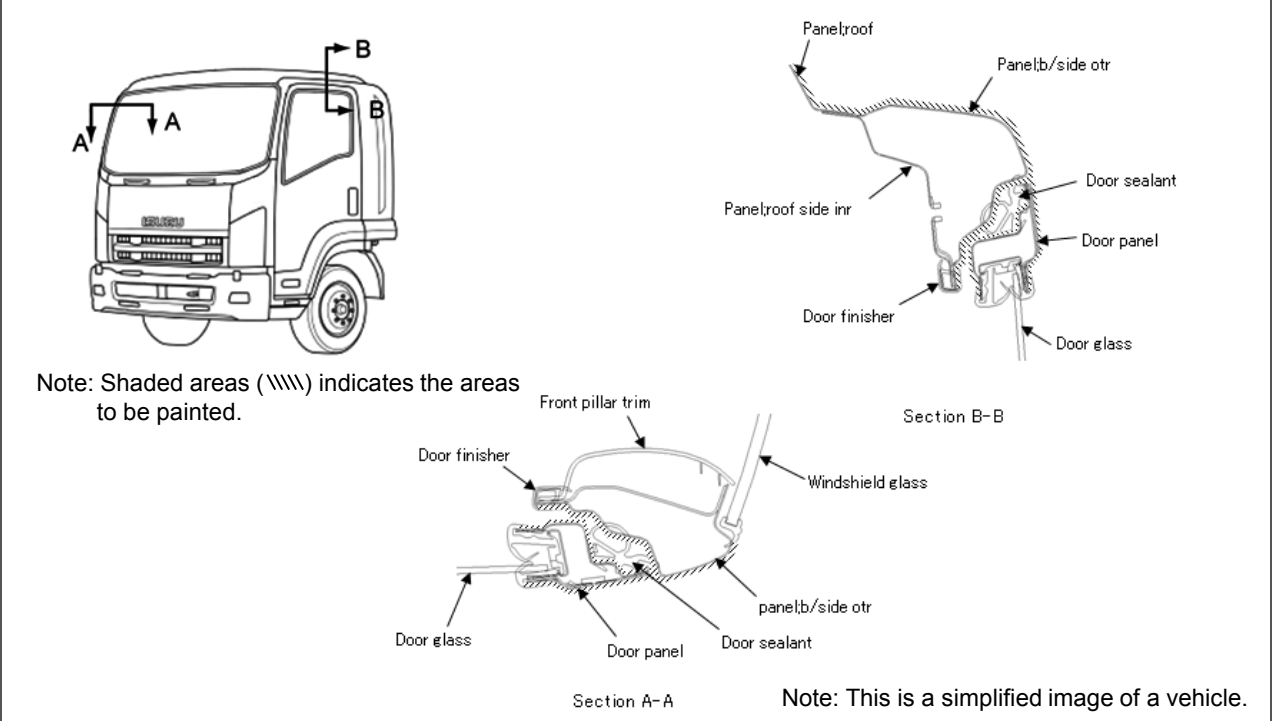
⑤ Paint boundary

- The range shown in the figure below is recommended as a range of the painting boundary.
- Shaded areas indicate the areas to be painted.

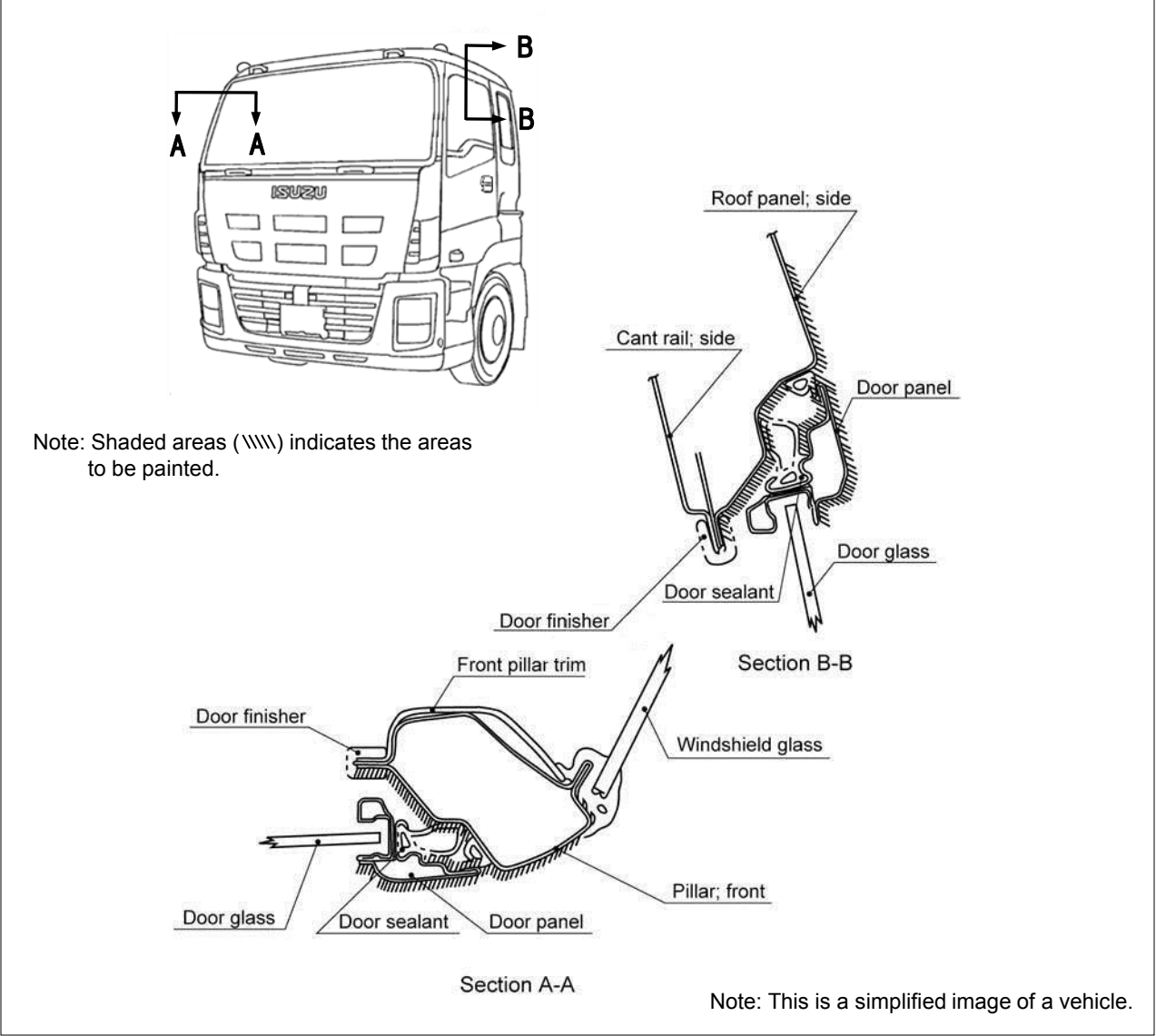
Paint boundary (example of light duty vehicle)



Paint boundary (example of medium duty vehicle)



Paint boundary (example of Heavy duty vehicle)

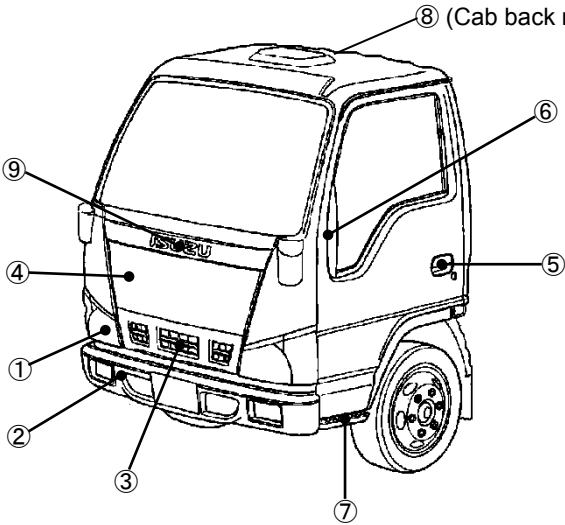


⑥ Paint drying

- When conducting high temperature forced drying, be sure that the atmospheric temperature of painted surface does not exceed 80°C.
 - Drying temperature over 80°C may deform rubber and plastic parts, and result in separation of adhesive-bonded joints.
- Remove bed mat.
- If the air cylinder for cab suspension is equipped, protect them from high temperature with heat insulation sheet, blanket, etc.
- If the drying temperature inevitably exceeds 80°C, follow the instructions below.
 - Remove components or protect them from high temperature with heat insulation sheet, blanket, etc.
 - Remove all the control units.

- Allowable temperature limit of the main plastic components

Light duty vehicle (NKR/QKR series) cab exterior



⑧ (Cab back right side)

⑨

④

①

②

③

⑦

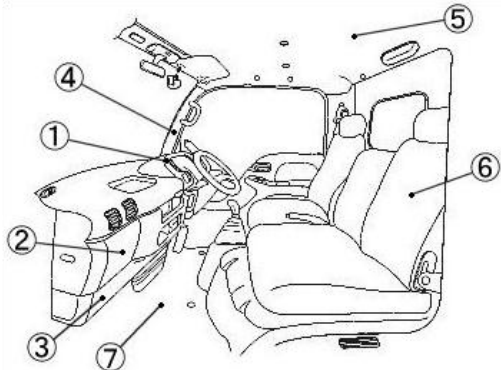
⑥

⑤

No.	Component name	Material	Allowable temperature limit (°C)
①	Head lamp	PC	80
②	Front bumper	PP	
③	Radiator grill	ASA/ABS	
④	Front ornament	ABS	
⑤	Door handle	PP-GF25	90
⑥	Door garnish	PP	
⑦	Support step	PP-GF40	80
⑧	Air duct	PP	90
⑨	Emblem	ABS	80

Note: This is a simplified image of a vehicle (cab).

Light duty vehicle (NKR/QKR series) cab interior



⑤

④

①

②

③

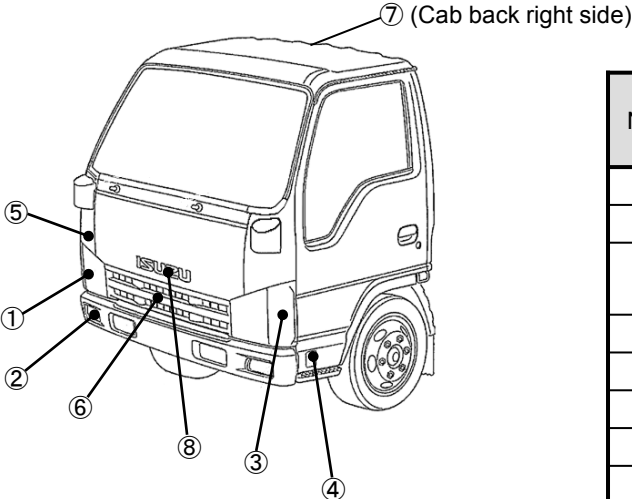
⑦

⑥

No.	Component name	Material	Allowable temperature limit (°C)	
①	Meter cluster	PPF	80	
②	Glove box			
③	Instrument panel cover			
④	Trim cover	PP		
⑤	Headlining	TPE/PEF/PP	80	
⑥	Seat cloth	PVC		
⑦	Floor mat			

Note: This is a simplified image of a cab interior.
Above Fig. shows right handle vehicle.
Left handle is mirrored image.

Light duty vehicle (standard cab) cab exterior



⑦ (Cab back right side)

⑤

①

②

⑥

⑧

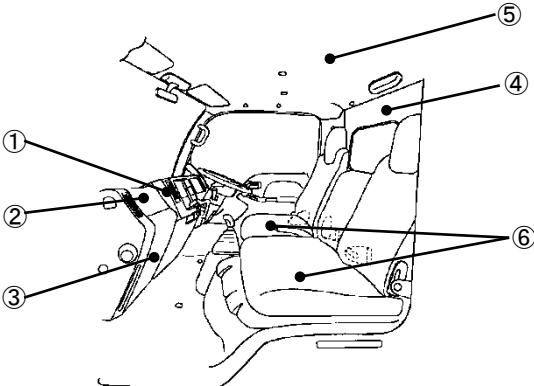
③

④

No.	Component name	Material	Allowable temperature limit (°C)	
①	Head lamp	PC	80	
②	Fog lamp	PP (housing)		
③	Front combination lamp	PMMA		
④	Side reflector			
⑤	Corner panel	ABS		
⑥	Grill	ASA	90	
⑦	Air duct	PP		
⑧	Emblem	ABS		

Note: This is a simplified image of a vehicle (cab).

Light duty vehicle (standard cab) cab interior



⑤

④

①

②

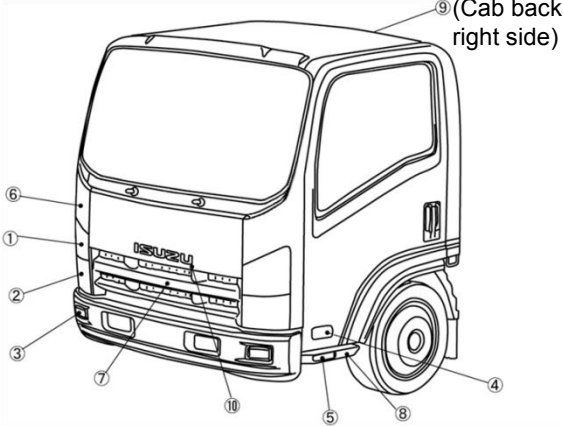
③

⑥

No.	Component name	Material	Allowable temperature limit (°C)
①	Meter cluster	PPF	80
②	Tray		
③	Instrument panel cover		
④	Trim cover	PP or PPF	90
⑤	Headlining	PET	
⑥	Seat assembly	PP or Other	80

Note: This is a simplified image of a cab interior.
Above Fig. shows right handle vehicle.
Left handle is mirrored image.

Light duty vehicle (high cab) cab exterior

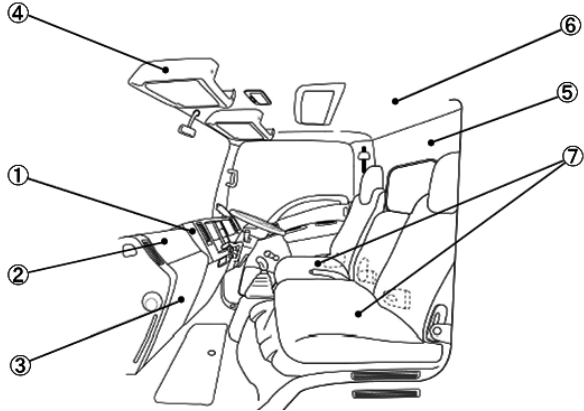


⑨ (Cab back right side)

No.	Component name	Material	Allowable temperature limit (°C)
①	Turn signal lamp (front)	PMMA	80
②	Head lamp	PC	90
③	Fog lamp	PP (housing)	
④	Turn signal lamp (side)	PC	80
⑤	Flank reflector	PMMA	
⑥	Corner panel	ABS	
⑦	Grill	ASA	90
⑧	Step closing	PP	
⑨	Air duct	80	
⑩	Emblem		ABS

Note: This is a simplified image of a vehicle (cab).

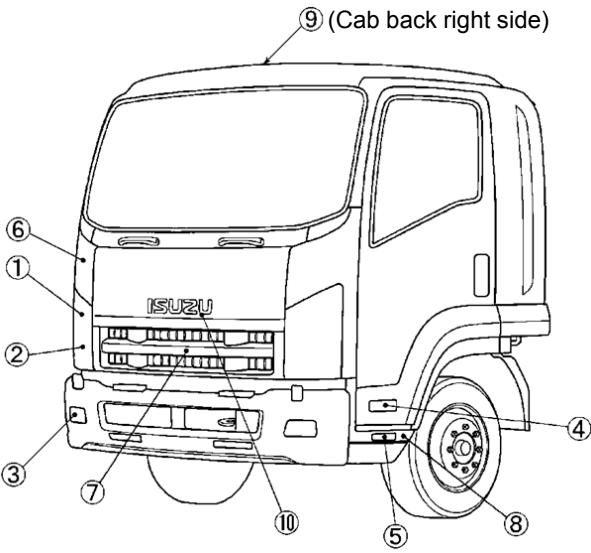
Light duty vehicle (high cab) cab interior



No.	Component name	Material	Allowable temperature limit (°C)	
①	Meter cluster	PPF	80	
②	Tray			
③	Instrument panel cover			
④	Header shelf	PP or PPF		
⑤	Trim cover			
⑥	Headlining	PET-PP or PET-PUR		
⑦	Seat assembly	PP or Other		

Note: This is a simplified image of a cab interior.
Above Fig. shows right handle vehicle.
Left handle is mirrored image.

Medium duty vehicle cab exterior

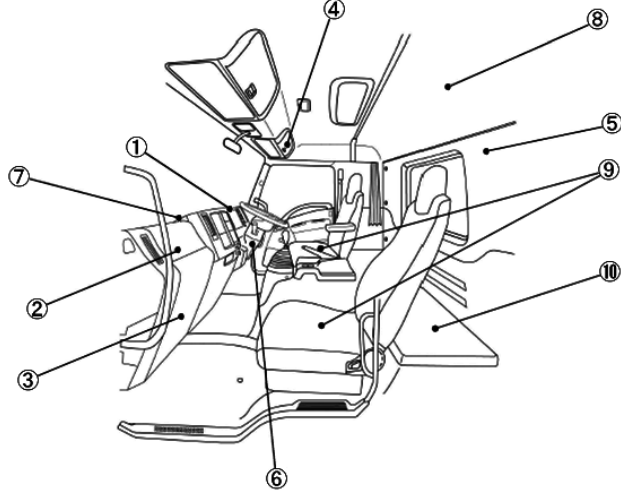


⑨ (Cab back right side)

Note: This is a simplified image of a vehicle (cab).

No.	Component name	Material	Allowable temperature limit (°C)	
①	Turn signal lamp (front)	PMMA or PC	80	
②	Head lamp	PC	90	
③	Fog lamp	PP		
④	Turn signal lamp (side) or flank signal lamp	PC	80	
⑤	Flank reflector	PMMA		
⑥	Corner panel	ABS		
⑦	Grill	ASA	90	
⑧	Step closing	PP		
⑨	Air duct			
⑩	Emblem	ABS	80	

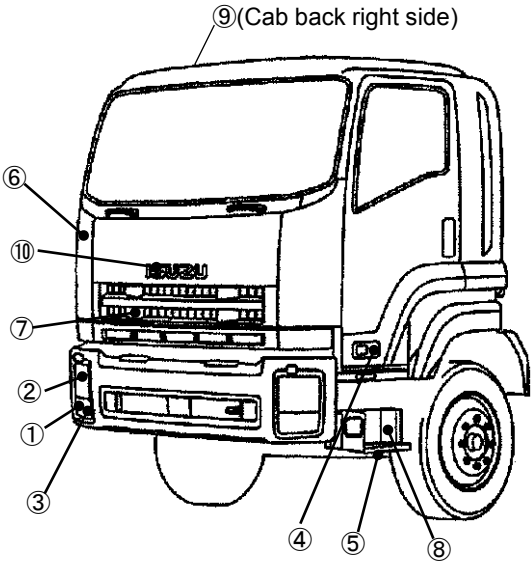
Medium duty vehicle cab interior



Note: This is a simplified image of a cab interior.
Above Fig. shows right handle vehicle.
Left handle is mirrored image.

No.	Component name	Material	Allowable temperature limit (°C)	
①	Meter cluster	PPF	80	
②	Tray			
③	Instrument panel cover			
④	Header shelf	PP or PPF		
⑤	Trim cover			
⑥	Steering cowl	PP-PVC		
⑦	Instrument panel pad	PPF		
⑧	Headlining	PET-PUR	70	
⑨	Seat assembly	PP or Other		
⑩	Bed mattress	PP or Other		

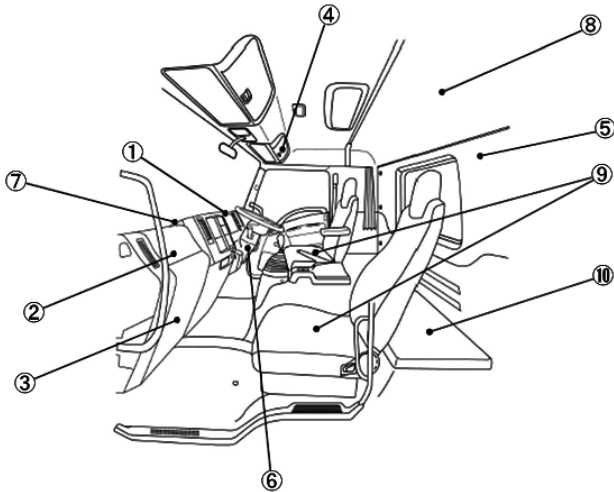
Heavy duty vehicle (FX/GX/FY series) cab exterior



No.	Component name	Material	Allowable temperature limit (°C)
①	Turn signal lamp (front)	PMMA or PC	80
②	Head lamp	PC	90
③	Fog lamp	PP	
④	Turn signal lamp (side) or flank signal lamp	PC	80
⑤	Flank reflector	PMMA	
⑥	Corner panel	ABS	
⑦	Grill	ASA	90
⑧	Step closing	PP	
⑨	Air duct		
⑩	Emblem	ABS	80

Note: This is a simplified image of a vehicle (cab).

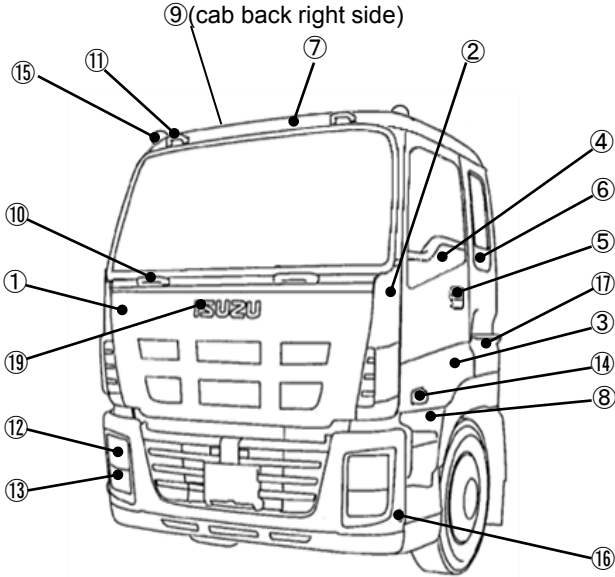
Heavy duty vehicle (FX/GX/FY series) cab interior



No.	Component name	Material	Allowable temperature limit (°C)
①	Meter cluster	PPF	80
②	Tray		
③	Instrument panel cover		
④	Header shelf	PP or PPF	
⑤	Trim cover		
⑥	Steering cowl	PP-PVC	
⑦	Instrument panel pad	PPF	
⑧	Headlining	PET-PUR	
⑨	Seat assembly	PP or Other	
⑩	Bed mattress	PP or Other	70

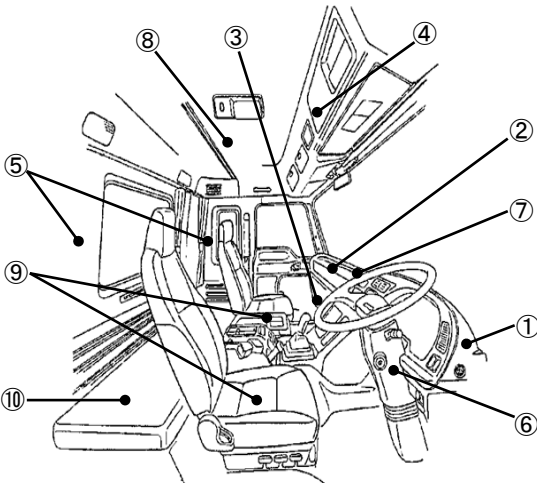
Note: This is a simplified image of a cab interior.
Above Fig. shows right handle vehicle.
Left handle is mirrored image.

Heavy duty vehicle (C&E series) cab exterior

	No.	Component name	Material	Allowable temperature limit (°C)
	①	Front lid	SMC	100
	②	Front corner panel	ABS	80
	③	Blister fender		
	④	Door ornament	ASA	90
	⑤	Door outer handle base plate		
	⑥	Rear quarter bent grill		80
	⑦	Roof ornament	ASA + PC	90
	⑧	Step closing	PP	80
	⑨	Air duct		
	⑩	Front grip and hinge		
	⑪	Roof assist grip	PC	90
	⑫	Head lamp		
	⑬	Turn signal lamp (front)		
	⑭	Turn signal lamp (side)	PMMA	80
	⑮	Roof marker lamp		
	⑯	Side reflector	ABS	
	⑰	Fender	PP	
	⑱	Emblem	ABS	

Note: This is a simplified image of a vehicle (cab).

Heavy duty vehicle (C&E series) cab interior

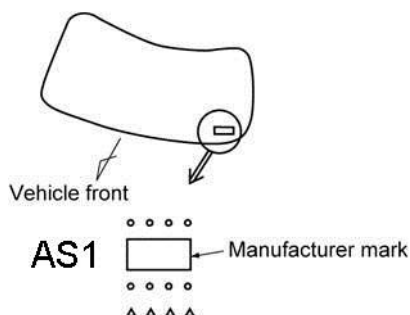
	No.	Component name	Material	Allowable temperature limit (°C)
	①	Meter cluster	PPF	80
	②	Fuse lid	PPF	
	③	Instrument panel cover	PPF	
	④	Header shelf	PP or PPF	
	⑤	Trim cover	PP or PPF	
	⑥	Steering cowl	PP	
	⑦	Instrument panel pad	JEC270C/PU R/PVC+ABS or PPF	
	⑧	Headlining	PET	
	⑨	Seat assembly	PP or Other	
	⑩	Bed mattress	PP or Other	70

Note: This is a simplified image of a cab interior.
Above Fig. shows right handle vehicle.
Left handle is mirrored image.

⑦ Laminated glass

- A laminated glass consists of two thermal-bonded glasses between which a polyvinyl butyral film is sandwiched. In hot and humid conditions during paint drying, follow the instructions below. Failure to follow these instructions may result in problems such as peeling of the polyvinyl butyral film and bubbles between the glasses.
- When conducting forced infrared drying, work limit within 30 minutes at temperature no higher than 120° C.
- If the temperature during such forced drying inevitably exceeds 120° C, before starting the forced drying, cover the glass to prevent the temperature of the glass surface from exceeding 120° C, or remove it.
- Laminated glasses can be identified by “LP”, “LS”, “L”, “AS1” or “LAMINATE” printed at the lower left corner. Partially tempered glasses have a “Z” print.

Location of the print on a laminated glass

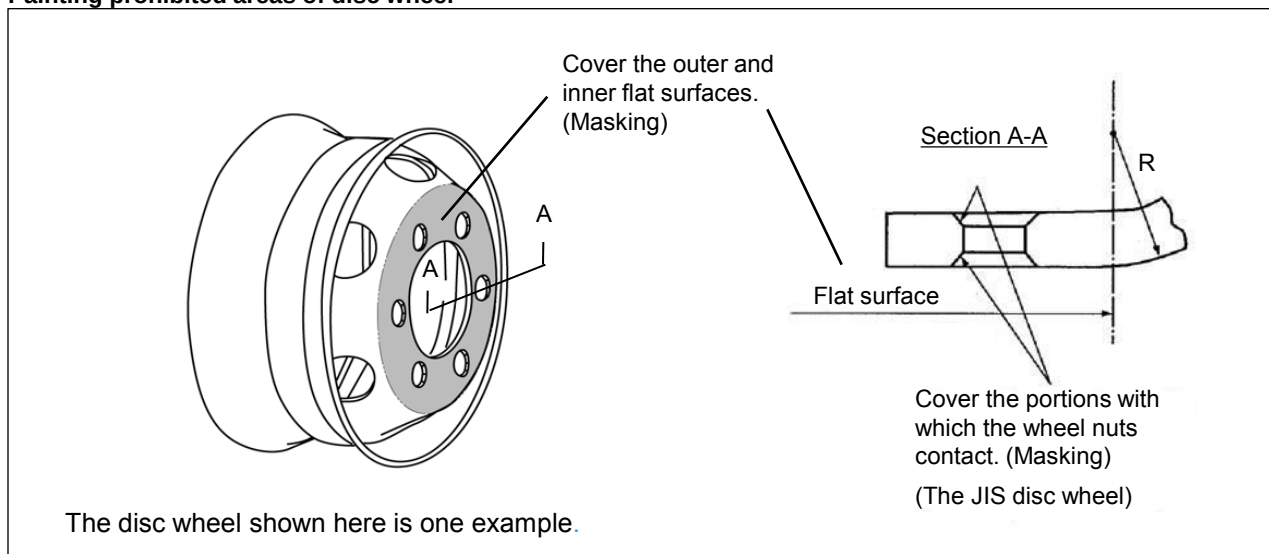


⑧ Painting-prohibited components or equipments

Following describes the common items that doesn't depend on the vehicle type and destination. There may be other prohibited components or equipments other than the following depending on the destination or specification, confirm "SPECIFIC INFORMATION" of the targeted vehicle type.

- Disc wheel
 - Do not apply additional painting on the contact faces of a mating disc wheel and a hub, and the portions with which wheel nuts contact because thickening the painting possibly slack or damage wheel nuts.

Painting prohibited areas of disc wheel



- Components related to braking system such as brake hose and brake booster
- Rubber hose, vinyl hose or connectors
- Wire harness and connectors
- Components and equipments related to suspension, steering, such as these made of rubber or plastic. Especially painting on the air bellows is strictly prohibited.
- Cab power tilt related components
- Caution plates and number plates
- Door sealing
- Rubber and plastic components around window
- Sliding part of check arm
- Inside the door lock (Be careful not to have painting in mechanism part.)
- Wiper blade and washer nozzle
- Reserve tanks for coolant and washer tank
- Regenerator such as condenser, radiator, and inter cooler
- Condenser fan
- Movable components such as compressor clutch
- Lattice part in radiator grill (black part)

2-14 EMBLEMS

① Precautions before affixing emblems

- If there is any extraneous substance on a bonding surface(cab, panel) such as water, oil and dust, emblems will not firmly stick to the surface. If there is any, wipe them off completely with a clean cloth soaked in IPA (isopropyl alcohol). Remove wax if the wax has adhered.
- The ideal ambient temperature for adhesion is 30°C to 50°C. Make the temperature of the bonding surface fall within this temperature range as much as possible.
- Follow the below instructions when using an organic solvent.
 - Work in an airy place.
 - Fire strictly prohibited.
 - Be careful do not get into eyes.
 - Stop working immediately when the physical condition worsens.
 - Follow instructions of the organic solvent used in detail.

② Precautions at affixing emblems

- As for affixing emblem, set the alignment that came with a chassis on the front lid of a cab, and then stick each letter sticker by using cut-outs of the alignment as a guide.
- The sticky side of emblem requires pressurization for bonding. To firmly stick the stickers, apply sufficient and uniform pressure on the emblem. Use of a roller is recommended as they provide uniform pressurization. The pressure of the roller should be no less than 5kg.
- Exposing the sticky side of emblem to the air for long hours weakens their adhesivity due to adhesion of dust, etc. To protect the sticky side, do not remove a release-paper until immediately before use.
- Since this sticky-side has very strong adhesive property, once emblems are affixed, peeling them off is not easy. Before sticking them, carefully check sticking position.

Location of emblem for various cab is described in “I -2-15 DECALS”, together with location of decal.

2-15 DECALS

The base material of a decal is thin, soft and stretchable vinyl chloride. Therefore, strictly follow the procedure to firmly affix decals without making wrinkles, bubbles or stretching.

① Precautions before affixing decals

- Confirm the material of bonding surface. There is possibility that the decals may not stick at all or may come off shortly after being affixed on following materials.
 - Low polarity plastic such as polyethylene, polypropylene, and teflon
 - Rough surfaces such as cloth, whetstone, sandpapers, and plastic emboss finished goods
 - When a mold lubricant, a plasticizer, and other additives to the surface of the plastic molding.
 - Hygroscopicity plastic such as ABS, noryl, and urethanes
- Remove wax if the wax has adhered.
- If there is any extraneous substance on a bonding surface (cab, panel) such as water, oil and dust, decals will not firmly stick to the surface. If there is any, wipe them off completely with a clean cloth soaked in IPA (isopropyl alcohol). If the substance cannot be removed, completely wipe it off with lacquer thinner, etc., and then wipe the surface again with IPA.

② Precautions at affixing decals

- The ambient temperature should be 10°C or more
- Begin working after leaving in atmosphere above 10°C for one hour or more, when the decal or the bonding body was kept at 10°C or less., etc., and then wipe the surface again with IPA.

③ How to affix a decal

	Dry type (Use paper carrier tape)	Wet type (Use neutral detergent solution)
Decal size	Small area 300 X 300mm or less Narrow width 50mm or less	Large area 300 X 300mm or more Wide width 50mm or more
Prepared apparatus and material	squeegee (spatula) cutter knife waste cloth needle IPA (iso-propyl alcohol) ※1 dryer for heating tape ※2	squeegee (spatula) cutter knife waste cloth needle IPA (iso-propyl alcohol) ※1 dryer for heating tape ※2 neutral detergent solution ※3 spray bottle

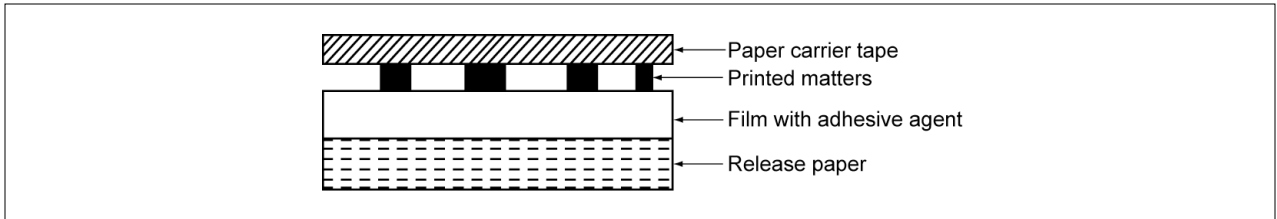
Note: * 1 When it is not possible to wipe off with IPA (iso-propyl alcohol) etc. used for the cleaning, it is necessary to prepare lacquer thinner.

* 2 Tape is used for alignment. Prepare NITTO paper adhesive tape No.720 or equivalent.

* 3 Use home liquid neutral detergent for the neutral detergent solution .

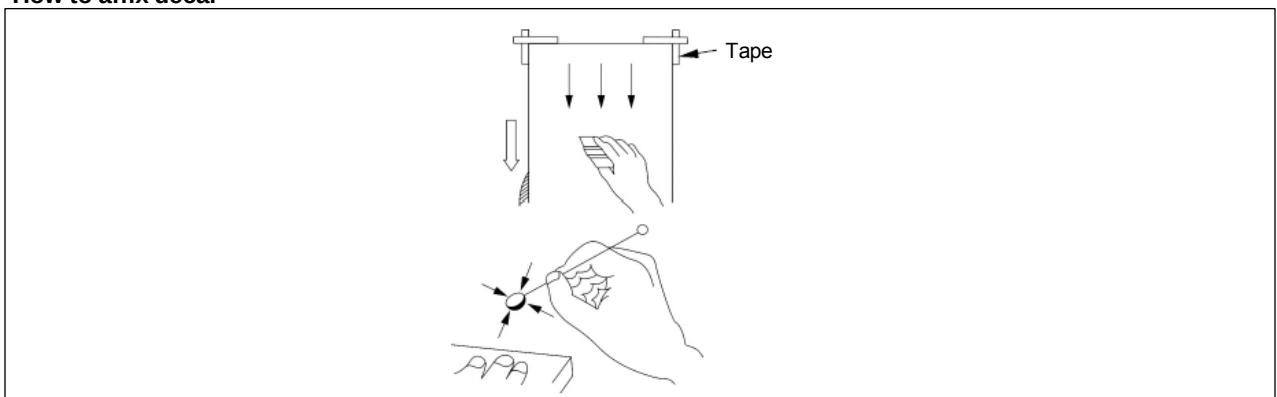
- Dry type (For sheet decals such as signature, mark, and stickers)
 - Prepare decal with paper carrier tape pasted to the printing surface.

Sectional view of sheet decal



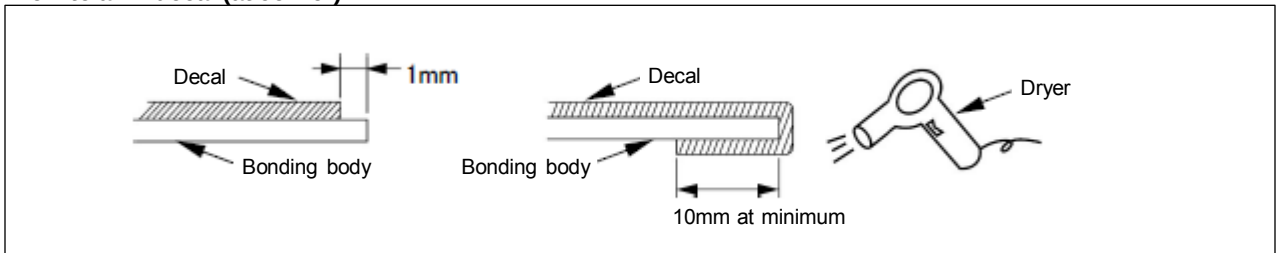
- Without removing a release paper, temporarily place the decal in position, and mark the position with tape.
- Peel off the release paper from the edge approximately 20mm, and then fold down the paper 180°.
- Align to the positioning tape, press and paste up the 20mm area with squeegee first. Temporarily fix the other edges with tape so that the decal will not move.
- Slowly peel off the release paper, and press the decal carefully.
- Basically, move the squeegee in same direction, downward or right to left.
- When you are sure with no meandering or mis-positioning, remove the positioning tapes.
- If there are bubbles trapped inside the decal, pop it with needle and press them out to flatten.
- Press the entire surface hard for the last time, and then carefully remove the paper carrier tape in 180° angle.

How to affix decal



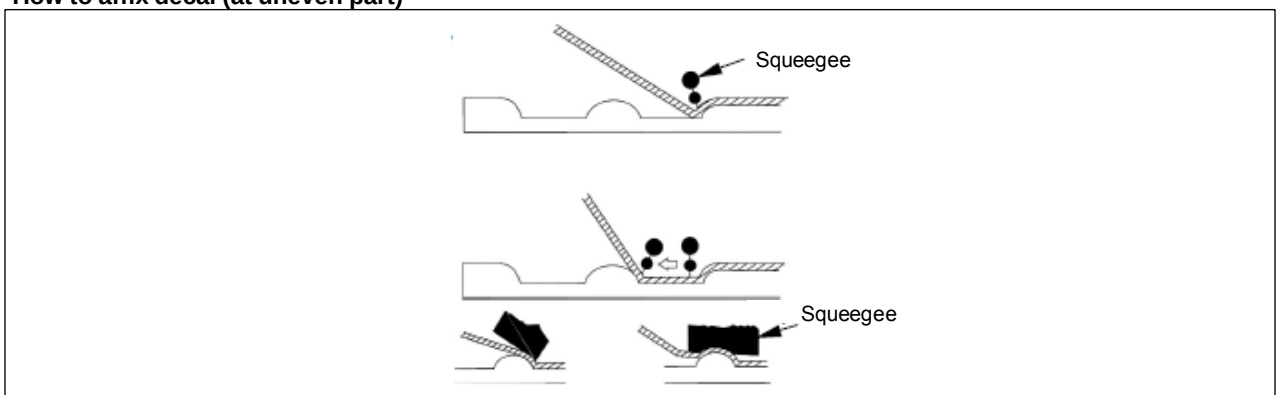
- Dry type (decals with cut characters or stripes)
 - Same way with the sheet decal.
By putting half pulling out slit in the released paper beforehand, it becomes easier to temporary fix the decal. Moreover, note that loosening is not caused in the decal when it puts on pressure by the squeegee enough when it is long. Press and stick the loosen decal, wrinkle and the bubble may occur.
- Dry type (fitting on corner area)
 - Cut or roll it inside. Make it 1mm shorter from the corner when cutting, or bend it 10mm or more when rolling it in, and press it heating with dryer.

How to affix decal (at corner)



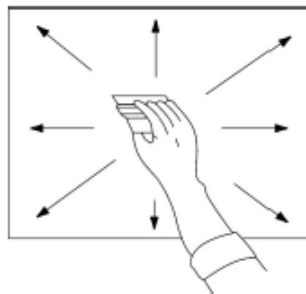
- Dry type (fitting on uneven part)
 - Press-fit concave part by using the edge of the squeegee. Please move right and left and work about the squeegee. Note that the decal does not loosen.
 - Press-fit toward next concave part by the squeegee. Note that the decal does not stretch.
 - Press-fit the convex part by using the squeegee as shown in figure. Press enough so that the squeegee deforms.
 - Fitting on a globular uneven part is impossible.

How to affix decal (at uneven part)



- Wet type (fitting on other than corner area)
 - Stretch prevention and the alignment of the decal become easy by using neutral detergent solution, the paper carrier tape is not needed. Fitting on a uneven part is impossible with wet type.
 - Put neutral detergent solution in spray bottle.
 - Without removing a release paper, temporarily place the decal in position, and fix the two corners of the decal with adhesive tape.
 - Spray neutral detergent solution evenly on adherence face and adhesive coated surface of decal after removing released paper.
 - Place the entire decal on the adherence body at the same time. Remove tape after confirming position.
 - To improve sliding of the squeegee, and to prevent the friction scar, spray the neutral detergent solution on the decal surface evenly.
 - Press-fit while pushing the spray liquid out from the center toward the edge by squeegee.
 - When the convex part is found after pushing out with the squeegee, the spray liquid remaining in the part. Make hole with a needle and push out the spray liquid.

How to affix decal (wet type)



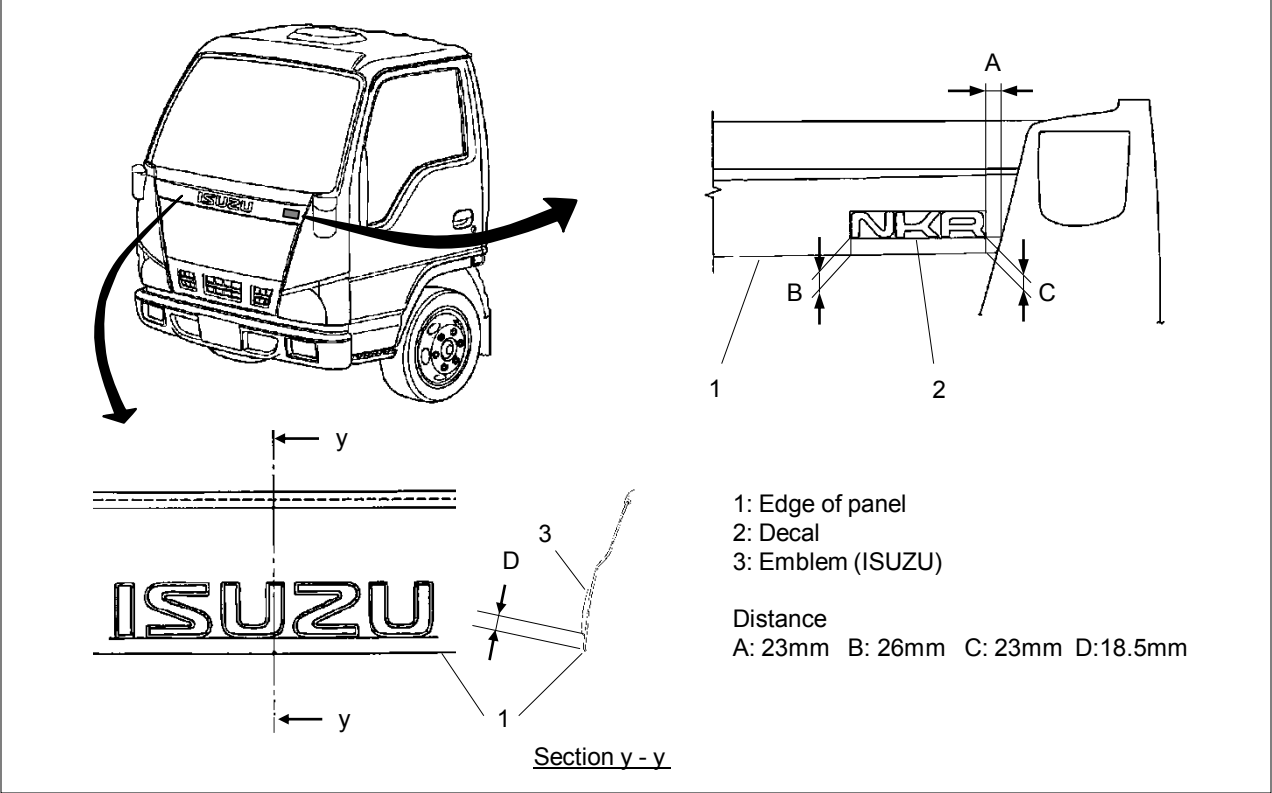
- Wet type (fitting on corner area)
 - Same way with the dry type.

④ Precautions for peeling off a decal

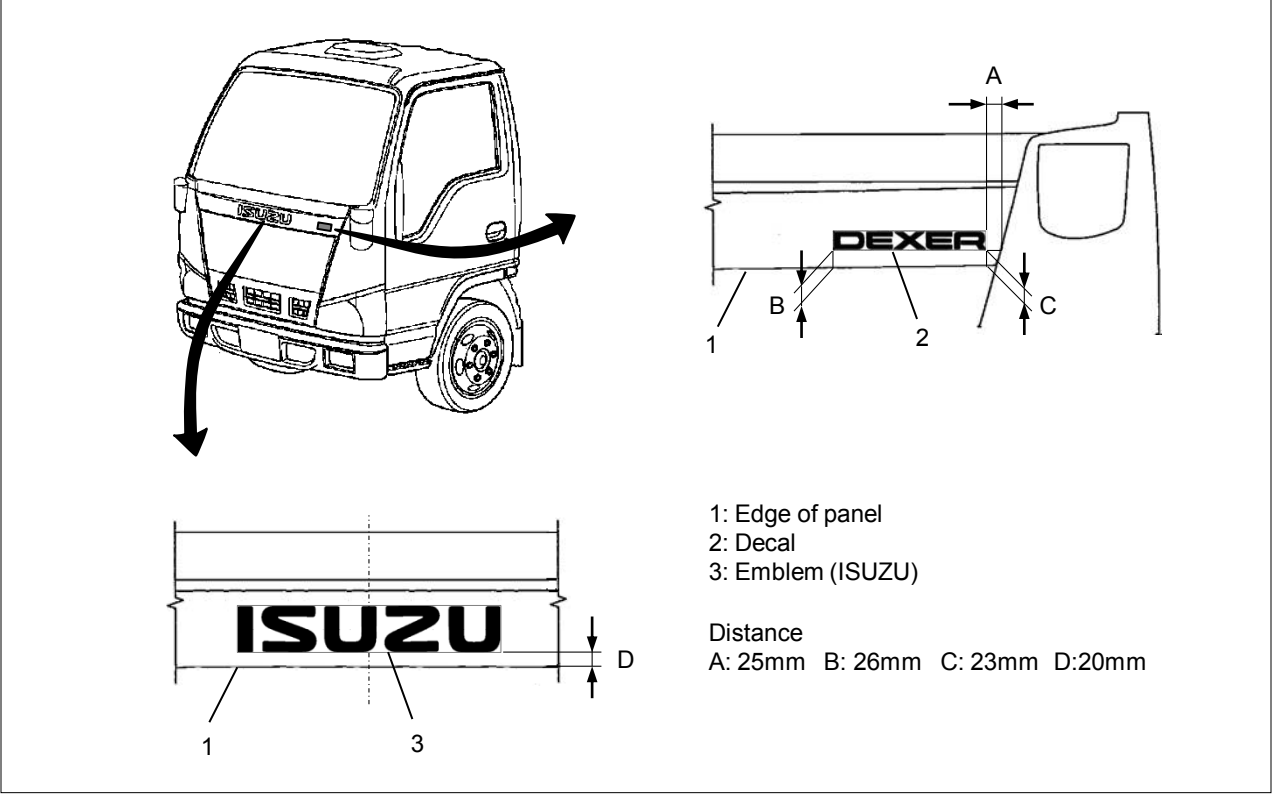
- How to peel off a decal (affixed less than one week)
 - Lift the edge of the decal with a spatula and pull it gradually.
 - After peeling the edge about 2cm, pull the decal slowly and gently.
 - Wipe with toluene to remove remaining adhesive agent on a bonding surface.
 - Wipe again with isopropyl alcohol after it dry, and wipe with waste cloth at the last.
- How to peel off a decal (affixed more than one week)
 - The degree of adhesion rises as for the decal that passes one week or more after affixing, peeling off is considerably difficult. Proceed work carefully.
 - Coat the decal with toluene to expand the decal.
 - Shave off the decal with a spatula.
 - Wipe with toluene to remove remaining adhesive agent on a bonding surface.
 - Wipe again with isopropyl alcohol after it dry, and wipe with waste cloth at the last.

Location of decal and emblem for various cab.

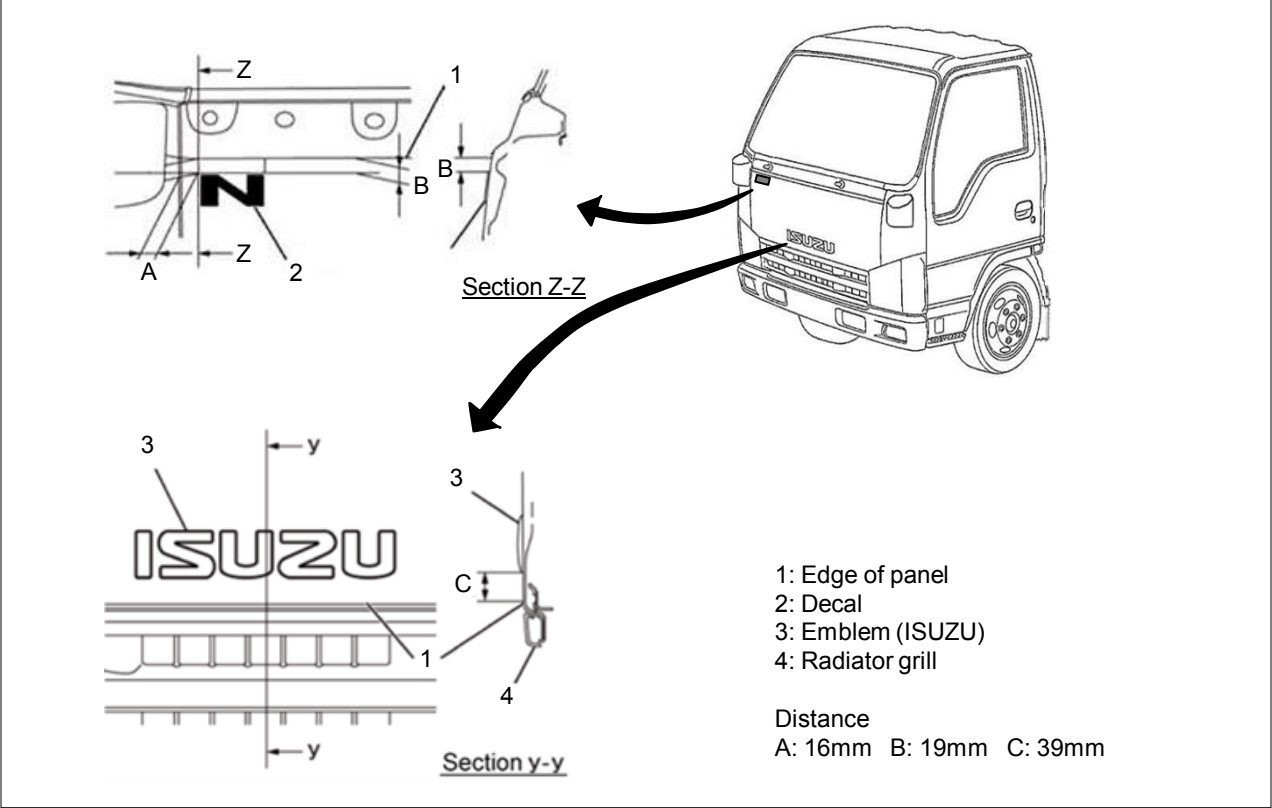
Light duty vehicle (NKR series)



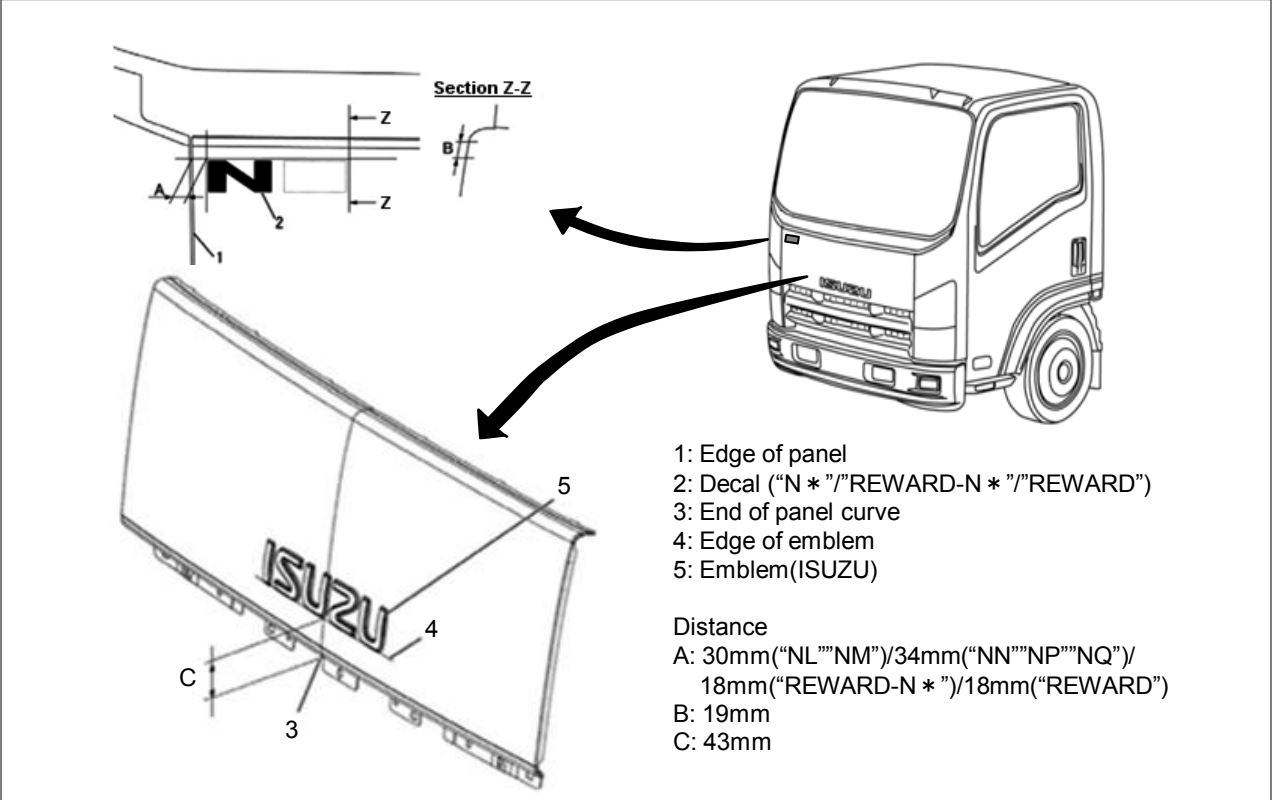
Light duty vehicle (QKR series)



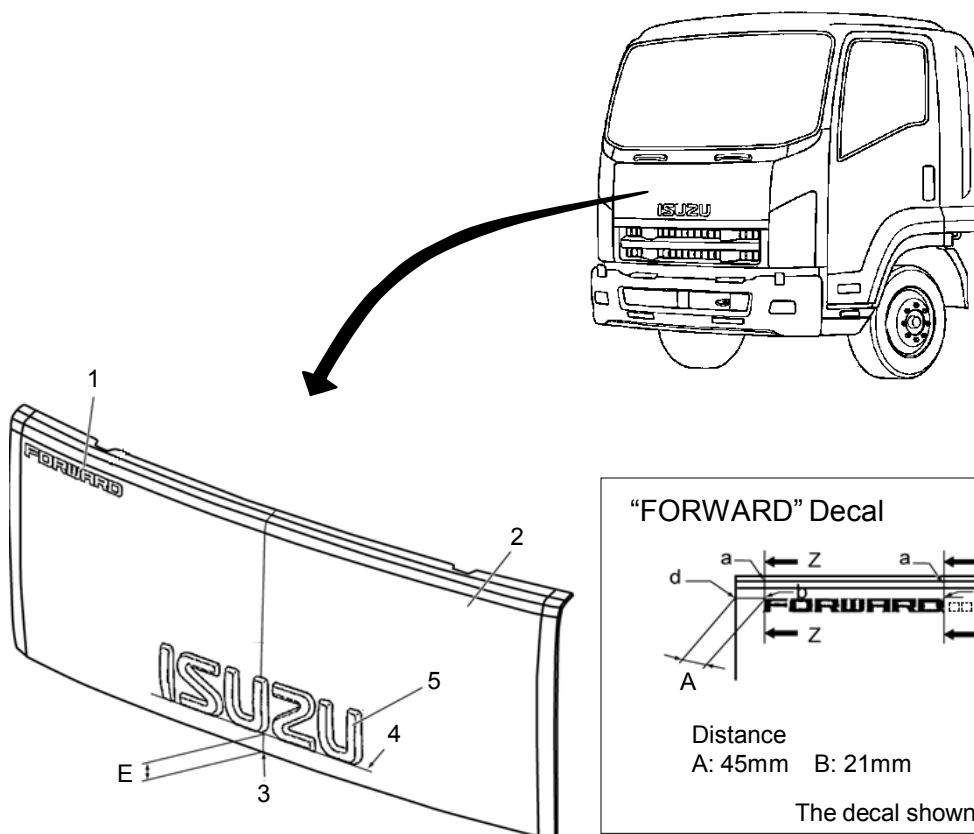
Light duty vehicle (standard cab)



Light duty vehicle (high cab)



Medium duty vehicle



1. "FORWARD" decal and "Model name" decal
 - a: End of panel curve
 - b: Upper left corner of the decal
 - c: Upper right corner of the decal
 - d: Edge of panel
2. Front lid
- 3: End of panel curve
- 4: Edge of emblem
- 5: Emblem (ISUZU)

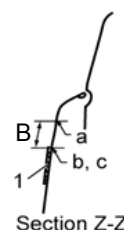
Distance
E: 40mm

"FORWARD" Decal

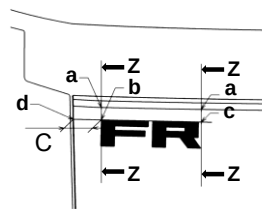


Distance
A: 45mm B: 21mm

The decal shown here is one example.

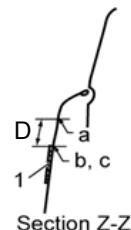


"Model name" decal

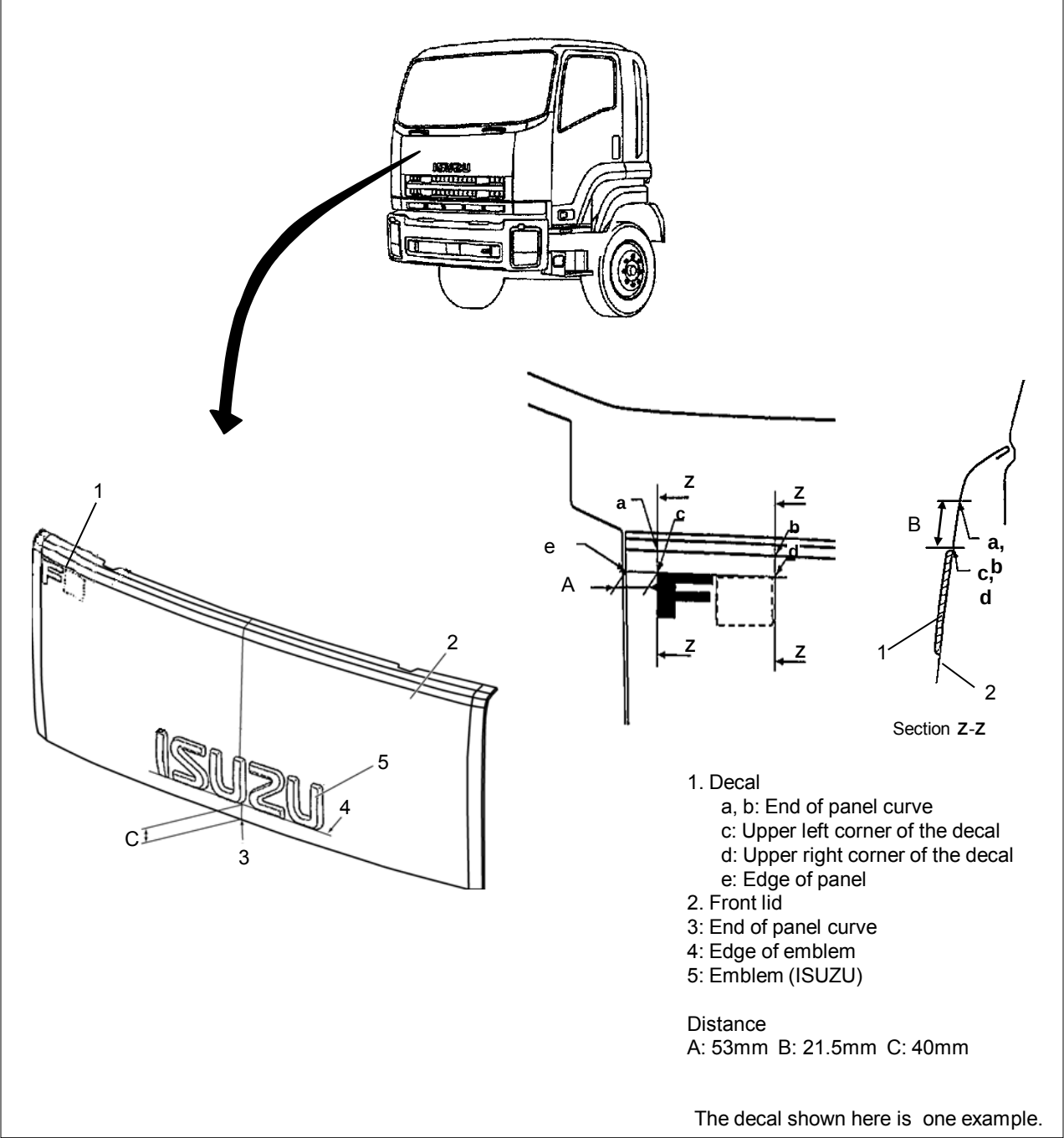


Distance
C: 46mm:STD cab / 53mm:Wide cab
D: 14mm:STD cab / 21.5mm:Wide cab

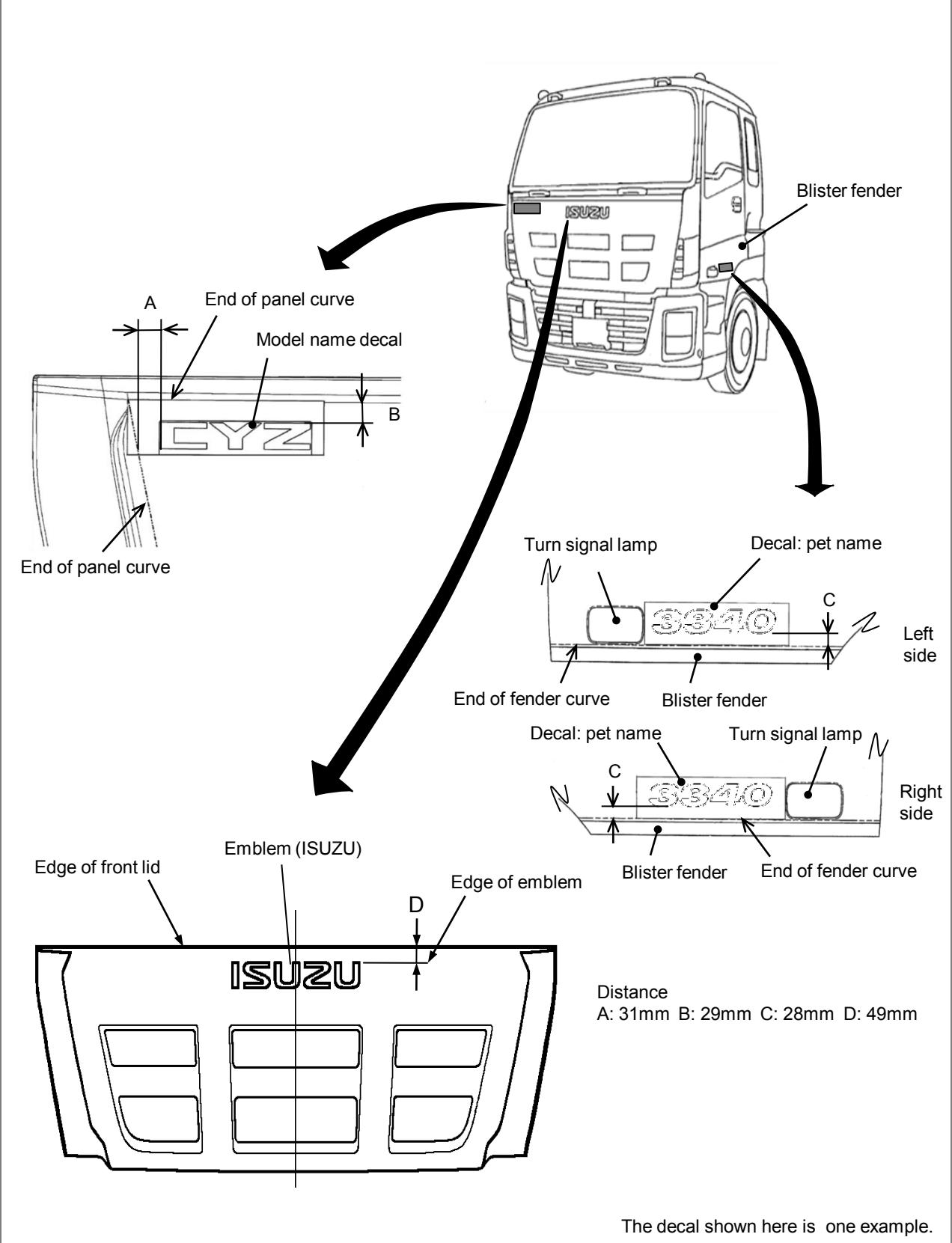
The decal shown here is one example.



Heavy duty vehicle (FX/GX/FY series)



Heavy duty vehicle (C&E series)



Chapter 3 REAR BODY MOUNTING

3-1 CONSIDERATION TO PROPER LOAD

3-2 CLEARANCE BETWEEN AN EQUIPMENT OR A COMPONENT ON A CAB CHASSIS AND A REAR BODY

3-3 SUB FRAME SIDE MEMBERS

3-4 FIXING OF A REAR BODY

3-1 CONSIDERATION TO PROPER LOAD

① Partial load and concentrated load

- When generation of partial load and concentric load in chassis frame is assumed due to structure of rear body or users loading, distribute the load by adding sub frame or main sill structure to prevent the transformation and the crack of chassis frame.
- When load is assumed to be concentrated partially such as at cab back crane, add reinforcements as needed to reduce stress.
- When installing a sub frame, allot the generated bend stress by calculating the combination beam. When installing reinforcement, do calculation same as the combination beam.

② Stress of chassis frame

- Follow the table below about the stress generated in chassis frame at time when the maximum payload was installed.

Unit: MPa {kgf/mm²}

	Light duty	Medium duty	Heavy duty
Smooth road use Flat bed, van, etc.	64 {6.5} at max	78 {8.0} at max	
Rough road use Tipper, mixer, garbage	44 {4.5} at max	64 {6.5} at max 44 {4.5} at max (Dump)	
Condition of frame tensile strength	440 {45} at min	540 {55} at min	

Note: The table is at 1G reaction force. A peculiar value to the frame tensile strength and the permissible stress value might be indicated according to the vehicle type and destination. Confirm the frame section coefficient diagram etc. of "SPECIFIC INFORMATION" of the vehicle type targeted without fail.

- Chassis frame is designed assuming uniformly distributed load. Frame stress may become maximum other than maximum load due to rear body feature, shipment, or rear body characteristics. Built suitable sub-frame or add reinforcement so the frame stress do not exceed limits referring to below examples.

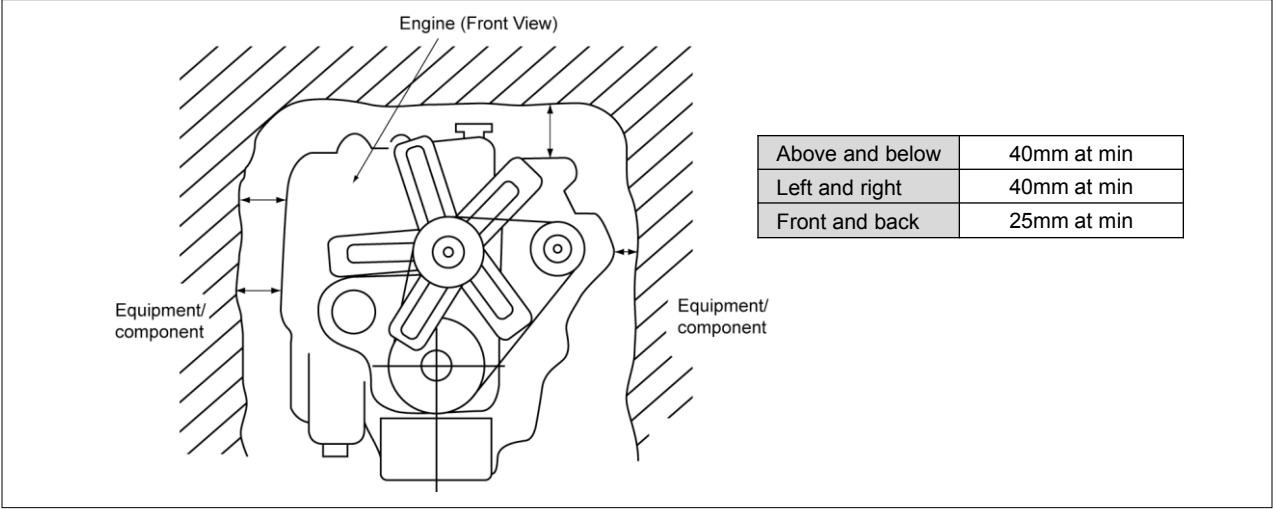
Rear body type	Load tendency
Construction equipment carrier	Load concentrate on one part depending to the construction machine mounted.
Lengthy cargo carrier	When the shipment exceed in front or rearward, the load concentrates on front axles or rear over hang.
Garbage compactor truck	Load concentrates on rear over hang for hopper weight. Further more, the load concentrates on tare meter load cell installation part.
Aerial work platform Earth auger	Load concentrates in front (cab back) part for bucket/boom weight.
Concrete mixer truck	Load concentrates on front and rear drum support.

3-2 CLEARANCE BETWEEN AN EQUIPMENT OR A COMPONENT ON A CAB
CHASSIS AND A REAR BODY

① Engine

- When installing any equipment or component around an engine, provide the following clearances from the engine.
- Do not place any object in front of the radiator or the inter cooler, otherwise cooling efficiency of the radiator or the inter cooler may be impaired.

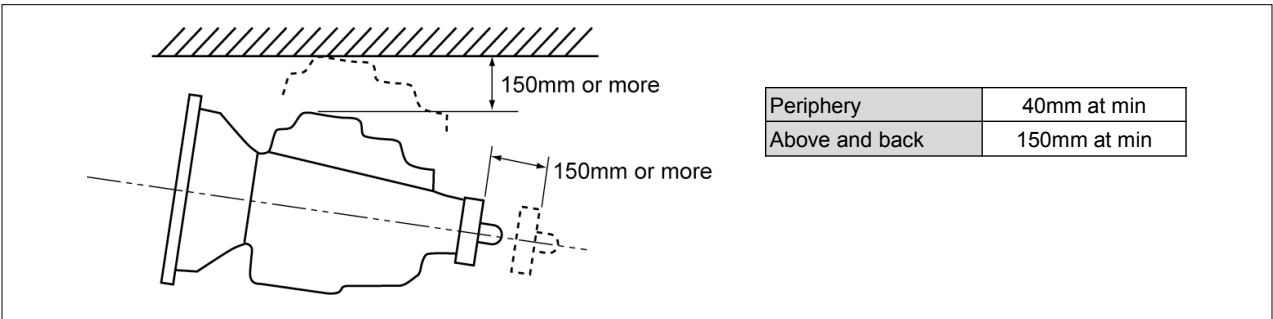
Clearance between engine and rear body



② Transmission

- Since transmission moves in unison with the engine, maintain a minimum of 40mm clearance around it.
- Transmission is to be retreated when removed. Therefore, not to contact anything, the transmission assembly must have a minimum of 150mm clearance behind it.
- Provide a minimum of 150mm clearance above the transmission assembly for check-ups and maintenance.

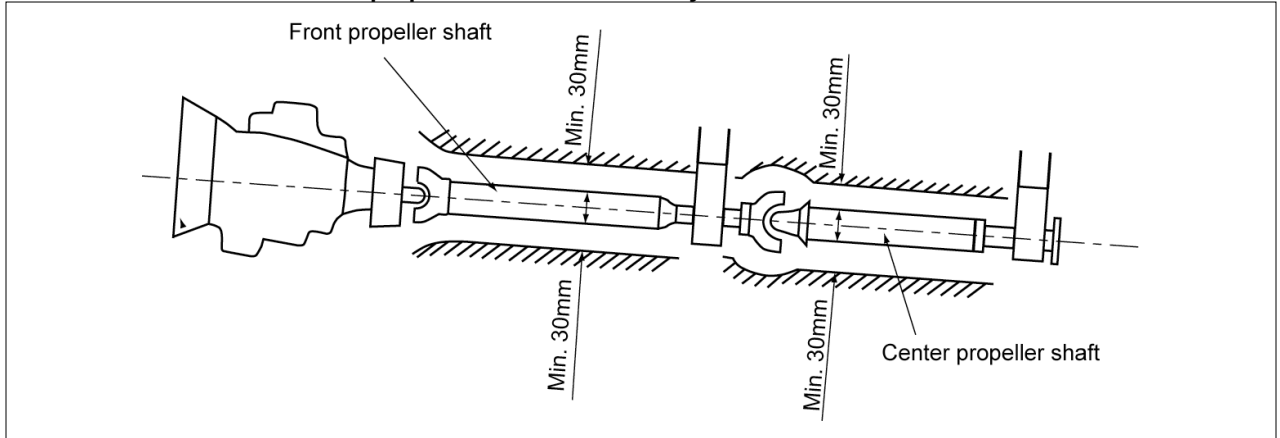
Clearance between transmission assembly and rear body



③ Front and center propeller shafts

- Provide a minimum of 30mm clearance around front and center propeller shafts.
- Equipment that emits high heat should be distant from the propeller shafts. If such equipment has to be placed near a propeller shaft, provide a heat insulation between them.

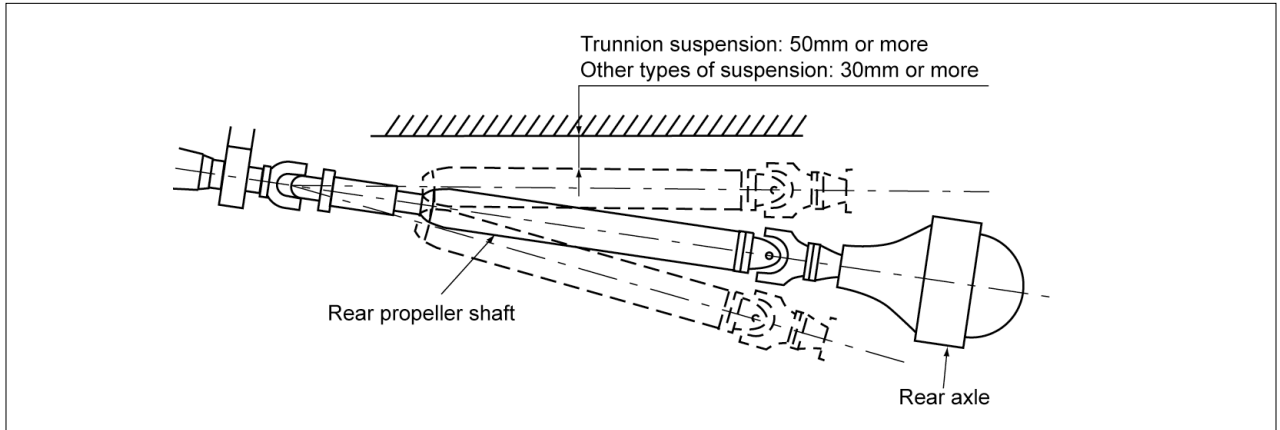
Clearance between front/center propellar shaft and rear body



④ Rear propeller shaft and rear axle

- There should be the following minimum clearance around rear propeller shaft and rear axle.

Clearance between rear propellar shaft/rear axle and rear body

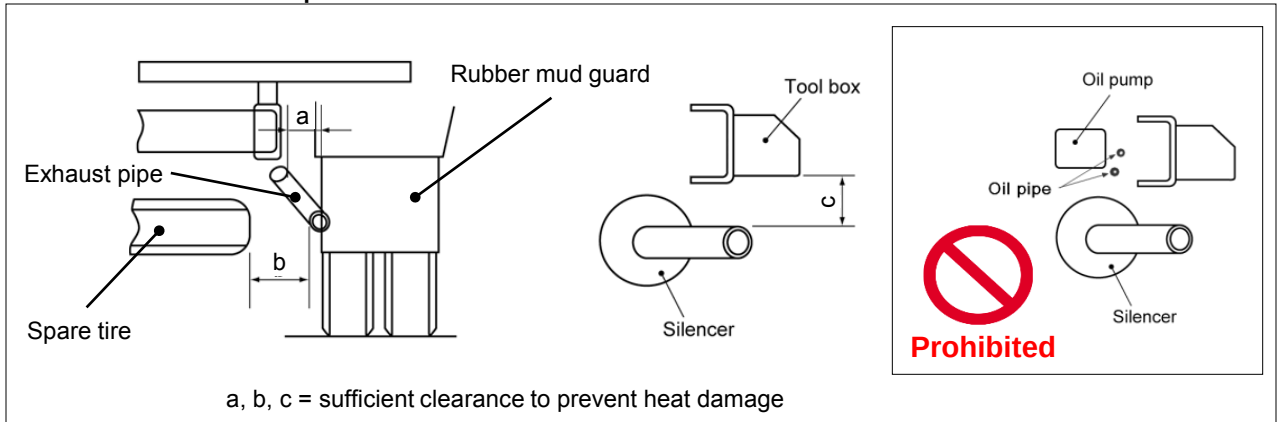


- Clearance between rear axle and chassis frame from spring deflection differs in each vehicle model. Refer to "SPECIFIC INFORMATION" of target vehicle model.

⑤ Exhaust system

- If a rear body is partially made of combustible materials such as wood and rubber, provide 100mm of clearance between the rear body and an exhaust pipe, a silencer or catalytic converter.
- If such clearance cannot be assured, provide a heat insulation between them.
- An oil pump or its piping for the equipments or the hydraulic system on a rear body must not be placed near exhaust system components such as a silencer or catalytic converter.
 - Smoking and the ignition when the leakage occurs in the hydraulic system.

Clearance of exhaust components

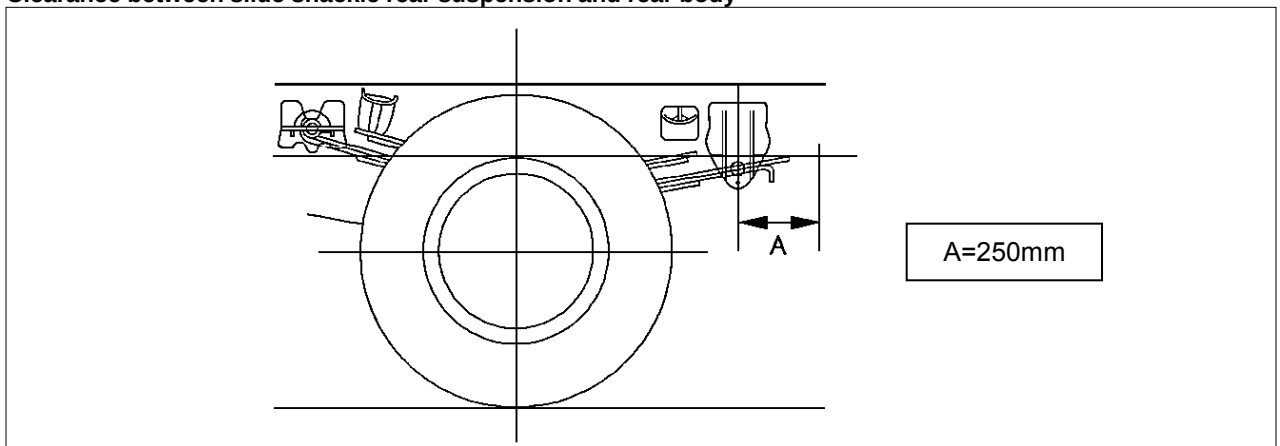


- The condition of exhaust emission device may be individually shown by the form and the vehicle type, and refer to "SPECIFIC INFORMATION" of the targeted vehicle type.

⑥ Slide shackle rear suspension

- As for slide shackle rear suspension, the rear end of the spring stretches out. Therefore, the rear body must not be mounted on the portion 'A' shown in the drawing below:

Clearance between slide shackle rear suspension and rear body



⑦ Pipes, cables, and wire harness

- Maintain the following clearance.

Between rear body and movable (engine, mission, etc.) wire harness and piping	40mm or more (most approached)
Between rear body and half fixed wiring and piping (wire harnesses cables, fuel piping)	10mm or more from most approached position
Between rear body and fixed wire harnesses and piping	battery cables: 15mm or more other wire harnesses: 10mm or more

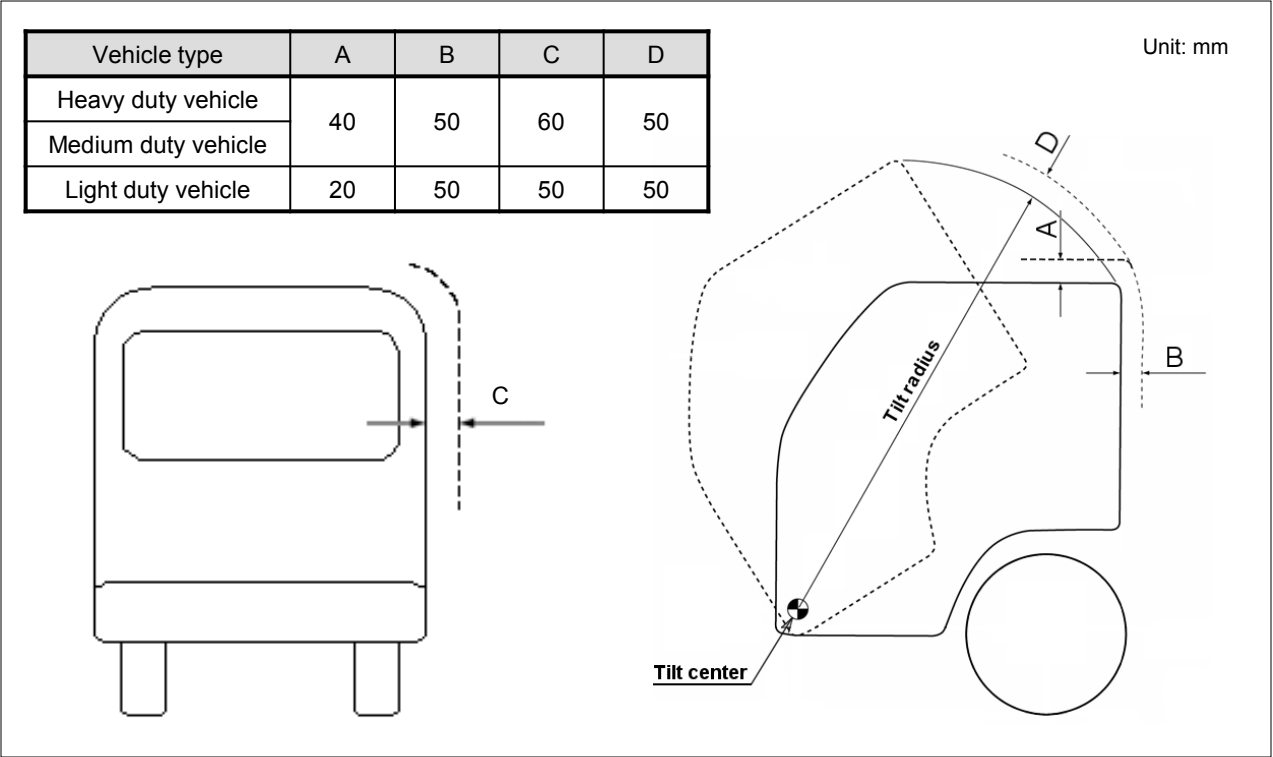
Note : When above condition is difficult to be satisfied, clip fix to avoid contact with rear body.

- Corrosion of pipes and cables promotes on tank truck, vacuum truck, etc. due to water and dew condensation at low-temperature part. Place protection board or keep pipes , cables and wires away from water dripping or dew condensation part.

⑧ Cab

- When the vehicle is moving, the whole cab moves in various directions in conjunction with cab suspension. Also, when a cab tilts, rear body may contact with the cab or cab chassis components such as an air inlet duct attached to the cab. Taking these into account, provide sufficient clearances between them as shown below to prevent the cab from contacting with anything.

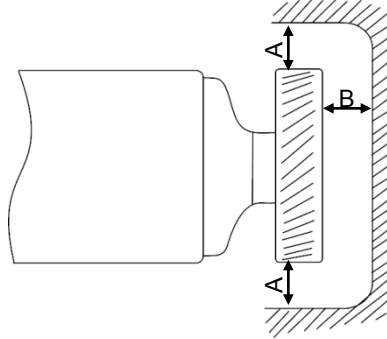
Clearance of cab



⑨ Retarder

- The retarder generate heat higher than 200°C. Maintain clearance of 100mm for both A and B from combustible materials such as wood and rubber.

Clearance between retarder and rear body

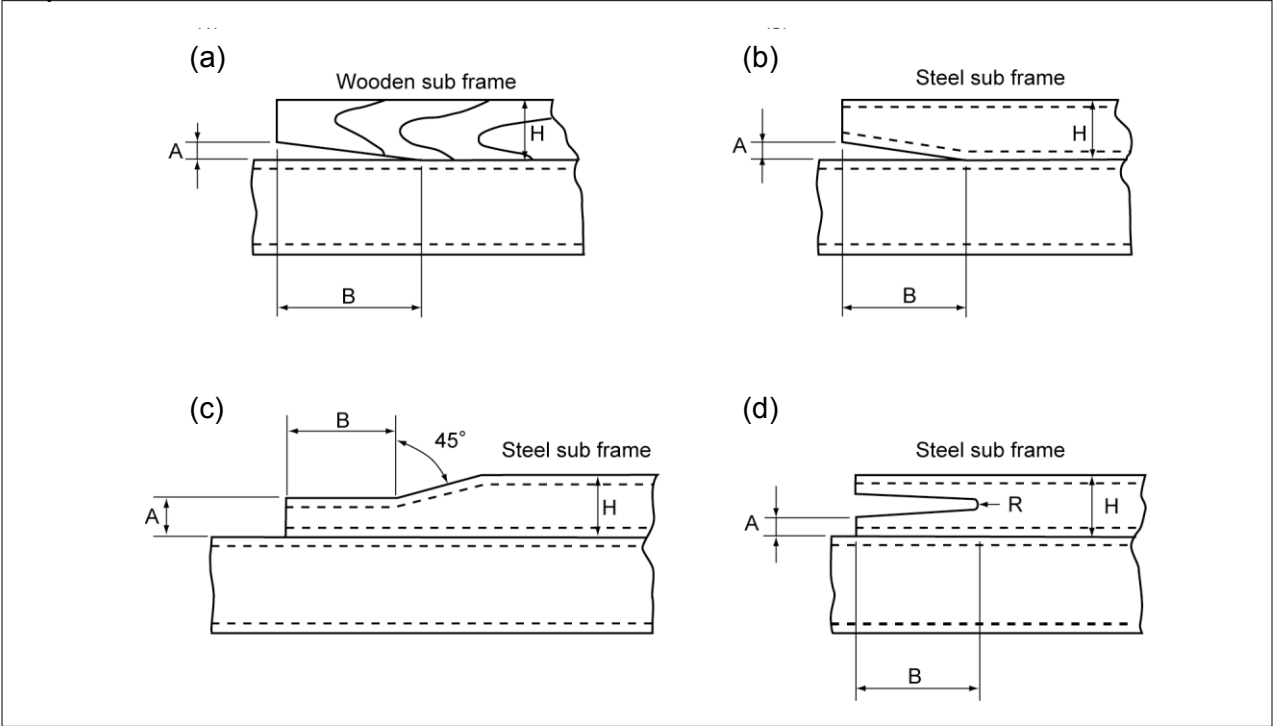


3-3 SUB FRAME SIDE MEMBERS

① Shape of the front end of sub frame side members

- The sub frame prevents load concentration in chassis frame when mounting rear body.
- Since load stress tends to concentrate in the area near the cross member behind cab back, appropriately proceed shape the ends of sub frame side members suited to rear body referring to the figure below so that the sub frame produces no acute change in the frame section.

Shape of the front end of sub frame



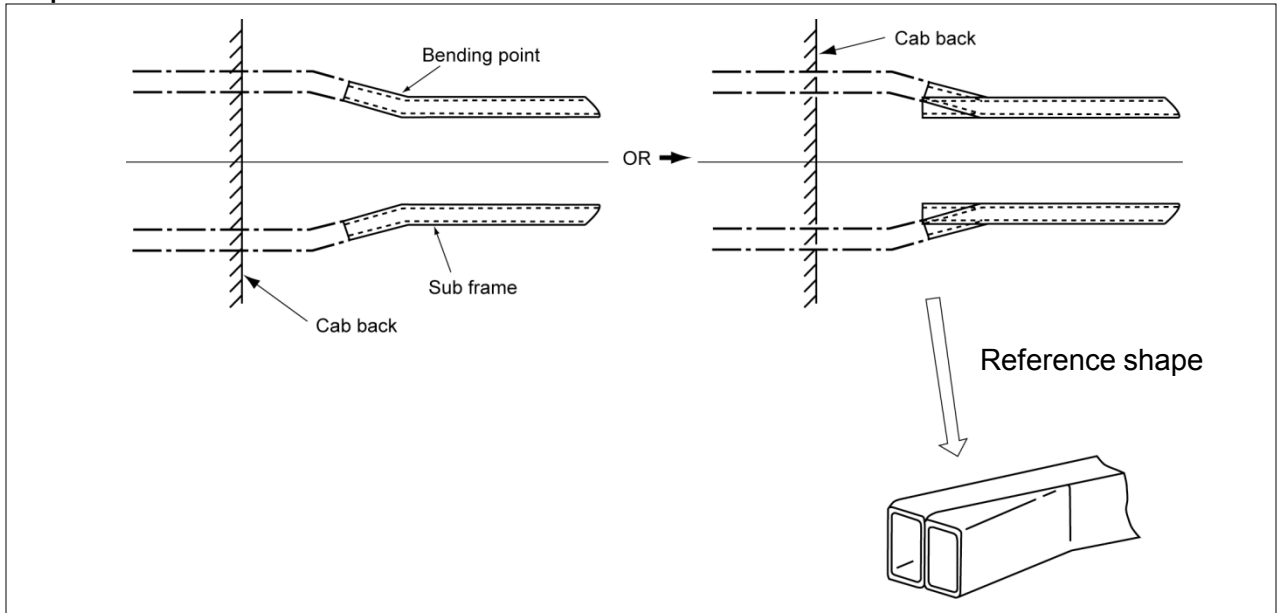
Unit: mm

	A	B	R
(a)	5	H/2 to H	—
(b)	5	H at min	—
(c)	H/3	H at min	—
(d)	H/4 to H/5	H to 1.5 times H	15 to 20

② Measure against sub frame

- When installing a sub frame on a chassis frame, covering the area where the width between chassis frame side members begin to change, sub frame side members need to be carefully formed so that they will follow the shape and the dimensions of the bending portion of the chassis frame side members.

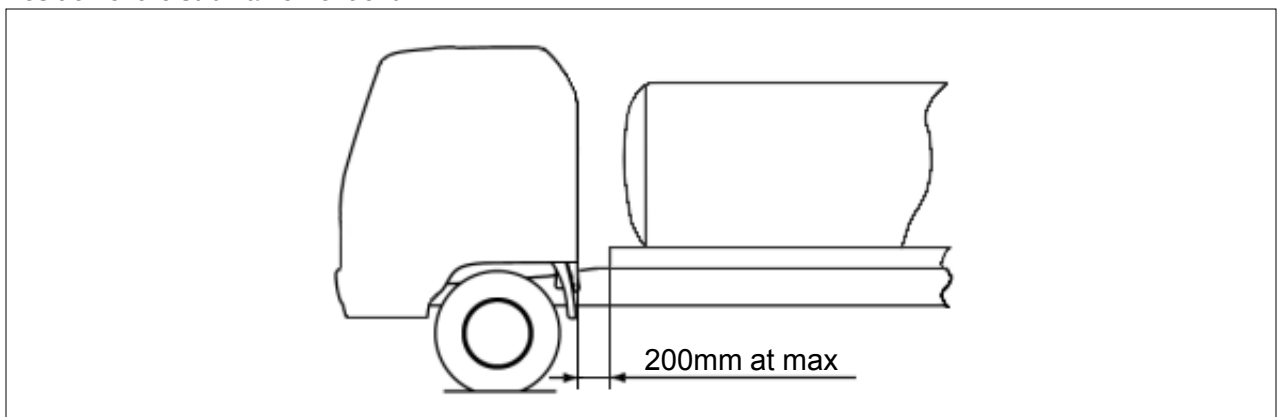
Shape of the chassis frame and sub frame



③ Front end of a sub frame

- Mislocation of the front end of a sub frame may result in cracks in a chassis frame in that area, or deterioration in ride comfort. The distance between the rear face of cab back and the front end of a sub frame should be no longer than 200mm.

Position of the sub frame front end



3-4 FIXING OF A REAR BODY

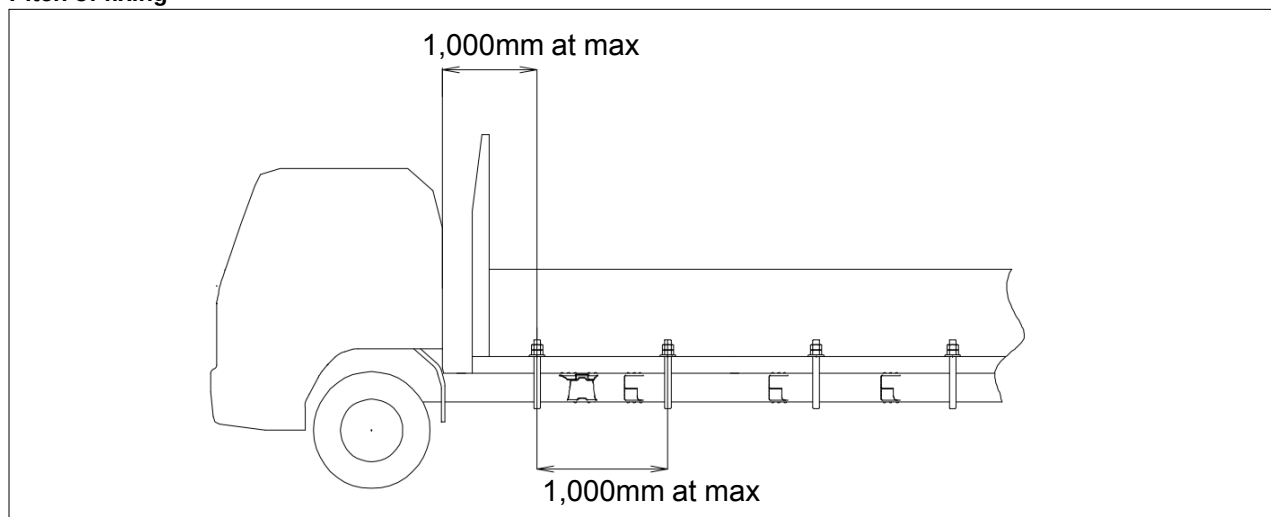
Fixing with U-bolt is recommended in ISUZU MOTORS.

Do not weld the sub frame with chassis frame directly or bolt on flange to fix rear body.

① Location and pitch of fixing

- The first fixing location
 - Fixing should be distant from the cab back by 1,000mm or less.
 - Follow the recommended location described in each vehicle model.
- Pitch of fixing
 - Fixing other than those in the rear axle area should be spaced no longer than 1,000mm. Securely fasten them on a chassis frame.

Pitch of fixing

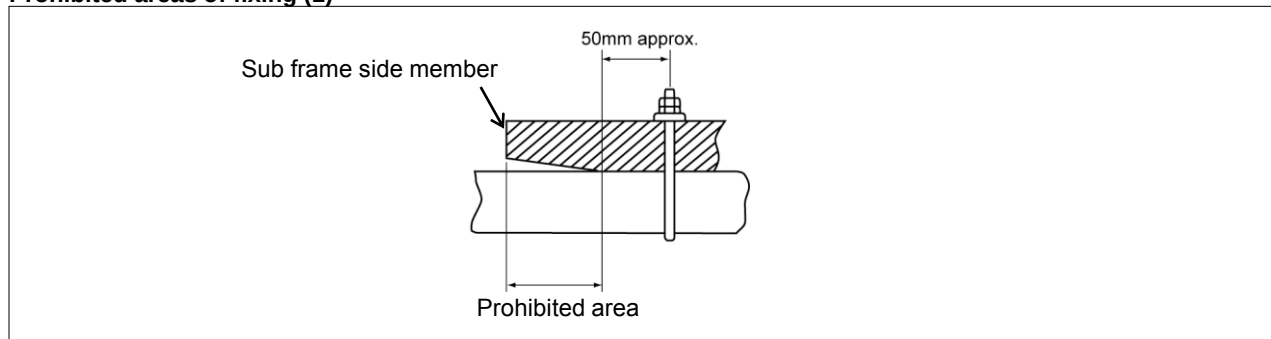


② Fixing prohibited areas

U-bolts and double-ended brackets are prohibited in the following areas.

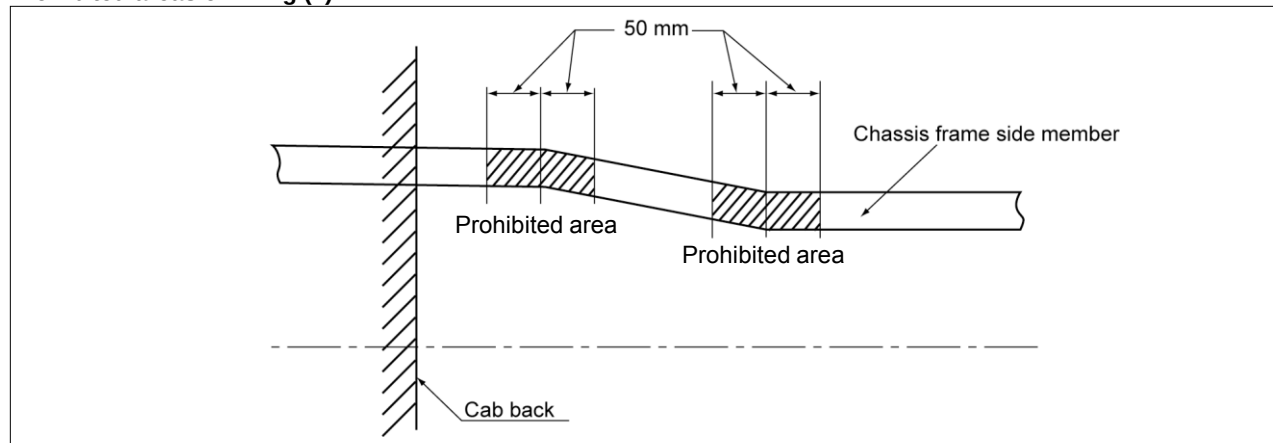
- Area from the front end of a sub frame side members to 50mm posterior to a kick-up point.

Prohibited areas of fixing (1)



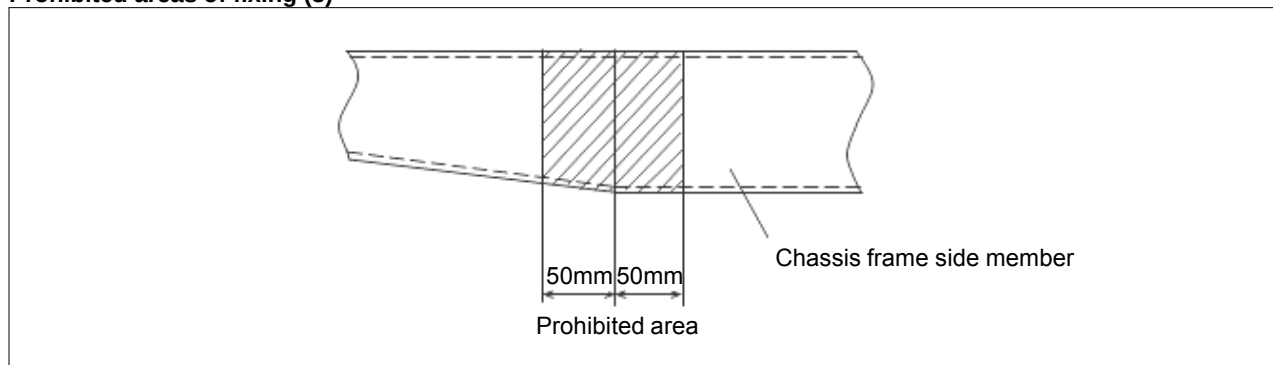
- Areas within 50mm anterior or posterior to the point where the width between chassis frame side members begin to change.

Prohibited areas of fixing (2)



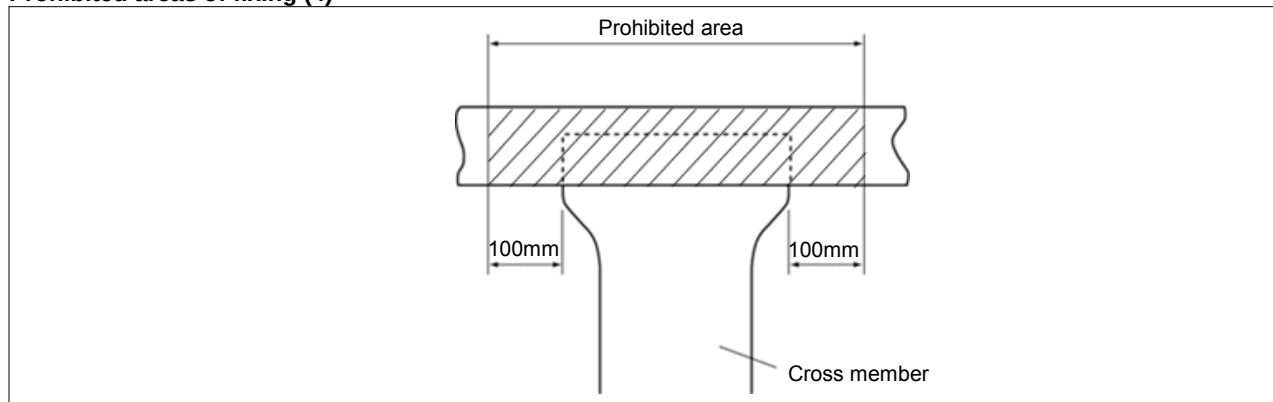
- Areas within 50mm anterior or posterior to the point where height of chassis member changes (frame kick point).

Prohibited areas of fixing (3)



- Areas within 100mm anterior or posterior to each cross member, gusset or spring bracket.

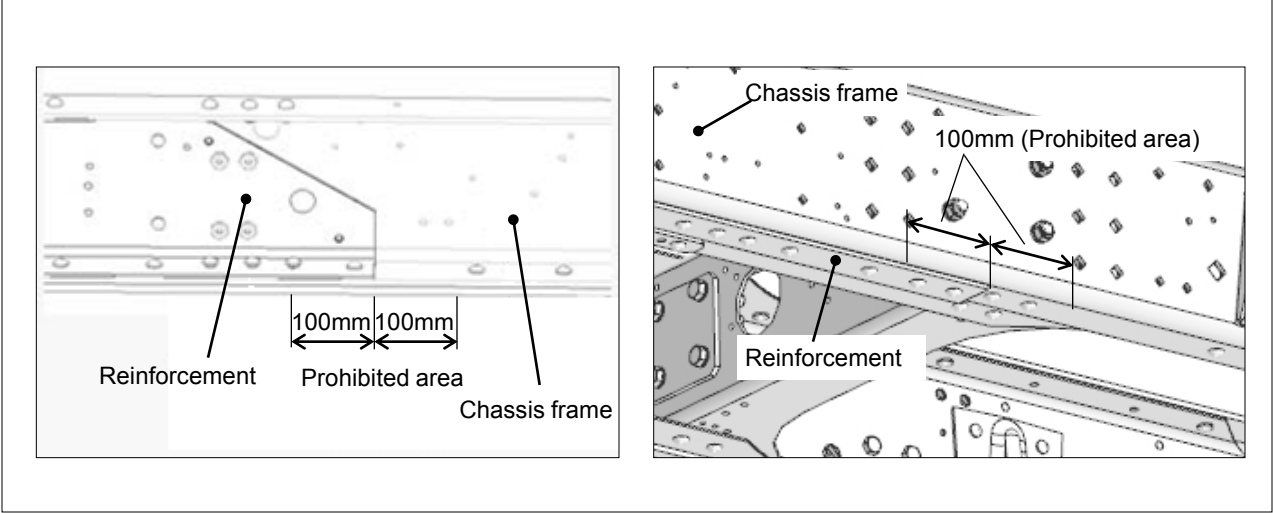
Prohibited areas of fixing (4)



- Check “SPECIFIC INFORMATION” of target vehicle for cross member shapes and details.
- Areas within 100mm anterior or posterior to rear suspension components or edge of its reinforcement.

- Areas within 100mm anterior or posterior to edge of inner or outer reinforcement.

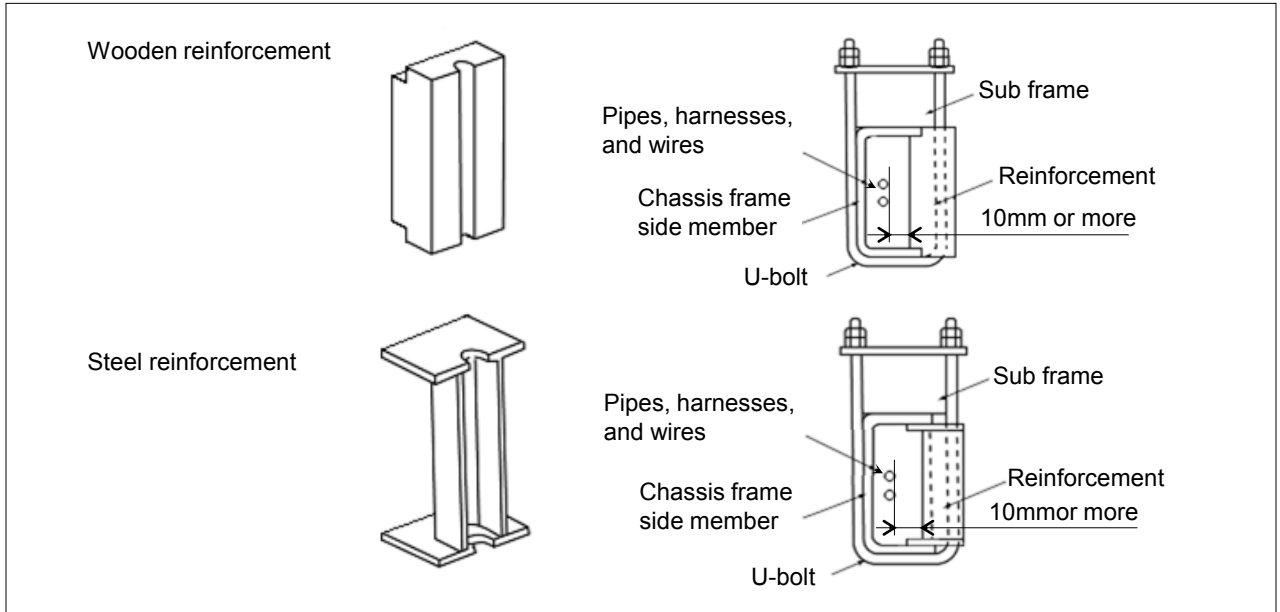
Prohibited areas of fixing (5)



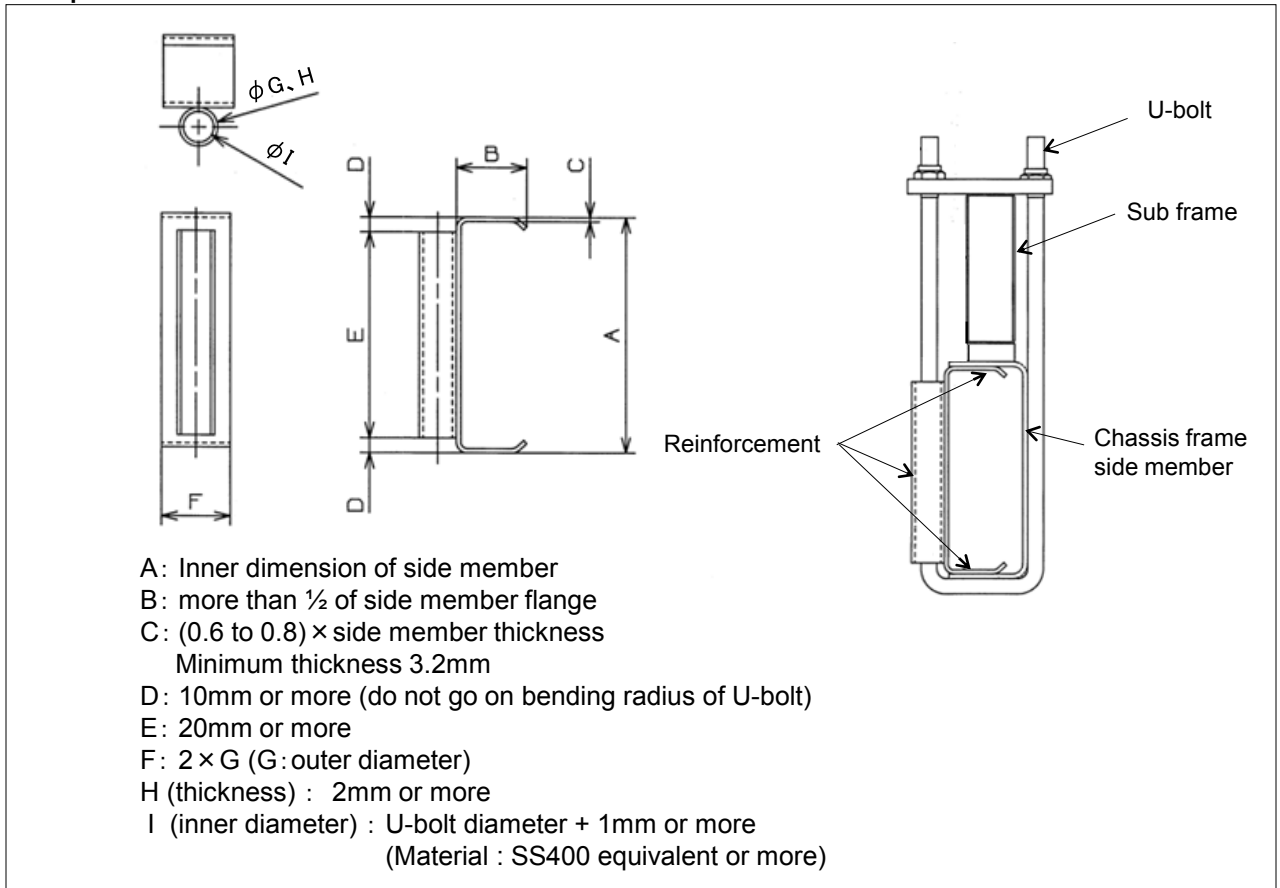
③ Caution at fixing with U-bolts

- When using U-bolts to fix a sub frame on a chassis frame, use a reinforcement on the inner side of chassis frame side members.

Installing of reinforcement

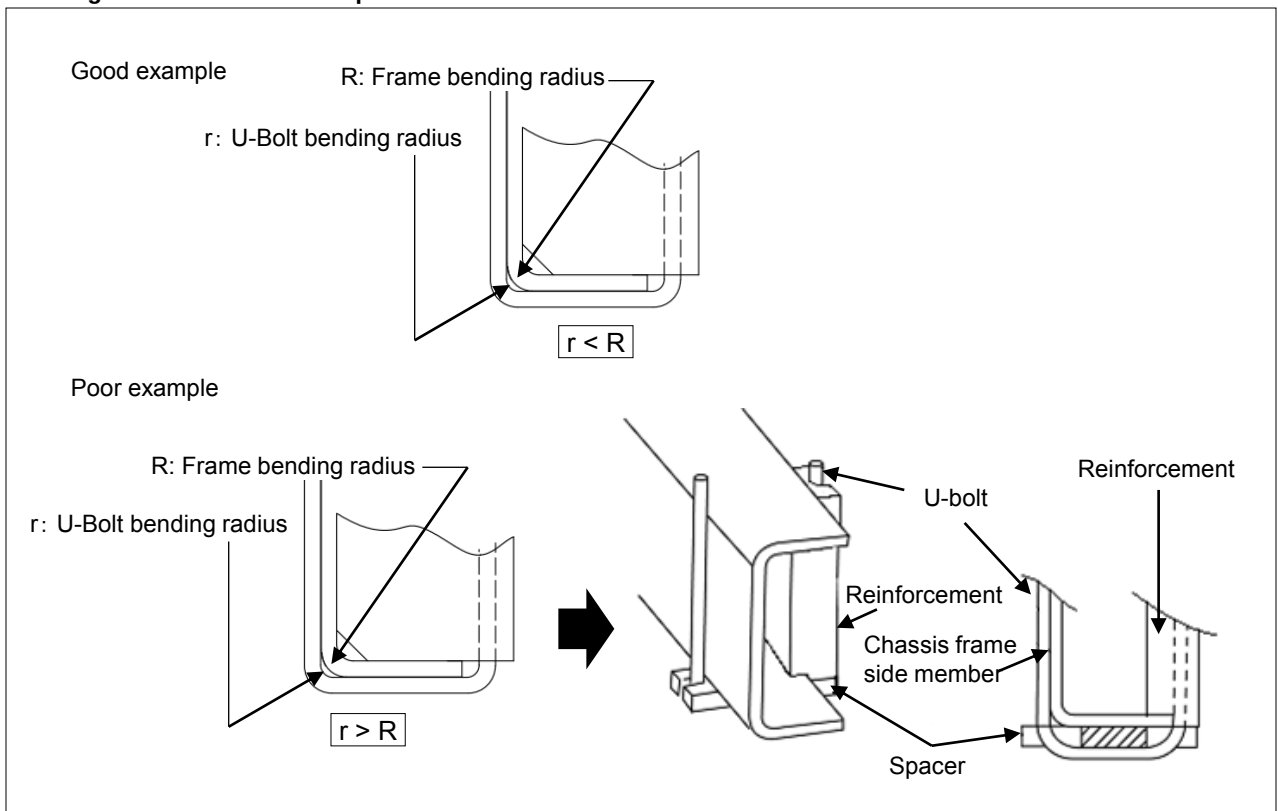


Example of reinforcement



- Welding reinforcement or U-bolt to frame is prohibited (include tack weld).
- The installation direction of U-bolts should be either downward or upward.
- It is recommended to use steel reinforcement for the area near exhaust system components. If wooden reinforcement are preferable, be sure to provide a sufficient clearance between a wooden block and an exhaust system component.
- If any pipe or harness is located near a reinforcement block, maintain a minimum of 10mm clearance between the block and fixed part or unmoving part of such pipe or harness, and a minimum of 40mm between the block and moving part of the pipe or the harness.
- U-bolts should be parallel to the side member flange.
 - Use U-bolts with bending radius smaller than that of chassis frame side member.
 - Insert a spacer as shown below if a U-bolt with bending radius smaller than that of chassis frame side member is difficult to obtain.

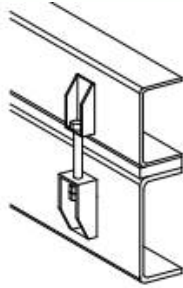
Bending radius of U-bolt and spacer installation



④ Caution at fixing with double-ended brackets

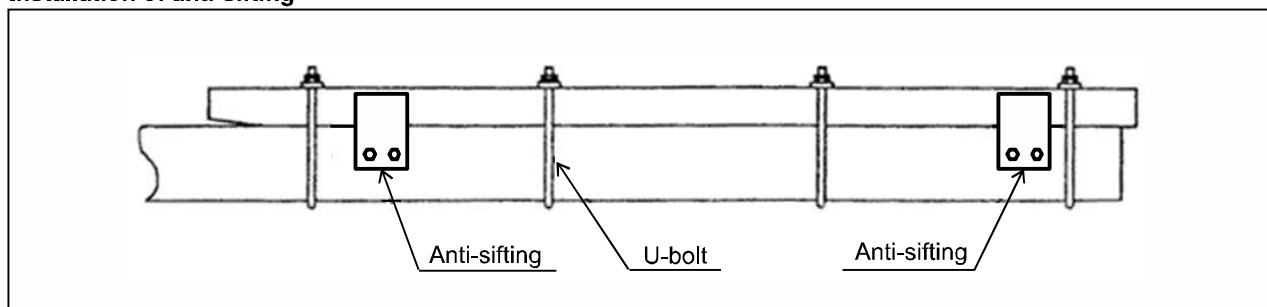
- For fixing of the rear body, the use of U-bolt is most recommended. Note the following when using the double-ended brackets due to various restrictions. Only fix with bolts on chassis frame. Do not weld.
- Follow " I -2-1 DRILLING OR WELDING A CHASSIS FRAME" for hole location, etc.
- For the location of fixing, see " I -3-4 ② Fixing prohibited areas" and do not drill holes or weld in that area.
- Use fastening bolt vertically as shown in below figure.

Fixing of double-ended brackets



⑤ Caution at using anti-sifting

Installation of anti-sifting



- Follow "I-2-1 DRILLING OR WELDING A CHASSIS FRAME" and "I-3-4 ② Fixing prohibited areas" for hole location, etc.
- The general characteristics and cautions of fastening device is shown below. Be aware of the characteristics, the mounting direction, and the method.

Example and characteristics of anti-sifting bracket

Method	Feature and notes
Double ended bracket	• Use in inverted v shape for anti-sifting forward and backward direction. • Bolt fix to chassis frame. • Tightening force decreases compare to vertical use. Use U-bolt as well for installing body.
"V" "I" type anti-sifting bracket for forward and backward direction	• Bolt fix to chassis frame. • Tightening with the body cannot be expected. Use U-bolt as well for installing body.
Anti-sifting bracket for right and left direction	• Bolt fix to chassis frame. • Tightening with the body cannot be expected. Use U-bolt as well for installing body.